National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Chronic Lymphocytic Leukemia/ Small Lymphocytic Lymphoma

NCCN Evidence Blocks™

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NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

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NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

See NCCN Categories of Preference.

NCCN Guidelines for Patients® available at www.nccn.org/patients

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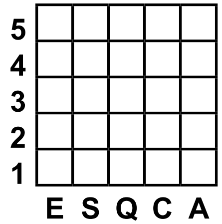
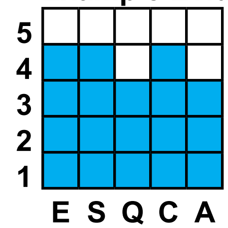
NCCN EVIDENCE BLOCKS CATEGORIES AND DEFINITIONS



E = Efficacy of Regimen/Agent
S = Safety of Regimen/Agent
Q = Quality of Evidence
C = Consistency of Evidence
A = Affordability of Regimen/Agent

Example Evidence Block



E = 4
S = 4
Q = 3
C = 4
A = 3

Efficacy of Regimen/Agent

5	**Highly effective:** Cure likely and often provides long-term survival advantage
4	**Very effective:** Cure unlikely but sometimes provides long-term survival advantage
3	**Moderately effective:** Modest impact on survival, but often provides control of disease
2	**Minimally effective:** No, or unknown impact on survival, but sometimes provides control of disease
1	**Palliative:** Provides symptomatic benefit only

Safety of Regimen/Agent

5	**Usually no meaningful toxicity:** Uncommon or minimal toxicities; no interference with activities of daily living (ADLs)
4	**Occasionally toxic:** Rare significant toxicities or low-grade toxicities only; little interference with ADLs
3	**Mildly toxic:** Mild toxicity that interferes with ADLs
2	**Moderately toxic:** Significant toxicities often occur but life threatening/fatal toxicity is uncommon; interference with ADLs is frequent
1	**Highly toxic:** Significant toxicities or life threatening/fatal toxicity occurs often; interference with ADLs is usual and severe

Note: For significant chronic or long-term toxicities, score decreased by 1

Quality of Evidence

5	**High quality:** Multiple well-designed randomized trials and/or meta-analyses
4	**Good quality:** One or more well-designed randomized trials
3	**Average quality:** Low quality randomized trial(s) or well-designed non-randomized trial(s)
2	**Low quality:** Case reports or extensive clinical experience
1	**Poor quality:** Little or no evidence

Consistency of Evidence

5	**Highly consistent:** Multiple trials with similar outcomes
4	**Mainly consistent:** Multiple trials with some variability in outcome
3	**May be consistent:** Few trials or only trials with few patients, whether randomized or not, with some variability in outcome
2	**Inconsistent:** Meaningful differences in direction of outcome between quality trials
1	**Anecdotal evidence only:** Evidence in humans based upon anecdotal experience

Affordability of Regimen/Agent (includes drug cost, supportive care, infusions, toxicity monitoring, management of toxicity)

5	**Very inexpensive**
4	**Inexpensive**
3	**Moderately expensive**
2	**Expensive**
1	**Very expensive**

DIAGNOSIS

ESSENTIAL:

- Hematopathology review of peripheral blood smear and all slides with at least one paraffin block representative of the tumor, if the diagnosis was made on a lymph node or bone marrow biopsy.
- Flow cytometry of blood is adequate for the diagnosis of chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL).
  - CLL diagnosis requires the presence of monoclonal B lymphocytes ≥5 x 10^9/L in peripheral blood based on flow cytometry[a]
    - ◊ Adequate immunophenotyping to establish diagnosis by flow cytometry[b]: kappa/lambda, CD19, CD20, CD5, CD23, CD10, CD200
    - ◊ If flow cytometry is used to establish diagnosis, also include cytospin for cyclin D1 or fluorescence in situ hybridization (FISH) for t(11;14); t(11q;v) to exclude mantle cell lymphoma (MCL).
  - Clonality of B cells should be confirmed by flow cytometry
  - SLL diagnosis requires presence of lymphadenopathy and/or splenomegaly with monoclonal B lymphocytes ≤5 x 10^9/L in peripheral blood. SLL diagnosis should be confirmed by histopathology evaluation of lymph node biopsy
    - ◊ Excisional or incisional biopsy, if lymph node is accessible. Fine-needle aspiration (FNA) biopsy alone is inadequate for the diagnosis. In certain circumstances, when a lymph node is not easily accessible for excisional or incisional biopsy, a combination of core needle biopsy (multiple biopsies may be required) in conjunction with appropriate ancillary techniques, such as FNA, may be sufficient for diagnosis.
    - ◊ Bone marrow aspirate with biopsy can be pursued if peripheral blood and lymph node biopsy material are nondiagnostic.
    - ◊ Adequate immunophenotyping to establish diagnosis by immunohistochemistry (IHC)[b]: CD3, CD5, CD10, CD20, CD23, cyclin D1, LEF1, SOX11[c]
- Absolute monoclonal B lymphocyte count[a]

INFORMATIVE FOR PROGNOSTIC AND/OR THERAPY DETERMINATION[d]:

- FISH to detect: +12; del(11q); del(13q); del(17p)
- *TP53* sequencing
- CpG-stimulated metaphase karyotype for complex karyotype (CK)
- Molecular analysis to detect: Immunoglobulin heavy chain variable region gene (IGHV) mutation status
- Beta-2-microglobulin

→ **CLL/SLL** → Workup (CSLL-2)

→ **Monoclonal B-cell lymphocytosis (MBL)**
- Absolute monoclonal B-lymphocyte count <5 x 10^9/L
- All palpable lymph nodes <1.5 cm
- No anemia
- No thrombocytopenia
- No palpable organomegaly
- No constitutional symptoms

→ Observe[e]

Footnotes on CSLL-1A

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
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FOOTNOTES

[a] Absolute monoclonal B lymphocyte count <5000/mm^3 that persists more than 3 months in the absence of palpable adenopathy or other clinical features of lymphoproliferative disorder is MBL. Cells of the same phenotype may be seen in reactive lymph nodes; therefore, diagnosis of SLL should only be made when effacement of lymph node architecture is seen. Bone marrow examination is not helpful for the diagnosis of MBL.

[b] Typical immunophenotype: CD5+, CD23+, CD43+/-, CD10-, CD19+, CD20 dim, sIg dim+, and cyclin D1-. Note: Some cases may be sIg bright+ or CD23- or dim; some MCL may be CD23+; cyclin D1 immunohistochemistry or FISH for t(11;14) should be considered in all cases, especially for those with an atypical immunophenotype (ie, CD23 dim or negative, CD20 bright, sIg bright). CD200 positivity may distinguish CLL from MCL, which is usually CD200-.

[c] LEF1 and SOX11 may be helpful in suspected cases of MCL that are cyclin D1-negative.

[d] Prognostic Variables for CLL/SLL (CSLL-A).

[e] Outside of clinical trials, CT scans are not required for diagnosis, serial monitoring, surveillance, routine monitoring of treatment response, or progression.

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WORKUP

ESSENTIAL:
- History and physical exam including measurement of size of liver and spleen and palpable lymph nodes
- Performance status
- B symptoms
- Complete blood count (CBC) with differential
- Comprehensive metabolic panel
- Beta-2-microglobulin
- Assess for distress[f]

USEFUL UNDER CERTAIN CIRCUMSTANCES:
- Quantitative immunoglobulins
- Serum protein electrophoresis (SPEP) with serum immunofixation electrophoresis (SIFE) to evaluate for monoclonal protein
- Reticulocyte count, haptoglobin, and direct antiglobulin test (Coombs)
- Chest/abdomen/pelvis (C/A/P) CT with contrast of diagnostic quality, if clinically indicated[g]
- Uric acid
- Lactate dehydrogenase (LDH)
- Unilateral bone marrow aspirate and biopsy (may be informative for the diagnosis of immune-mediated or disease-related cytopenias)
- Hepatitis B[h] and C testing if treatment is contemplated
- Pregnancy testing in patients of childbearing age if systemic therapy or RT is planned
- Discussion of fertility preservation[i]
- Whole body FDG-PET/CT scan to direct nodal biopsy, if histologic transformation is suspected. See HT-1.

→ SLL/Localized (Lugano Stage I) (CSLL-3)

→ CLL (Rai Stages 0–IV) or SLL (Lugano Stage II–IV) (CSLL-3)

[f] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
[g] Outside of clinical trials, CT scans are not required for diagnosis, serial monitoring, surveillance, routine monitoring of treatment response, or progression. CT scans may be warranted for the evaluation of symptoms of bulky disease or for the assessment of risk for TLS prior to initiating venetoclax.
[h] Hepatitis B testing is indicated because of the risk of reactivation during treatment (eg, immunotherapy, chemoimmunotherapy [CIT], chemotherapy, targeted therapy). See Treatment and Viral Reactivation (CSLL-C 1 of 6). Tests include hepatitis B surface antigen (HBsAg) and core antibody for a patient with no risk factors. For patients with risk factors or previous history of hepatitis B, add e-antigen. If positive, check viral load and consult with a gastroenterologist.
[i] Fertility preservation options include: sperm banking, in vitro fertilization (IVF), or ovarian tissue or oocyte cryopreservation.

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PRESENTATION[j]

SLL/Localized (Lugano Stage I)[k] → **Locoregional involved-site radiation therapy (ISRT)[l] (if indicated)** → **Observe[e]**

SLL (Lugano Stages II–IV)[k] →
CLL (Rai Stage 0–IV)[k] →
Observe →

Evaluate for indications for treatment[m]:
- Eligible for clinical trial
- Significant disease-related symptoms:
  - Fatigue (severe)
  - Drenching night sweats
  - Unintentional weight loss (≥10% in previous 6 months)
  - Fever without infection
- Threatened end-organ function
- Progressive, symptomatic, or bulky disease (spleen >6 cm below costal margin, lymph nodes >10 cm)
- Progressive thrombocytopenia[n,o]
- Progressive anemia[n]
- Steroid-refractory autoimmune cytopenias

→ **No indication** → Observe[e]

→ **Indication present** →
- Prior to initiation of treatment, evaluate[p]
  - FISH for del(17p)
  - *TP53* mutation status
  - CpG-stimulated metaphase karyotype
- IGHV mutation status[q]
- Imaging as appropriate
- Bone marrow aspirate and biopsy as clinically indicated

→ **First-line therapy (CSLL-4)**

Histologic transformation (Richter) → HT-1

[e] Outside of clinical trials, CT scans are not required for diagnosis, serial monitoring, surveillance, routine monitoring of treatment response, or progression.
[j] Supportive Care for Patients with CLL/SLL (CSLL-C).
[k] See Rai and Binet Classification Systems (CSLL-B 1 of 2) and Lugano Modification of Ann Arbor Staging System (CSLL-B 2 of 2).
[l] See NCCN Guidelines for B-Cell Lymphomas, Principles of Radiation Therapy for additional details.
[m] Absolute lymphocyte count (ALC) alone is not an indication for treatment in the absence of leukostasis, which is rarely seen in CLL.
[n] Platelet counts >100,000/µL are typically not associated with clinical risk.
[o] Select patients with mild, stable cytopenia (ANC <1000/µL, Hgb <11 g/dL, or platelet <100,000/µL) may continue to be observed. Other causes of anemia/thrombocytopenia (eg, autoimmune disorders, vitamin/iron deficiency) should be excluded.
[p] Consider next generation sequencing (NGS)-based clonality testing for measurable residual disease (MRD) assessment if MRD-guided treatment is planned and MRD flow is not being used for MRD assessment. See CSLL-E.
[q] IGHV mutation status does not change over time and analysis does not need to be repeated if previously done prior to initiation of treatment.

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FIRST-LINE THERAPY[r,s]	RESPONSE TO THERAPY[u]	SECOND-LINE THERAPY[r,s,v,z,aa]	THIRD-LINE THERAPY[r,s,v,z,aa]

Suggested Regimens (CSLL-D 2 of 6)

Time-limited treatment[t]

BCL2 inhibitor (BCL2i) + anti-CD20 mAb See Suggested Regimens (CSLL-D 1 of 6)
- **BCL2i intolerance**[v] → **cBTKi** → Relapsed or Refractory disease after prior therapy with cBTKi or ncBTKi-[y] and BCL2i-containing regimens → CSLL-5A
- **Response after completion of treatment** → **Observe until relapse with indication for treatment** → **BCL2i-containing regimens**[bb] → **cBTKi** → Relapsed or Refractory disease … → CSLL-5A; **or cBTKi** → Relapsed or Refractory disease … → CSLL-5A
- **Progression on treatment**[w,x] → **cBTKi** → Relapsed or Refractory disease … → CSLL-5A

or

BCL2 inhibitor (BCL2i) + cBTKi ± anti-CD20 mAb See Suggested Regimens (CSLL-D 1 of 6)
- **BCL2i intolerance**[v] → **cBTKi** → Relapsed or Refractory disease … → CSLL-5A
- **cBTKi intolerance**[v] → **BCL2i + alternate cBTKi or BCL2i ± anti-CD20 mAb** → Relapsed or Refractory disease … → CSLL-5A
- **Response after completion of treatment** → **Observe until relapse with indication for treatment** → **BCL2i-containing regimens**[bb] **or cBTKi** → Relapsed or Refractory disease … → CSLL-5A
- **Progression on BCL2i + cBTKi**[w,x,y] → **Noncovalent (reversible) BTKi (ncBTKi)** → Relapsed or Refractory disease … → CSLL-5A

or

Continuous treatment

Covalent BTK inhibitor (cBTKi-based regimens) ± anti-CD20 mAb → CSLL-5

Relapsed or Refractory disease after prior therapy with cBTKi or ncBTKi-[y] and BCL2i-containing regimens → CSLL-5A

Sequencing therapy following CIT as first-line therapy (See CSLL-D 1 of 6 for suggested treatment regimens): Observation until relapse with indications for treatment is recommended for response after completion of CIT. Time-limited treatment with BCL2i-containing regimens or cBTKi-based regimens are options for relapse after a period of remission or progression on treatment. See CSLL-D 2 of 6 for suggested treatment regimens for second-line and subsequent therapy.

Footnotes on CSLL-5A

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FIRST-LINE THERAPY[r,s] | **RESPONSE TO THERAPY[u]** | **SECOND-LINE THERAPY[r,s,v,z,aa,bb]** | **THIRD-LINE THERAPY[r,s,v,z,aa,bb]**

Suggested Regimens (CSLL-D 2 of 6)

Continuous treatment

Covalent BTK inhibitor (cBTKi-based regimens) ± anti-CD20 mAb
See Suggested Regimens (CSLL-D 1 of 6)

- **Intolerance[v]** → **Alternate cBTKi or ncBTKi** → **BCL2i-containing regimens**
  - or **BCL2i-containing regimens** → **Alternate cBTKi or ncBTKi**
- **Response** → **Continue treatment with the same cBTKi[cc] until intolerance and/or progression** (→ Intolerance; → Progression while on treatment)
- **Progression while on treatment[w,x,y]** → **BCL2i-containing regimens or ncBTKi**

All third-line paths / BCL2i-containing regimens or ncBTKi → **Relapsed or Refractory disease after prior therapy with cBTKi or ncBTKi-[y] and BCL2i-containing regimens** → CSLL-5A

Sequencing therapy following CIT as first-line therapy (See CSLL-D 1 of 6 for suggested treatment regimens): Observation until relapse with indications for treatment is recommended for response after completion of CIT. Time-limited treatment with BCL2i-containing regimens or cBTKi-based regimens are options for relapse after a period of remission or progression on treatment. See CSLL-D 2 of 6 for suggested treatment regimens for second-line and subsequent therapy.

Footnotes on CSLL-5A

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RELAPSED OR REFRACTORY THERAPY[r,s]

Relapsed or Refractory disease after prior therapy with cBTKi or ncBTKi-[y] and BCL2i-containing regimens →

Clinical trial
or
CAR T-cell Therapy (CSLL-D 2 of 6)
or
ncBTKi (if not previously given) (CSLL-D 2 of 6)
or
Other Recommended Regimens (CSLL-D 2 of 6; Therapy for relapsed or refractory disease after BTKi- and BCL2i-containing regimens)
or
Consider allogeneic hematopoietic cell transplant (HCT) if without significant comorbidities

FOOTNOTES FOR CSLL-4 and CSLL-5

[r] Clinical trial is preferred when available.
[s] Supportive Care for Patients with CLL/SLL (CSLL-C).
[t] Time-limited treatment with BCL2i + cBTKi ± CD20 mAb may be fixed-duration or MRD-guided (based on undetectable MRD [uMRD]-stopping criteria as described in clinical trials). See MRD Assessment (CSLL-E).
[u] See Discussion for iwCLL response criteria.
[v] Adverse reactions can be managed by following the recommended dose modifications in the FDA-approved package insert (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm).
[w] If progression with indication for subsequent therapy: Re-evaluate with FISH for del(17p)/TP53 mutation status and CpG-stimulated metaphase karyotype, prior to initiation of subsequent therapy.
[x] Consider the possibility of histologic transformation in patients with progressive disease. Biopsy is recommended to confirm histologic transformation. If histologic transformation see HT-1.
[y] Testing for BTK and PLCG2 mutations may be useful in patients with disease progression or no response while on BTKi therapy including if poor adherence is considered as a possible cause. BTK and PLCG2 mutation status alone is not an indication to change treatment in absence of disease progression. Alternate cBTKi (acalabrutinib, ibrutinib, or zanubrutinib) could be considered for intolerance in absence of disease progression. ncBTKi can be considered for resistance or intolerance to prior cBTKi-based regimens.
[z] In patients with disease responding to therapy: Continue the same BTKi until progression and/or intolerance. At time of disease progression, transition to next therapy as soon as possible. Treatment-free interval should be as short as possible. It is safe to overlap with venetoclax while on a BTKi. If treated with BCL2i-containing time-limited treatment, observe until relapse with indications for retreatment.
[aa] Patients with CLL/SLL without del(17p)/*TP53* mutation who discontinue a cBTKi for intolerance can remain off treatment (if there is no disease progression) for an extended period of time and alternate BTKi can be reinitiated when symptomatic.
[bb] Venetoclax ± anti-CD20 mAb (obinutuzumab preferred) or venetoclax + cBTKi is a treatment option for relapse after a period of remission. Single agent venetoclax (continuous) can also be considered as a treatment option for relapse after a period of remission.
[cc] In patients with no intolerance, cBTKi can be continued until disease progression while following recommended dose modification guidance as needed.

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PROGNOSTIC VARIABLES IN PATIENTS WITH CLL/SLL

Table 1: Cytogenetics and gene mutations

Method of Detection	Prognostic Variable	Clinical Association[a]
Interphase cytogenetics (FISH)	Del(13q) as a sole abnormality	Favorable prognosis: longest median overall survival (OS) in patients treated with chemoimmunotherapy (CIT).[1]
	Trisomy 12	Intermediate prognosis for time-to-first treatment (TTFT) in patients with newly diagnosed CLL; and intermediate OS in patients treated with chemoimmunotherapy.[1]
	None	Intermediate prognosis for TTFT in patients with newly diagnosed CLL; and intermediate OS in patients treated with chemoimmunotherapy.[1]
	Del(11q)	Shortened TTFT, and shorter median OS with chemoimmunotherapy.[1]
	Del(17p)[b]	• Inferior response, shorter treatment-free interval, and inferior survival with chemoimmunotherapy.[1-3] • Good responses but shorter progression-free survival (PFS) and OS with both time-limited venetoclax-based regimens[4,5] and covalent BTK inhibitor (cBTKi).[6-10]
DNA sequencing	IGHV[c] Mutated: >2% mutation or <98% homology with germline gene sequence Unmutated: ≤2% of mutation or ≥98% homology with germline gene sequence)	• Unmutated IGHV is associated with shorter TTFT and PFS with chemoimmunotherapy and time-limited venetoclax-based regimens, compared to mutated IGHV.[3,5,11-15] • Overall response rates are not correlated with IGHV mutation status in patients treated with cBTKi or time-limited venetoclax-based regimens.[6-10] • PFS and OS are not correlated with IGHV mutation status in patients treated with cBTKi.[6,7,16]
	TP53 mutation[d] with del(17p)	• Shorter PFS and OS with chemoimmunotherapy,[e,17-20] time-limited venetoclax-based regimens[4,5] and cBTKi.[6-10] • Increased resistance to chemoimmunotherapy and targeted therapies; significantly shortened TTFT and time-to-next treatment (TTNT) and potentially increased risk for Richter transformation. • Low burden variant allele frequencies (VAF, <10%) may behave similar to wild type (WT).[21,22]
	TP53 mutation[d] without del(17p)	*TP53* mutation in the absence of del(17p) is associated with shortened PFS and OS. However, PFS and OS outcomes on treatment with TP53 mutation but in absence of del(17p) may be more favorable than with concurrent *TP53* mutation and del(17p).[18,23]
	Other Significant Mutations	See Table 2 and Table 3 (CSLL-A 3 of 5)

Footnotes

References on CSLL-A (4 of 5) and (5 of 5)

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PROGNOSTIC VARIABLES IN PATIENTS WITH CLL/SLL

Table 1: Cytogenetics and gene mutations (continued)

Method of Detection	Prognostic Variable	Clinical Association[a]
CpG-stimulated metaphase karyotype[f]	Complex karyotype (CK) (≥3 unrelated clonal chromosome abnormalities in more than one cell on karyotype)	• Inferior survival and shorter time to disease progression with chemoimmunotherapy, time-limited venetoclax based regimens, and cBTKi.[24-27] • High CK (≥5 unrelated chromosomal abnormalities) is an adverse prognostic factor independent of clinical stage, IGHV mutation status, del(17p) and/or *TP53* mutation.[28,29]
Serum test	Beta-2 Microglobulin (B2M) (can be affected by renal dysfunction in a CLL-independent manner)	• Correlates with CLL/SLL disease burden and retains independent prognostic significance in multivariate models[30,31] • Elevated levels[g] are associated with overall response, treatment free intervals, and OS in patients treated with frontline chemoimmunotherapy[32] • Reduction in levels associated with prolonged PFS in patients treated with ibrutinib[33,34]

References on CSLL-A (4 of 5) and (5 of 5)

[a] This table provides the impact of molecular and cytogenetic variables on time-to-first treatment and survival for patients treated with chemoimmunotherapy and/or small molecule inhibitors. Prognostic markers and scoring systems (including CLL-IPI) can aid in integrating multiple variables, but OS can be underestimated for cohorts treated prior to the availability of current targeted/novel therapies.
[b] Prognostic significance may be dependent on the percentage of malignant cells with the deletion; prognosis is more favorable when the percentage of cells with del(17p) is low (Tam CS, et al. Blood 2009;114:957-964; Van Dyke DL, et al. Br J Haematol 2016;173:105-113). Patients with del(17p) often also have a concurrent *TP53* mutation.
[c] IGHV gene usage involving VH3-21 is considered higher risk regardless of the IGHV mutation status (Tobin G, et al. Blood 2002;99:2262-2264; Oscier DG, et al. Blood 2002;100:1177-1184; Oscier D, et al. Haematologica 2010;95:1705-1712). IGHV4–39 usage is associated with a higher risk for Richter's transformation (Rossi D, et al. Blood 2007;11: Abstract 3086).
[d] Patients can have *TP53* mutations independent of del(17p). *TP53* mutation status also provides additional prognostic information in addition to the cytogenetic abnormalities detected by FISH.
[e] Hazard ratios (HR) for PFS and OS with *TP53* mutation on treatment versus *TP53* WT are higher with chemoimmunotherapy compared to targeted therapies.
[f] Conventional metaphase FISH is difficult in CLL due to the very low in vitro proliferative activity of the leukemic cells. CK is based on results of metaphase karyotyping of CpG-stimulated CLL cells. Interphase FISH is the standard method to detect specific chromosomal abnormalities that may have prognostic significance. CpG oligonucleotide stimulation can be utilized to enhance metaphase cytogenetics.
[g] Cutoff values vary between institutions and studies. CLL-IPI utilizes a cutoff of >3.5 mg/L; other studies use a cutoff of 4 mg/L in patients treated with chemoimmunotherapy.

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PROGNOSTIC VARIABLES IN PATIENTS WITH CLL/SLL

Table 2: Baseline and acquired gene mutations

Mutated Gene	Clinical Association
ATM[35-38]	• Shorter TTFT, PFS, TTNT and OS with CIT compared to WT
BIRC3[36,39,40]	• Shorter TTNT, PFS with CIT compared to WT • May be more sensitive to BCL2i-containing regimens
NOTCH1[4,36,41,42]	• Shorter TTFT and PFS irrespective of the IGHV mutation status and shorter TTNT with CIT compared to WT • Shorter OS in historical cohorts • Correlated with higher risk for Richter transformation
SF3B1[36,43]	• Shorter TTFT and OS with CIT compared to WT
TP53	• See Table 1 (CSLL-A 1 of 5)

Table 3: Gene mutations associated with resistance to targeted therapies

Mutated Gene	Clinical Association
***BTK* (C481, L528, A428D, and T474)**[27,44-46]	• *BTK* C481 and *BTK* A428D mutations have been detected in patients with disease progressing on cBTKi • *BTK* L528 or T474 mutations have been detected in patients with disease progressing on cBTKi and noncovalent (reversible) BTKi (ncBTKi)
CARD11[47]	• Identified in patients with disease progressing on BTK inhibitors
PLCG2[27,44-47]	
***BCL2* (G101V and D103Y)**[48,49]	• Identified in patients with disease progressing on venetoclax

References on CSLL-A (4 of 5) and (5 of 5)

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All recommendations are category 2A unless otherwise indicated.

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CLL STAGING SYSTEMS

Rai System[a]

Stage	Description	Modified Risk Status
0	Lymphocytosis, lymphocytes in blood >5 x 10^9/L clonal B cells and/or >40% lymphocytes in the bone marrow	Low
I	Stage 0 with enlarged node(s)	Intermediate
II	Stage 0–I with splenomegaly, hepatomegaly, or both	Intermediate
III	Stage 0–II with hemoglobin <11.0 g/dL or hematocrit <33%	High
IV	Stage 0–III with platelets <100,000/mm^3	High

Binet System[b]

Stage	Description
A	Hemoglobin ≥10 g/dL and Platelets ≥100,000/mm^3 and <3 enlarged areas
B	Hemoglobin ≥10 g/dL and Platelets ≥100,000/mm^3 and ≥3 enlarged areas
C	Hemoglobin <10 g/dL and/or Platelets <100,000/mm^3 and any number of enlarged areas

[a] This research was originally published in Blood. Rai KR, Sawitsky A, Cronkite EP, et al. Clinical staging of chronic lymphocytic leukemia. Blood 1975;46:219-234. © The American Society of Hematology.
[b] From: Binet JL, Auquier A, Dighiero G, et al. A new prognostic classification of chronic lymphocytic leukemia derived from a multivariate survival analysis. Cancer 1981;48:198-206.

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SLL STAGING SYSTEM

Lugano Modification of Ann Arbor Staging System[c] (for primary nodal lymphomas)

Stage[d]	Involvement[f]	Extranodal (E) Status
Limited		
Stage I	One node or a group of adjacent nodes	Single extranodal lesions without nodal involvement
Stage II	Two or more nodal groups on the same side of the diaphragm	Stage I or II by nodal extent with limited contiguous extranodal involvement
Stage II bulky[e]	II as above with "bulky" disease	Not applicable
Advanced		
Stage III	Nodes on both sides of the diaphragm	Not applicable
	Nodes above the diaphragm with spleen involvement	
Stage IV	Additional non-contiguous extralymphatic involvement	Not applicable

Reprinted with permission. © 2014 American Society of Clinical Oncology. All rights reserved. Cheson B, Fisher R, Barrington S, et al. Recommendations for Initial Evaluation, Staging and Response Assessment of Hodgkin and Non-Hodgkin Lymphoma – the Lugano Classification. J Clin Oncol 2014;32:3059-3068.

[c] Extent of disease is determined by FDG-PET/CT for avid lymphomas and CT for non-avid histologies.
[d] Categorization of A versus B has been removed from the Lugano Modification of Ann Arbor Staging System.
[e] Whether stage II bulky is treated as limited or advanced disease may be determined by histology and a number of prognostic factors.
[f] Note: Tonsils, Waldeyer's ring, and spleen are considered nodal tissue.

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SUPPORTIVE CARE FOR PATIENTS WITH CLL/SLL

Anti-infective Prophylaxis

- Recommended during treatment and thereafter (if tolerated) for patients receiving phosphoinositide 3-kinase (PI3K) inhibitors, purine analog- or bendamustine-based CIT, and/or alemtuzumab
  - Herpes virus prophylaxis with acyclovir or equivalent
  - Pneumocystis jiroveci pneumonia (PJP) prophylaxis with sulfamethoxazole/trimethoprim or equivalent
- Consider PJP and varicella zoster virus (VZV) prophylaxis in patients at increased risk for opportunistic infection and receiving BTKi therapy. Monitor for fungal infection.
- Monitor blood counts and consider growth factor support or fluoroquinolone and/or fungal prophylaxis for the management of venetoclax-induced neutropenia.
- Hepatitis B virus (HBV) and cytomegalovirus (CMV) prophylaxis and monitoring is recommended for patients at high risk. See Viral Reactivation below.

Viral Reactivation

HBV:

- Hepatitis B surface antigen (HBsAg) and Hepatitis B core antibody (HBcAb) testing for all patients receiving therapy
  - Quantitative hepatitis B viral load by quantitative RT-PCR (qPCR) and surface antibody only if one of the screening tests is positive
- Patients receiving intravenous immunoglobulin (IVIG) may be HBcAb-positive as a consequence of IVIG therapy.
- Prophylactic antiviral therapy with entecavir is recommended for any patient who is HBsAg-positive and receiving treatment. If there is active disease (qPCR+), it is considered treatment/management and not prophylactic therapy. In cases of HBcAb positivity, prophylactic antiviral therapy is preferred; however, if there is a concurrent high-level hepatitis B surface antibody, these patients may be monitored with serial hepatitis B viral load.
  - Entecavir is preferred.[a] Avoid lamivudine due to risks of resistance development.
  - Other antivirals including adefovir, telbivudine, and tenofovir are proven active treatments and are acceptable alternatives.

Viral Reactivation (continued)

  - Monitor hepatitis B viral load with qPCR monthly through treatment and every 3 months thereafter.
    - ◊ If viral load is consistently undetectable, treatment is considered prophylactic.
    - ◊ If viral load does not drop or previously undetectable qPCR becomes positive, consult hepatologist.
  - Maintain prophylaxis up to 12 months after oncologic treatment ends.
    - ◊ Consult with hepatologist for duration of therapy in patient with active HBV.

Hepatitis C virus (HCV):

- Evidence from large epidemiology studies, molecular biology research, and clinical observation supports an association of HCV and B-cell non-Hodgkin lymphoma (NHL). Direct-acting antiviral (DAA) agents for chronic carriers of HCV with genotype 1 demonstrated a high rate of sustained viral responses.
  - Low-grade B-cell NHL
    - ◊ According to the American Association for the Study of Liver Diseases, combined therapy with DAA should be considered in asymptomatic patients with HCV genotype 1 since this therapy can result in regression of lymphoma.

CMV reactivation in patients with previous CMV infection (seropositive):

- Clinicians must be aware of the high risk of CMV reactivation in patients receiving PI3K inhibitors or alemtuzumab. The current recommendations for appropriate screening are controversial. CMV viremia should be measured by PCR at least every 4 weeks. Some clinicians use ganciclovir (oral or IV) pre-emptively if viremia is present; others use ganciclovir only if viral load is rising. Consultation with an infectious disease expert may be necessary.

John Cunningham virus (JCV):

- Progressive multifocal leukoencephalopathy (PML) related to JCV can be seen in patients receiving treatment.

[a] Huang YH, et al. J Clin Oncol 2013;31:2765-2772; Huang H, et al. JAMA 2014;312:2521-2530.

Continued

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SUPPORTIVE CARE FOR PATIENTS WITH CLL/SLL

Precautions when initiating treatment or transitioning to next therapy for disease progression while on treatment

- Workup prior to new/next treatment to assess tumor lysis syndrome (TLS) risk and additional management needs:
  - CBC with ALC
  - Staging CT scan to evaluate extent of lymphadenopathy if using a BCL2i-based regimen or if concern for bulky adenopathy not accessible by physical exam and currently using a BTKi-based regimen
  - Bone marrow evaluation if clinically indicated

- Agent considerations
  - BTK-inhibitor (covalent or noncovalent BTKi; cBTKi or ncBTKi)
    - Potential for significant transient increase in ALC on initiation due to egress of lymphocytes from tissue microenvironment into peripheral blood
    - Low risk for TLS
    - If transitioning from cBTKi to BCL2i or ncBTKi, can see flare with abrupt discontinuation and time off treatment, hence should minimize time off treatment between drugs; and when transitioning from a cBTKi or ncBTKi to BCL2i, consider overlapping of BTKi and BCL2i during the BCL2i ramp-up until disease control is achieved
  - BCL2-inhibitor (BCL2i)
    - TLS risk – see venetoclax prescribing information for risk assessment, mitigation, and management (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm)
    - For patients with progression on prior BTKi, continue the BTKi through the venetoclax ramp-up to reduce the risk of tumor flare and consider administering obinutuzumab prior to venetoclax for cytoreduction
    - TLS prophylaxis for hyperuricemia with allopurinol, febuxostat, and rasburicase according to prescribing information and institutional standards
    - Consider hospitalization for TLS monitoring, mitigation, and management according to venetoclax prescribing information
    - Potassium may be falsely elevated due to pseudohyperkalemia, particularly in patients with high ALC; this should be evaluated and clarified prior to initiation of therapy. Venetoclax should not be initiated unless a reliable measurement of potassium can be obtained. Methods to avoid pseudohyperkalemia include a fresh stick without a tourniquet, sample maintained on ice and taken directly to the lab, use of a blood gas tube, obtaining potassium result from a blood gas, or whole blood potassium.
  - Anti-CD20 mAb (obinutuzumab)
    - TLS risk similar in type to chemoimmunotherapy
    - First-infusion reaction can be mitigated by administration of corticosteroids for 1 to 3 days prior to the initiation of obinutuzumab
    - High-dose methylprednisolone (eg, 1000 mg daily for 3 doses) has been effective at reducing obinutuzumab first-infusion-related reactions and reducing bulk of disease
    - Infection risk
      - See obinutuzumab prescribing information for further guidance (https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm), including guidance on holding antihypertensive medications surrounding the initial dose

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SUPPORTIVE CARE FOR PATIENTS WITH CLL/SLL

Tumor Lysis Syndrome (TLS)

- Laboratory hallmarks of TLS:
  - High serum creatinine
  - High potassium
  - High uric acid
  - High phosphorous
  - Low calcium
  - High LDH

- Symptoms of TLS:
  - Nausea and vomiting, shortness of breath, irregular heartbeat, clouding of urine, lethargy, and/or joint discomfort

- TLS features
  - Consider TLS prophylaxis for patients with the following risk factors:
    - Patients receiving treatment with venetoclax, CIT, lenalidomide, and obinutuzumab
    - Progressive disease after small-molecule inhibitor therapy
    - Bulky (>5 cm) lymph nodes
    - Spontaneous TLS
    - Elevated white blood cell (WBC) count
    - Pre-existing elevated uric acid
    - Renal disease or renal involvement by tumor

- Treatment of TLS:
  - For BCL2i-associated TLS, see venetoclax prescribing information
  - TLS is best managed if anticipated and management is started prior to treatment.
  - Centerpiece of treatment includes:
    - Rigorous hydration
    - Management of hyperuricemia
    - Frequent monitoring of electrolytes and aggressive correction (essential)
  - First-line and at retreatment for hyperuricemia
    - Allopurinol, febuxostat, and rasburicase according to prescribing information and institutional standards
    - Glucose-6-phosphate dehydrogenase (G6PD) testing is required prior to use of rasburicase. Rasburicase is contraindicated in patients with a history consistent with G6PD. In these patients, rasburicase should be substituted with allopurinol.
  - Rasburicase is indicated for patients with any of the following risk factors (Doses of 3–6 mg are usually effective.[b] One dose of rasburicase is frequently adequate. Re-dosing should be individualized):
    - Urgent need to initiate therapy in a patient with bulky disease
    - Situations where adequate hydration may be difficult or impossible
    - Acute renal failure
  - If TLS is untreated, its progression may cause acute kidney failure, cardiac arrhythmias, seizures, loss of muscle control, and death.

[b] There are data to support that fixed-dose rasburicase is very effective in adult patients.

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SUPPORTIVE CARE FOR PATIENTS WITH CLL/SLL

Autoimmune Cytopenias

- **Autoimmune hemolytic anemia (AIHA): Diagnosis with reticulocyte count, haptoglobin, and direct antiglobulin test (Coombs).**
  - **AIHA that develops in the setting of treatment with fludarabine: Stop, treat, and avoid subsequent fludarabine.**
- **Immune thrombocytopenic purpura (ITP): Evaluate bone marrow for cause of low platelets.**
- **Pure red cell aplasia (PRCA): Consider bone marrow evaluation and testing for parvovirus B19, herpesviruses, and drug effects.**
- **Treatment: Corticosteroids, rituximab, IVIG, cyclosporin A, splenectomy, eltrombopag, or romiplostim (ITP), or BTKi-based therapy for steroid-refractory or recurrent AIHA.**

Blood Product Support

- **Transfuse according to institutional or published standards.**
- **Irradiate all blood products to avoid transfusion-associated graft-versus-host disease (GVHD).**

Cancer Screening

- **Patients with CLL/SLL have a higher risk of developing secondary cancers, including melanoma and non-melanoma skin cancers.[c]**
- **Risk factors for skin cancers include inability to tan, fair skin that sunburns easily, and a history of intensive sun exposure at a young age. Annual dermatologic skin screening is recommended.**
- **Age appropriate screening guidelines should be closely followed for breast, cervical, colon, and prostate cancers.**

Complications of mAb Therapy

- **Obinutuzumab infusion-related reactions (initial reaction and reactions in patients with high ALC [>100,000 µL]) can be severe and patients should be monitored closely. Consider premedication with corticosteroid, antihistamine, and acetaminophen. Monitoring and prophylaxis for TLS is recommended for patients with high ALC.**
- **Rare complications such as mucocutaneous reactions including paraneoplastic pemphigus, Stevens-Johnson syndrome, lichenoid dermatitis, vesiculobullous dermatitis, and toxic epidermal necrolysis can occur. Consultation with a dermatologist is recommended for management of these complications. Re-challenge with the same mAb in such settings is not recommended. It is unclear if re-challenge with alternative anti-CD20 mAb poses the same risk of recurrence. An alternative anti-CD20 mAb could be used for patients with intolerance (including those experiencing severe hypersensitivity reactions requiring discontinuation of chosen anti-CD20 mAb).**

Rituximab Rapid Infusion and Subcutaneous Administration

- **If no severe infusion reactions were experienced with prior cycle of rituximab, a rapid infusion over 90 minutes can be used.**
- **Rituximab and hyaluronidase human injection for subcutaneous use may be used in patients who have received at least one full dose of a rituximab product by intravenous route.**

[c] Mehrany K, et al. Dermatol Surg 2005;31:38-42; Mehrany K, et al. Arch Dermatol 2004;140:985-988; Mehrany, K et al. J Am Acad Dermatol 2005;53:1067-1071.

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SUPPORTIVE CARE FOR PATIENTS WITH CLL/SLL

Recurrent Sinopulmonary Infections (requiring IV antibiotics or hospitalization)
- Antimicrobials as appropriate
- Evaluate serum IgG, if <500 mg/dL
  - Begin monthly IVIG 0.3–0.5 g/kg or may substitute a subcutaneous immunoglobulin (SCIG) product given weekly at appropriately adjusted equivalent doses
  - Adjust dose/interval to maintain nadir level of approximately 500 mg/dL

Thromboprophylaxis
- Recommended for prevention of thromboembolic events in patients receiving lenalidomide:
  - Aspirin 81 mg PO daily if platelets above 50 x 10^{12}/L
  - Patients already on anticoagulants, such as warfarin, do not need aspirin
- Note that the above may differ from the NCCN Guidelines for Cancer-Associated Venous Thromboembolic Disease in which the recommendations with lenalidomide pertain only to patients with multiple myeloma

Tumor Flare Reactions
- Management of tumor flare is recommended for patients receiving lenalidomide
- Painful lymph node enlargement or lymph node enlargement with evidence of local inflammation, occurring with treatment initiation; may also be associated with spleen enlargement, low-grade fever, and/or rash
- Treatment:
  - Steroids (eg, prednisone 25–50 mg PO daily for 5–10 days)
  - Antihistamines for rash and pruritus
- Prophylaxis:
  - Consider in patients with bulky lymph nodes (>5 cm)
  - Steroids (eg, prednisone 20 mg PO daily for 5–7 days followed by rapid taper over 5–7 days)

Bleeding and Hemorrhage Risk with BTKi
- Increased risk for bleeding and bruising with cBTKI and ncBTKi. Hold 3 days before and after a minor surgical procedure and 7 days before and after a major surgical procedure.
- Consider the benefit-risk in patients requiring antiplatelet or anticoagulant therapies; concomitant use of ≥3 anti-platelet agents not recommended (eg, BTKi, aspirin or other anti-platelet agents, direct oral anticoagulants).
- Thrombocytopenia (platelets <100,000/μL) and increased risk for bleeding should also be taken into consideration.

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SUPPORTIVE CARE FOR PATIENTS WITH CLL/SLL

Immunizations

- **Avoid all live vaccines including live attenuated influenza vaccine**
- **Recommended vaccines**
  - **Annual influenza vaccine[d]**
  - **Pneumococcal vaccine: see CDC Guidelines for Pneumococcal Vaccination**
  - **Zoster vaccine recombinant, adjuvanted for all patients treated with BTKi**
  - **Respiratory syncytial virus (RSV) vaccine: single dose vaccination for patients with CLL/SLL, including patients age <60 y**
  - **COVID-19 vaccine for all patients with CLL/SLL. See CDC COVID-19 Vaccination Clinical & Professional Resources**
    - **See Management of Concurrent COVID-19 and Cancer in Patients in the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections**

[d] In patients who have received rituximab, B-cell recovery occurs by approximately 9 months. Prior to B-cell recovery, patients generally do not respond to influenza vaccine and if given should not be considered vaccinated.

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

SUGGESTED TREATMENT REGIMENS[a,b,c,d]

CLL/SLL without del(17p)/*TP53* Mutation (alphabetical by category)

FIRST-LINE THERAPY[e,f,g]

Preferred
- BCL2i-containing regimens
  - Venetoclax/Acalabrutinib[j] ± Obinutuzumab (fixed duration)[h] (category 1)
  - Venetoclax + Obinutuzumab (fixed duration) (category 1)
- cBTKi-based regimens[i,j]
  - Acalabrutinib (continuous) ± Obinutuzumab (category 1)
  - Zanubrutinib (continuous) (category 1)

Other Recommended
- BCL2i-containing regimen
  - Venetoclax/Ibrutinib[j] (category 1; fixed duration)[k]
  - Venetoclax/Ibrutinib[j] (category 1; MRD-guided)[l]

Useful in Certain Circumstances
- BCL2i-containing regimen
  - Venetoclax/Zanubrutinib[j] (MRD-guided)[m]
- cBTKi-based regimens[i,j]
  - Ibrutinib[n] (continuous) (category 1)
  - Ibrutinib (continuous) + anti-CD20 mAb (category 2B)[o]
- High-dose methylprednisolone (HDMP) + anti-CD20 mAb[o] (category 2B; category 3 for patients <65 y without significant comorbidities)
- Consider only when cBTKi and BCL2i are not available or not feasible
  - Bendamustine[p] + anti-CD20 mAb[o,q]
  - FC (Fludarabine/Cyclophosphamide)[r,s,t] + Rituximab
  - Chlorambucil[u] + Obinutuzumab
  - Obinutuzumab

CLL/SLL with del(17p)/*TP53* Mutation (alphabetical by category)

FIRST-LINE THERAPY[e,f,g]

Preferred
- BCL2i-containing regimens
  - Venetoclax + Obinutuzumab (fixed duration)
  - Venetoclax/Acalabrutinib + Obinutuzumab (MRD-guided)[v]
  - Venetoclax/Zanubrutinib[j] (MRD-guided)[m]
- cBTKi-based regimens[i,j]
  - Acalabrutinib (continuous) ± Obinutuzumab
  - Zanubrutinib (continuous)

Other Recommended
- BCL2i-containing regimen
  - Venetoclax/Ibrutinib[j,w] (fixed duration)

Useful in Certain Circumstances
- cBTKi-based regimen[i,j]
  - Ibrutinib[n] (continuous)
- HDMP + anti-CD20 mAb[o]
- Consider only when cBTKi and BCL2i are not available or not feasible
  - Obinutuzumab

See Evidence Blocks on EB-1

Suggested Regimens for Second-Line and Subsequent Therapy (CSLL-D 2 of 5)
Therapy for Relapsed or Refractory Disease After Prior BTKi-Based and BCL2i-Containing Regimens (CSLL-D 2 of 5)

Footnotes
References

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-1.
Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.
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SUGGESTED TREATMENT REGIMENS[a,b,c,d]
CLL/SLL with or without del(17p)/*TP53* Mutation
(alphabetical by category)

SECOND-LINE OR SUBSEQUENT THERAPY[e,f]	
Preferred	**Other Recommended**
• BCL2i-containing regimen* ‣ Venetoclax + Obinutuzumab • cBTKi-based regimens[i,j,x] ‣ Acalabrutinib (continuous) (category 1) ‣ Zanubrutinib (continuous) (category 1) • ncBTKi-based regimen: ‣ Pirtobrutinib (continuous) (category 1) (resistance or intolerance to prior cBTKi-based regimens)	• BCL2i-containing regimens* ‣ Venetoclax + Rituximab (category 1) ‣ Venetoclax* ‣ Venetoclax/Ibrutinib[j,y,z] (category 2B) • cBTKi-based regimen[i,j] ‣ Ibrutinib[n] (continuous) (category 1)

* Venetoclax ± anti-CD20 mAb (obinutuzumab preferred) or venetoclax + cBTKi is a treatment option for relapse after a period of remission. Single agent venetoclax (continuous) can also be considered a treatment option for relapse after a period of remission.

THERAPY FOR RELAPSED OR REFRACTORY DISEASE AFTER PRIOR cBTKi-BASED AND BCL2i-CONTAINING REGIMENS[e,f]	
Preferred	**Other Recommended**
• Chimeric antigen receptor (CAR) T-cell therapy ‣ Lisocabtagene maraleucel (CD19-directed)[aa] ± Ibrutinib[j] • ncBTKi-based regimen: ‣ Pirtobrutinib (continuous) (category 1) (if not previously given)	• PI3Ki-based regimens ‣ Duvelisib ‣ Idelalisib[bb] ± Rituximab • FC + Rituximab[s,t,**] • Lenalidomide[cc] ± Rituximab • Obinutuzumab • Alemtuzumab[dd] ± Rituximab (with del(17p)/*TP53* mutation) • Bendamustine[p] + Rituximab[q,**] (category 2B for patients ≥65 y or patients <65 y with significant comorbidities) • HDMP + anti-CD20 mAb[o] (category 2A with del[17p]; category 2B without del[17p])

** Not recommended for CLL/SLL with del(17p)/*TP53* mutation.

See Evidence Blocks on EB-2 and EB-3

Footnotes
References

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

SUGGESTED TREATMENT REGIMENS
FOOTNOTES

[a] See references for regimens on CSLL-D 4 of 5 and CSLL-D 5 of 5.
[b] Supportive Care for Patients with CLL/SLL (CSLL-C).
[c] Rituximab and hyaluronidase human injection for subcutaneous use may be used in patients who have received at least one full dose of a rituximab product by intravenous route.
[d] Re-challenge with the same mAb is not recommended in patients experiencing rare complications (eg, mucocutaneous reactions including paraneoplastic pemphigus, Stevens-Johnson syndrome, lichenoid dermatitis, vesiculobullous dermatitis, and toxic epidermal necrolysis). It is unclear whether re-challenge with alternative anti-CD20 mAbs poses the same risk of recurrence.
[e] An FDA-approved biosimilar is an appropriate substitute for rituximab.
[f] Please refer to package insert for full prescribing information, dose modifications, and monitoring for adverse reactions: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm.
[g] Time-limited treatment with BCL2i + cBTKi ± CD20 mAb may be fixed-duration or MRD-guided (based on uMRD-stopping criteria as described in clinical trials). See MRD Assessment (CSLL-E).
[h] AMPLIFY: Brown JR, et al. N Engl J Med 2025;392:748-762.
[i] TKIs are available in different formulations, dosage forms, and strengths that are subject to different administration instructions. The choice of cBTKi should be made by the treating clinician based on patient-specific factors (eg, age, functional status, comorbid conditions) and the agent's toxicity profile. These products are not interchangeable.
[j] A baseline cardiovascular risk assessment should be considered prior to initiation of cBTKi. Awan F, et al. Blood Adv 2022;18:5516.
[k] See publication for duration of treatment (GLOW: Munir T, et al. J Clin Oncol 2023;41:3689-3699; Niemann CU, et al. Blood 2024;144:Abstract 1871).
[l] See the publication for details on MRD assessments and management, including criteria for treatment discontinuation (FLAIR: Munir T, et al. N Engl J Med 2025;393:1177-1190). See CSLL-E.
[m] See the publication for details on MRD assessments and management, including criteria for treatment discontinuation (SEQUOIA [Arm D]: Shadman M, et al. J Clin Oncol 2025;43:2409-2417). See CSLL-E.
[n] The Panel consensus to not list ibrutinib as a preferred regimen is based on the toxicity profile.
[o] Anti-CD20 mAbs include: obinutuzumab or rituximab.
[p] For patients aged ≥65 y or patients aged <65 y with significant comorbidities (CrCl <70 mL/min) dose is 70 mg/m^2 in cycle 1 with escalation to 90 mg/m^2 if tolerated.
[q] Not recommended for patients who are frail.
[r] Data from the CLL10 study confirmed the superiority of FCR over bendamustine + rituximab (BR) in younger patients. For patients >65 y, the outcome was similar for both regimens with less myelosuppression and infection for BR. FCR was associated with improved PFS (with a plateau in PFS beyond 10-year follow-up) in patients with mutated IGHV without del (17p)/*TP53* mutation (Thompson P, et al. Blood 2016;127:303-309).
[s] AIHA should not preclude the use of combination therapy containing fludarabine; however, patients should be observed carefully and fludarabine should be avoided in those where a history of fludarabine-associated AIHA is suspected.
[t] Recommended only for patients aged <65 y without significant comorbidities.
[u] Recommended only for patients aged ≥65 y or patients aged <65 y with significant comorbidities (creatinine clearance [CrCl] <70 mL/min).
[v] See the publication for details on MRD assessments and management, including criteria for treatment discontinuation (Davids MS, et al. J Clin Oncol 2025;43:788-799). See CSLL-E.
[w] CAPTIVATE: Allan JN, et al. Clin Cancer Res 2023;29:2593 -2601; Ghia P, et al. J Clin Oncol 2025;43:Abstract 7036.
[x] Acalabrutinib or zanubrutinib has not been shown to be effective for ibrutinib-refractory CLL with BTK C481S mutations. Patients with ibrutinib intolerance have been successfully treated with acalabrutinib or zanubrutinib without recurrence of symptoms.
[y] CLARITY: Hillmen P, et al. J Clin Oncol 2019;37:2722-2729.
[z] Venetoclax containing combinations with ibrutinib are not appropriate for patients who are intolerant to or have disease progression on ibrutinib.
[aa] See CAR T-Cell–Related Toxicities in the NCCN Guidelines for Management of CAR T-Cell and Lymphocyte Engager-Related Toxicities for the management of cytokine release syndrome (CRS) and neurologic toxicity management.
[bb] Indicated for patients for whom rituximab monotherapy would be considered appropriate due to the presence of other comorbidities (reduced renal function as measured by CrCl <60 mL/min, or NCI CTCAE grade ≥3 neutropenia or grade ≥3 thrombocytopenia resulting from myelotoxic effects of prior therapy with cytotoxic agents).
[cc] Lenalidomide can be given as continuous or intermittent dosing for patients with CLL. Growth factors and/or dose adjustment may be needed to address cytopenias, without necessitating holding treatment. See Andritsos L, et al. J Clin Oncol 2008;26:2519-2525; Wendtner C, et al. Leuk Lymphoma 2016;57:1291-1299.
[dd] While alemtuzumab is no longer commercially available for CLL, it may be obtained for clinical use. Alemtuzumab is less effective for bulky (>5 cm) lymphadenopathy; monitor for CMV reactivation. See Viral Reactivation (CSLL-C 1 of 4).

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All recommendations are category 2A unless otherwise indicated.

SUGGESTED TREATMENT REGIMENS
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Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

Continued

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Lisocabtagene ± ibrutinib

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Venetoclax + ibrutinib

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Venetoclax + Zanubrutinib

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MEASURABLE RESIDUAL DISEASE (MRD) ASSESSMENT

- **Evidence from clinical trials suggests that undetectable MRD in the peripheral blood after the end of time-limited treatment is an important predictor of efficacy.[a-e]**
- **Allele-specific oligonucleotide polymerase chain reaction (ASO-PCR) and six-color flow cytometry (MRD flow) are the two validated methods used for the detection of MRD at the level of 10^{-4} to 10^{-5}.[f,g] Next-generation sequencing (NGS)-based assays have been shown to be more sensitive, thus allowing for the detection of MRD at the level of 10^{-6}.[h-j]**
- **MRD evaluation should be performed using an assay with a sensitivity of 10^{-4} according to the standardized European Research Initiative on CLL (ERIC) method or standardized NGS method.**
- **For MRD-guided treatment, serial MRD evaluations using either flow cytometry (MRD flow) or NGS with a limit of detection (sensitivity) of at least 10^{-4} CLL cells in blood or bone marrow may be performed every 3 to 6 months during active treatment as directed in the published trial results that support MRD-guided treatment regimens. See the publications for details on MRD assessments and management, including criteria for treatment discontinuation (CSLL-E 2 of 2).**
- **Either flow cytometry (MRD flow) or NGS are acceptable assays as long as the limit of detection (sensitivity) is achieved.**
- **When assessing MRD in blood, be aware that anti-CD20 monoclonal antibody administration within the prior 3 months can lead to false undetectable MRD (uMRD) status, particularly in blood.**

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All recommendations are category 2A unless otherwise indicated.

MEASURABLE RESIDUAL DISEASE (MRD) ASSESSMENT

Table 1: MRD-guided treatment regimens

Trial	First-line Therapy	Regimen
FLAIR[k]	CLL/SLL without del(17p) or *TP53* mutation	• Venetoclax/Ibrutinib[n] (category 1; other recommended)
SEQUOIA (Arm D)[l]	CLL/SLL without del(17p) or *TP53* mutation	• Venetoclax/Zanubrutinib[o] (category 2A; useful in certain circumstances)
	CLL/SLL with del(17p) or *TP53* mutation	• Venetoclax/Zanubrutinib[o] (category 2A; preferred)
Phase II study[m]	CLL/SLL with del(17p) or *TP53* mutation	• Venetoclax/Acalabrutinib + Obinutuzumab[p] (category 2A; preferred)

[k] Munir T, Girvan S, Cairns DA, et al. Measurable residual disease-guided therapy for chronic lymphocytic leukemia. N Engl J Med 2025:Online ahead of print.
[l] Shadman M, Munir T, Ma S, et al. Zanubrutinib and venetoclax for patients with treatment-naive chronic lymphocytic leukemia/small lymphocytic lymphoma with and without del(17p)/tp53 mutation: SEQUOIA Arm D Results. J Clin Oncol 2025;43:2409-2417.
[m] Davids MS, Ryan CE, Lampson BL, et al. Phase II study of acalabrutinib, venetoclax, and obinutuzumab in a treatment-naive chronic lymphocytic leukemia population enriched for high-risk disease. J Clin Oncol 2025;43:788-799.
[n] The duration of combined ibrutinib + venetoclax was determined based on serial MRD monitoring and treatment discontinuation in patients meeting with the uMRD-guided stopping criteria.
[o] Discontinuation of treatment (venetoclax after a minimum of 12 cycles or zanubrutinib after a minimum of 27 cycles) was allowed in patients meeting with the uMRD-guided stopping criteria.
[p] Patients could discontinue treatment if CR with uMRD in the bone marrow was achieved at the start of cycle 16.

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DIAGNOSIS

ESSENTIAL:
- **An excisional or incisional biopsy of the suspected lymph node/lesion with an increased standardized uptake value (SUV) on FDG-PET/CT scan (particularly if discordant with the remaining disease).[a]**
  - **Diffuse large B-cell lymphoma (DLBCL): Sheets of confluent large B cells that are not part of a proliferation center are sufficient to diagnose a Richter transformation to DLBCL.[b,c,d]**
  - **Classic Hodgkin lymphoma (CHL): Rare transformation to CHL demonstrates large Reed-Sternberg (RS) cells that express CD30, CD15, and PAX-5 but lack strong, uniform CD20 and CD45 (also lack co-expression of both OCT-2 and BOB.1). The background lymphocytes in those CHL cases are CD3+ T cells with a varying degree of admixed eosinophils, histiocytes, and plasma cells.[e]**
- **Bone marrow aspirate with biopsy can be pursued if excisional or incisional biopsy material is nondiagnostic.**
- **Hematopathology review of all slides with at least one paraffin block representative of the tumor.**
- **Molecular analysis to establish clonal relatedness between CLL and DLBCL cells.[f]**

USEFUL UNDER CERTAIN CIRCUMSTANCES (should be done with CLL/SLL specimen - blood or bone marrow aspirate):
- **FISH to detect +12; del(11q); del(13q); del(17p)**
- **CpG-stimulated metaphase karyotype for CK**
- ***TP53* sequencing**

→ **Workup** (HT-2)

[a] Biopsy of lesions with SUV ≥5 is recommended if there is high clinical suspicion or the lesion has a differentially higher uptake than the remaining disease.
[b] While occasionally an increase in proliferative rate can be shown with Ki-67, this is not considered diagnostic of a transformation.
[c] Proliferation centers in CLL may express c-MYC and/or cyclin D1. This does not change the diagnosis.
[d] Accelerated CLL or CLL with expanded proliferation centers may be diagnosed in cases where proliferation centers in CLL are expanded or fused together (>20x field or 0.95 mm^2) AND show Ki-67 proliferative rate >40% or >2.4 mitoses/proliferation center. Progression to CLL with increased prolymphocytes may occur when there are increased prolymphocytes in the blood (>15%). Neither of these findings should be considered as Richter transformation, but rather as progression of CLL associated with more aggressive disease and poorer outcome (Gine E, et al. Haematologica 2010;95:1526-1533; Ciccone M, et al. Leukemia 2012;26:499-508; Campo E, et al. Blood 2022;140:1229-1253; Alaggio R, et al. Leukemia 2022;36:1720-1748). Optimal management for these cases has not been established.
[e] If morphologic RS cells are identified but the background cells are still the B cells of CLL, an EBV stain such as Epstein-Barr virus-encoded RNA (EBER) should be performed. EBV infection of CLL can produce RS-like proliferations, but the background cells are still CLL and not the reactive mix typically seen in Hodgkin lymphoma. These cases should NOT be considered a Richter transformation event.
[f] Immunoglobulin gene rearrangement studies of CLL and histologically transformed tissue may be performed to establish the clonal relationship. Comparative clonality testing for immunoglobulin heavy chain (IGH) gene rearrangement should be done on defined CLL tissue and biopsy specimen of transformed tissue to establish clonal relatedness between CLL/SLL and DLBCL cells (either using commercial NGS-based assay or lab-based assay for IGH clonality).

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All recommendations are category 2A unless otherwise indicated.

WORKUP

ESSENTIAL:
- Whole body FDG-PET/CT scan (preferred) or C/A/P CT with contrast of diagnostic quality (if PET/CT scan not feasible)
- An excisional or incisional biopsy of the suspected lymph node/lesion with an increased SUV on FDG-PET/CT scan (particularly if discordant with the remaining disease).[a]
- FNA biopsy alone is inadequate for the diagnosis of histologic transformation. In certain circumstances, when a lymph node/lesion is not easily accessible for excisional or incisional biopsy, core needle biopsy (multiple biopsies may be required) in conjunction with appropriate ancillary techniques (ie, IHC, flow cytometry) may be sufficient for diagnosis.
- History and physical exam with attention to node-bearing areas, including Waldeyer's ring, and the size of liver and spleen
- Performance status
- B symptoms
- CBC with differential
- Comprehensive metabolic panel
- LDH, uric acid
- Molecular analysis to establish clonal relatedness between CLL and DLBCL cells[f]
- Assess for distress[g]

USEFUL IN SELECTED CASES:
- Unilateral bone marrow aspirate and biopsy
- Multigated acquisition (MUGA) scan/echocardiogram if anthracycline-based regimen is indicated
- Hepatitis B[h] and C testing
- Epstein-Barr virus (EBV) evaluation by EBV-latent membrane protein 1 (LMP1) or Epstein-Barr virus-encoded RNA in situ hybridization (EBER-ISH)
- Pregnancy testing in patients of childbearing age
- Discussion of fertility preservation[i]
- Human leukocyte antigen (HLA) typing

CLINICAL PRESENTATION / TREATMENT[j]

- Transformation to DLBCL (Richter transformation) → HT-3
- Transformation to CHL → Clinical trial (preferred) or See NCCN Guidelines for Hodgkin Lymphoma

[a] Biopsy of lesions with SUV ≥5 is recommended if there is high clinical suspicion or the lesion has a differentially higher uptake than the remaining disease.

[f] Immunoglobulin gene rearrangement studies of CLL and histologically transformed tissue may be performed to establish the clonal relationship. Comparative clonality testing for IGH gene rearrangement should be done on defined CLL tissue and biopsy specimen of transformed tissue to establish clonal relatedness between CLL/SLL and DLBCL cells (either using commercial NGS-based assay or lab-based assay for IGH clonality).

[g] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).

[h] Hepatitis B testing is indicated because of the risk of reactivation during treatment (eg, immunotherapy, CIT, chemotherapy, targeted therapy). See Viral Reactivation (CSLL-C 1 of 6). Tests include HBsAg and core antibody for a patient with no risk factors. For patients with risk factors or previous history of hepatitis B, add e-antigen. If positive, check viral load and consult with gastroenterologist.

[i] Fertility preservation options include: sperm banking, in IVF, or ovarian tissue or oocyte cryopreservation.

[j] Supportive Care for Patients with CLL/SLL (CSLL-C).

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CLINICAL PRESENTATION | **TREATMENT[j]** | **RESPONSE ASSESSMENT[m]** | **ADDITIONAL THERAPY[j]**

Richter transformation → **Clonality testing compared to CLL[f] (initiation of therapy should not be delayed for clonality testing)**

- **Untreated CLL or Clonally unrelated** → **Clinical trial or CIT regimens (HT-A)**
  - **Complete response** → **Observation**
  - **Partial response or Refractory disease or Progression while on treatment** → **Clinical trial or CAR T-cell therapy (HT-A) (if eligible, after at least 1 prior systemic therapy regimen) or Non-CIT regimens (other than CAR T-cell therapy) (HT-A)** → **If CR, consider allogeneic HCT[n]**
- **Previously treated CLL and clonally related or Clonal relation unknown[k,l]** → **HT-4**

[f] Immunoglobulin gene rearrangement studies of CLL and histologically transformed tissue may be performed to establish the clonal relationship. Comparative clonality testing for IGH gene rearrangement should be done on defined CLL tissue and biopsy specimen of transformed tissue to establish clonal relatedness between CLL/SLL and DLBCL cells (either using commercial NGS-based assay or lab-based assay for IGH clonality).

[j] Supportive Care for Patients with CLL/SLL (CSLL-C).

[k] Richter transformation to DLBCL (clonally related or clonal relation unknown) is generally managed with treatment regimens recommended for DLBCL. However, these regimens typically result in poor responses and optimal first-line therapy is not established. The regimens listed on HT-A are used at NCCN Member Institutions based on published data.

[l] It is recommended that clonal relatedness be determined; however, if is not feasible and CLL has not been previously treated, consider treating as clonally unrelated.

[m] PET/CT scan is required for response assessment and should be interpreted via the PET 5-Point Scale (5-PS). See Lugano Response Criteria for Non-Hodgkin Lymphoma (NHODG-C) in the NCCN Guidelines for B-Cell Lymphomas.

[n] Herrera AF, et al. Blood Adv 2021;5:3528-3539; Kharfan-Dabaja MA, et al. Biol Blood Marrow Transplant 2016;22:2117-2125.

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CLINICAL PRESENTATION	TREATMENT[j]	RESPONSE ASSESSMENT[m]	ADDITIONAL THERAPY[j]
Previously treated CLL and clonally related or Clonal relation unknown[k,l] →	Clinical trial (preferred) or Treatment with non-CIT regimens (other than CAR T-cell therapy)[o] (preferred) (HT-A) or CIT regimens (HT-A) →	Complete response →	Observation or Consider allogeneic HCT[n] and/or evaluate eligibility for CAR T-cell therapy or Continue treatment with non-CIT regimen until progression
		Partial response →	CAR T-cell therapy (HT-A) (if eligible, after at least 1 prior systemic therapy regimen) or Non-CIT regimens (other than CAR T-cell therapy) (not previously used) (HT-A) → If CR, consider allogeneic HCT[n] or If not a candidate for HCT or CAR T-Cell therapy: • Continue the same non-CIT regimen until progression or switch to an alternate non-CIT regimen (HT-A)
		Refractory disease or Progression while on treatment →	Clinical trial or CAR T-cell therapy (HT-A) (if eligible, after at least 1 prior systemic therapy regimen) or Non-CIT regimens (other than CAR T-cell therapy) (not previously used) (HT-A)

[j] Supportive Care for Patients with CLL/SLL (CSLL-C).
[k] Richter transformation to DLBCL (clonally related or clonal relation unknown) is generally managed with treatment regimens recommended for DLBCL. However, these regimens typically result in poor responses and optimal first-line therapy is not established. The regimens listed on HT-A are used at NCCN Member Institutions based on published data.
[l] It is recommended that clonal relatedness be determined; however, if is not feasible and CLL has not been previously treated, consider treating as clonally unrelated.
[m] PET/CT scan is required for response assessment and should be interpreted via the PET 5-Point Scale (5-PS). See Lugano Response Criteria for Non-Hodgkin Lymphoma (NHODG-C) in the NCCN Guidelines for B-Cell Lymphomas.
[n] Herrera AF, et al. Blood Adv 2021;5:3528-3539; Kharfan-Dabaja MA, et al. Biol Blood Marrow Transplant 2016;22:2117-2125.
[o] In patients with previously treated CLL, options depend on treatment received for CLL prior to transformation.

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SUGGESTED TREATMENT REGIMENS

RICHTER TRANSFORMATION[a,b,c,d]
(Regimens listed by alphabetical order and category of evidence)

Chemoimmunotherapy (CIT)[e]
• CHOP (Cyclophosphamide/Doxorubicin/Vincristine/Prednisone) + Rituximab ± Venetoclax* (category 2B with Venetoclax) • Dose-Adjusted EPOCH (Etoposide/Prednisone/Vincristine/Cyclophosphamide/Doxorubicin) + Rituximab • HyperCVAD (Cyclophosphamide/Vincristine/Doxorubicin/Dexamethasone) + Rituximab alternating with Cytarabine/High-Dose Methotrexate • OFA (Oxaliplatin/Fludarabine/Cytarabine) + Rituximab • Patients who are frail or patients with poor left ventricular ejection fraction (LVEF): See NCCN Guidelines for B-Cell Lymphomas (First-line therapy for DLBCL (BCEL-C): Patients with Poor LVEF or Very Frail Patients and Patients >80 Years of Age with Comorbidities)

NON-CIT BASED REGIMENS[d]			
ICI-based regimens[f]	**BTKi (monotherapy)**	**Bispecific Antibody Therapy**	**CAR T-cell Therapy (CD19-directed; after at least 1 prior systemic therapy regimen)[i]**
• Venetoclax* + Atezolizumab[g] + Obinutuzumab • Nivolumab[h] ± Ibrutinib† • Pembrolizumab ± Ibrutinib† • Zanubrutinib† + Tislelizumab-jsgr	• Pirtobrutinib** • Acalabrutinib† (category 2B)	• Epcoritamab-bysp • Glofitamab-gxbm	• Axicabtagene ciloleucel • Lisocabtagene maraleucel • Tisagenlecleucel

See Evidence Blocks on EB-4

* BCL-2-inhibitor (BCL2i)
** Noncovalent (reversible) BTKi (ncBTKi)
† Covalent BTKi (cBTKi)

Footnotes
References

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SUGGESTED TREATMENT REGIMEN FOOTNOTES

[a] Rituximab and hyaluronidase human injection for subcutaneous use may be used in patients who have received at least one full dose of a rituximab product by intravenous route. An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.
[b] For BCL2i-associated TLS, see venetoclax prescribing information.
[c] Refer to package insert for full prescribing information, dose modifications, and monitoring for adverse reactions: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm.
[d] The panel acknowledged that there is a paucity of data for the use of these regimens in patients with Richter transformation refractory to chemotherapy or in patients with a del(17p)/*TP53* mutation; however, these regimens may be considered given the limited options available for this patient population. Additional data will be forthcoming.
[e] It is reasonably safe to combine cBTKi (given continuously for disease control) with CIT regimens without the addition of other targeted agents.
[f] ICI-based combination regimens with cBTki used for prior therapy are not appropriate for patients who are intolerant to or have disease progression on the same cBTKi.
[g] Atezolizumab and hyaluronidase-tqjs subcutaneous injection may be substituted for IV atezolizumab. Atezolizumab and hyaluronidase-tqjs has different dosing and administration instructions compared to atezolizumab for intravenous infusion.
[h] Nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.
[i] See also CAR T-Cell–Related Toxicities in the NCCN Guidelines for Management of Immunotherapy-Related Toxicities for the management of cytokine release syndrome (CRS) and neurologic toxicity.

REFERENCES

Richter Transformation

Chemoimmunotherapy

CHOP + Rituximab ± Venetoclax

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HyperCVAD + Rituximab

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OFAR

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Continued

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SUGGESTED TREATMENT REGIMENS
REFERENCES

Richter Transformation (continued)

ICI-based regimens

Venetoclax/Atezolizumab + Obinutuzumab

Tedeschi A, Frustaci AM, Condoluci A, et al. Atezolizumab, venetoclax, and obinutuzumab combination in Richter transformation diffuse large B-cell lymphoma (MOLTO): A multicentre, single-arm, phase 2 trial. Lancet Oncol 2024;25:1298-1309.

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Pembrolizumab

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Zanubrutinib + Tislelizumab-jsgr

Al-Sawaf O, Ligtvoet R, Robrecht S, et al. Tislelizumab plus zanubrutinib for Richter transformation: the phase 2 RT1 trial. Nat Med 2024;30:240-248.

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BTKi (monotherapy)

Acalabrutinib

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Bispecific Antibody Therapy

Epcoritamab-bysp

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CAR T-Cell Therapy

Axicabtagene ciloleucel

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All recommendations are category 2A unless otherwise indicated.

EVIDENCE BLOCKS FOR CLL/SLL WITHOUT del(17p)/*TP53* MUTATION: FIRST-LINE THERAPY

Preferred Regimens
Acalabrutinib
Acalabrutinib + Obinutuzumab
Venetoclax + Obinutuzumab
Venetoclax/Acalabrutinib
Venetoclax/Acalabrutinib + Obinutuzumab
Zanubrutinib
Other Recommended Regimens
Venetoclax/Ibrutinib (fixed duration)
Venetoclax/Ibrutnib (MRD-Guided)

Useful in Certain Circumstances
Venetoclax/Zanubrutinib (MRD-guided)
Ibrutinib
Ibrutinib + Obinutuzumab
Ibrutinib + Rituximab
Fludarabine/Cyclophosphamide + Rituximab (FCR)
Bendamustine + Obinutuzumab
Bendamustine + Rituximab
Chlorambucil + Obinutuzumab
Obinutuzumab
HDMP + Obinutuzumab
HDMP + Rituximab

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR CLL/SLL WITHOUT del(17p)/*TP53* MUTATION: SECOND-LINE OR SUBSEQUENT THERAPY

Preferred Regimens	
Acalabrutinib	
Zanubrutinib	
Venetoclax + Obinutuzumab	
Pirtobrutinib	

Other Recommended Regimens	
Ibrutinib	
Venetoclax/Ibrutinib	
Venetoclax	
Venetoclax + Rituximab	

Therapy for Relapsed or Refractory disease after prior cBTKi-based and BCL2i-containing regimens

Preferred Regimens	
Liscabtagene maraleucel	
Lisocabtagene maraleucel/Ibrutinib	
Pirtobrutinib	

Other Recommended Regimens	
Duvelisib	
Idelalisib	
Idelalisib + Rituximab	
Fludarabine/Cyclophosphamide + Rituximab (FCR)	
Lenalidomide	
Lenalidomide + Rituximab	
Obinutuzumab	
Bendamustine + Rituximab	
HDMP + Obinutuzumab	
HDMP + Rituximab	

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR CLL/SLL WITH del(17p)/*TP53* MUTATION

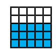
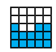
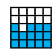
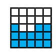
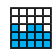
First-line Therapy
Preferred Regimens
Acalabrutinib
Acalabrutinib + Obinutuzumab
Zanubrutinib
Venetoclax + Obinutuzumab
Venetoclax/Acalabrutinib + Obinutuzumab (MRD-guided)
Venetoclax/Zanubrutinib (MRD-guided)
Other Recommended Regimens
Venetoclax + Ibrutinib
Useful in Certain Circumstances
Ibrutinib
HDMP + Obinutuzumab
HDMP + Rituximab
Obinutuzumab

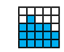
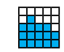
Second-line or Subsequent Therapy
Preferred Regimens
Acalabrutinib
Zanubrutinib
Venetoclax + Obinutuzumab
Pirtobrutinib
Other Recommended Regimens
Ibrutinib
Venetoclax
Venetoclax + Rituximab
Venetoclax/Ibrutinib

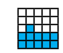
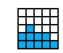
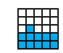
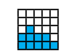
Therapy for Relapsed or Refractory disease after prior tcBTKi-based and BCL2i-containing regimens
Preferred Regimens
Liscabtagene maraleucel
Lisocabtagene maraleucel/Ibrutinib
Pirtobrutinib
Other Recommended Regimens
Duvelisib
Idelalisib
Idelalisib + Rituximab
Alemtuzumab
Alemtuzumab + Rituximab
HDMP + Obinutuzumab
HDMP + Rituximab
Lenalidomide
Lenalidomide + Rituximab

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR RICHTER'S TRANSFORMATION TO DLBCL
(Clonally related or unknown clonal status)

Chemoimmunotherapy	
CHOP + Rituximab	
Dose-adjusted EPOCH + Rituximab	
HyperCVAD + Rituximab	
OFAR	
Venetoclax/CHOP + Rituximab	

BTKi monotherapy	
Pirtobrutinib	
Acalabrutinib	

ICI-based regimens	
Nivolumab	
Nivolumab/Ibrutinib	
Pembrolizumab	
Pembrolizumab/Ibrutinib	
Zanubrutinib + Tislelizumab	
Venetoclax + Atezolizumab + Obinutuzumab	

CAR T-cell Therapy	
Lisocabtagene maraleucel	
Axicabtagene ciloleucel	
Tisagenlecleucel	

Bispecific antibody therapy	
Epcoritamab-bysp	
Glofitamab-gxbm	

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

ABBREVIATIONS

AIHA	autoimmune hemolytic anemia
ALC	absolute lymphocyte count
ANC	absolute neutrophil count
ASO-PCR	allele-specific oligonucleotide polymerase chain reaction
BCL2i	BCL2-inhibitor
BTKi	BTK inhibitor
C/A/P	chest/abdomen/pelvis
CAR	chimeric antigen receptor
CBC	complete blood count
cBTKi	covalent BTK inhibitor
CHL	classic Hodgkin lymphoma
CIT	chemoimmunotherapy
CK	complex karyotype
CMV	cytomegalovirus
CrCl	creatinine clearance
CRS	cytokine release syndrome
DAA	direct-acting antiviral
DLBCL	diffuse large B-cell lymphoma
EBER	Epstein-Barr virus-encoded RNA
EBER-ISH	Epstein-Barr virus-encoded RNA in situ hybridization
EBV	Epstein-Barr virus
ERIC	European Research Initiative on CLL
FISH	fluorescence in situ hybridization
FNA	fine-needle aspiration
G6PD	glucose-6-phosphate dehydrogenase
GVHD	graft-versus-host disease
HBcAb	hepatitis B core antibody
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HCT	hematopoietic cell transplant
HCV	hepatitis C virus
HLA	human leukocyte antigen
HR	hazard ratio
IHC	immunohistochemistry
ISRT	involved-site radiation therapy
ITP	immune thrombocytopenic purpura
IVF	in vitro fertilization
IVIG	intravenous immunoglobulin
JCV	John Cunningham virus
LDH	lactate dehydrogenase
LVEF	left ventricular ejection fraction
mAb	monoclonal antibody
MBL	monoclonal B-cell lymphocytosis
MCL	mantle cell lymphoma
MRD	minimal residual disease
MUGA	multigated acquisition
ncBTKi	noncovalent BTK inhibitor
NGS	next-generation sequencing
NHL	non-Hodgkin lymphoma
OS	overall survival
PFS	progression-free survival
PI3Ki	Pi3K inhibitor
PJP	pneumocystis jirovecii pneumonia
PML	progressive multifocal leukoencephalopathy
PRCA	pure red cell aplasia
qPCR	quantitative RT-PCR
RS	Reed-Sternberg
RSV	respiratory syncytial virus
RT-PCR	reverse transcriptase polymerase chain reaction
SCIG	subcutaneous immunoglobulin
SIFE	serum immunofixation electrophoresis
SPEP	serum protein electrophoresis
SUV	standardized uptake value
TLS	tumor lysis syndrome
TTFT	time-to-first treatment
TTNT	time-to-next treatment
uMRD	undetectable MRD
VAF	variant allele frequency
VZV	varicella zoster virus
WBC	white blood cell
WT	wild type

NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

This discussion corresponds to the NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma. Last updated: December 22, 2025.

Discussion

Overview

Chronic lymphocytic leukemia (CLL) and small lymphocytic lymphoma (SLL) are characterized by progressive accumulation of leukemic cells in the peripheral blood, bone marrow, and lymphoid tissues.[1] Morphologically, these leukemic cells appear as small, mature lymphocytes that can be found admixed with occasional larger or atypical cells, or prolymphocytes. CLL remains the most prevalent adult leukemia in Western countries. In 2025, an estimated 23,690 people will be diagnosed with CLL in the United States, and an estimated 4460 people will die from the disease.[2]

CLL and SLL are essentially different manifestations of the same disease that are similarly managed.[1] The major difference is that in CLL, a significant number of the abnormal lymphocytes are found circulating in blood in addition to being resident in bone marrow and lymphoid tissue, while in SLL, the bulk of disease is in lymph nodes, bone marrow, and other lymphoid tissues and there are few (if any) abnormal lymphocytes circulating in blood.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of this version of the NCCN Clinical Practice Guidelines (NCCN Guidelines®) for CLL/ SLL, an electronic search of the PubMed database was performed to obtain key literature in CLL and SLL published since the previous Guidelines update using the following search terms: chronic lymphocytic leukemia/small lymphocytic lymphoma, Richter syndrome, and histologic transformation. The PubMed database was chosen as it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[3]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Randomized Controlled Trial; Clinical Trial, Phase II; Clinical Trial, Phase III; Clinical Trial; Guideline; Meta-Analysis; Systematic Reviews; and Validation Studies.

The data from key PubMed articles as well as articles from additional sources deemed relevant to these guidelines as discussed by the Panel during the Guidelines update were included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation. NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms *men*, *women*, *female*, and *male* when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Staging

The Rai and Binet systems are the two staging systems currently used for the evaluation of patients with CLL, both in routine practice and clinical trial settings.[4,5] Both staging systems rely on physical assessments (ie, presence of lymph node involvement, enlarged spleen and/or liver) and blood parameters (presence of anemia or thrombocytopenia) to evaluate the degree of tumor burden.

The modified Rai classification stratifies patients into three risk groups: low-risk disease (Rai stage 0), intermediate-risk disease (Rai stage I–II), and high-risk disease (Rai stage III–IV) with historic median survival times of 150 months, 71 to 101 months, and 19 months, respectively, in the era of chemotherapy- and chemoimmunotherapy-based treatment.[4] Survival times in the current era of targeted therapy will most assuredly be improved and will become available with longer follow-up for patients who received targeted therapies.

The Binet staging system stratifies patients into three prognostic groups based on the number of involved areas and the level of hemoglobin and platelets and, like the Rai staging system, provides meaningful correlation with clinical outcome.[5]

The Lugano Modification of the Ann Arbor Staging System is used for patients with SLL.[6]

Prognostic Factors

The prognostic significance of molecular and cytogenetic variables may vary depending on the patient population, treatment regimens, and clinical outcomes being evaluated. The impact of these variables on the clinical outcome is discussed below.

Immunoglobulin Heavy Chain Variable Region Gene Mutation

A cut-off level of ≤2% deviation from germline immunoglobulin heavy chain variable (IGHV) sequence is routinely used in clinical practice to differentiate IGHV unmutated CLL from IGHV mutated CLL.[7-9] Percent deviation from the germline sequence was studied and higher levels were incrementally associated with favorable progression-free survival (PFS) and overall survival (OS) following treatment with the FCR (fludarabine, cyclophosphamide, and rituximab), suggesting that IGHV mutation percentage is a continuous variable.[10] In addition, IGHV gene usage involving VH3-21 is associated with poor outcomes regardless of the IGHV mutation status (as defined by percent homology with germline sequence).[11]

IGHV-unmutated status (≤2% of mutation or ≥98% homology with germline gene sequence) and/or the VH3-21 gene usage is associated with lower response rates and shorter time-to-first treatment (TTFT) and significantly shorter PFS and OS compared to IGHV-mutated in patients receiving chemoimmunotherapy, independent of the stage of the disease.[12-17]

Continuous treatment with a covalent BTKi (cBTKi; ibrutinib, acalabrutinib, or zanubrutinib) with or without anti-CD20 monoclonal antibody (mAb) results in high response rates and survival independent of the IGHV mutation status.[18-23]

IGHV unmutated status remains a prognostic factor for shorter survival after time-limited treatment with BCL2 inhibitor (BCL2i)-containing regimens (venetoclax + anti-CD20 mAb or venetoclax/cBTKi with or without anti-CD20 mAb).[24-34] IGHV-unmutated status and/or the VH3-21 gene usage was also an independent predictor of shorter treatment-free interval and/or survival after time-limited treatment with BCL2i-containing regimens even when high-risk genetic abnormalities were included in the multivariable regression models.[14,15]

BTKi-Based Regimens

The ELEVATE-TN trial showed that acalabrutinib ± obinutuzumab resulted in greater PFS benefit compared to chlorambucil + obinutuzumab both in IGHV-unmutated and IGHV-mutated CLL; however, in patients with IGHV-mutated CLL, the PFS benefit was significant only for combined acalabrutinib + obinutuzumab.[19]

The results of the SEQUOIA study confirmed that PFS was significantly better for zanubrutinib (compared to BR) in patients with IGHV-unmutated and IGHV-mutated CLL.[22]

In the ECOG-ACRIN Cancer Research Group (E1912) study, ibrutinib + rituximab resulted in superior PFS compared to FCR in patients with IGHV-unmutated CLL (hazard ratio [HR], 0.27; $P < .001$) and IGHV-mutated CLL (HR, 0.27; $P < .001$).[23]

BCL2i-Containing Regimens

Venetoclax + anti-CD20 mAb

The extended follow-up data from the CLL14 study showed that venetoclax + obinutuzumab (VenO) resulted in longer PFS for patients with IGHV-mutated CLL compared to those with IGHV unmutated CLL (after a median follow-up of 76 months, the median PFS was not reached for patients in the IGHV-mutated group compared to 65 months for those in the IGHV-unmutated group).[24]

In the phase III randomized GAIA/CLL13 trial, although VenO with or without ibrutinib resulted in significant PFS benefit compared to chemoimmunotherapy among patients with IGHV unmutated CLL as well as in those with IGHV mutated CLL, in the exploratory subgroup analysis IGHV-unmutated was a predictor of shorter PFS across all treatment groups.[25,26]

In the CLL17 trial (N = 909; venetoclax–obinutuzumab, n = 303; venetoclax–ibrutinib, n = 305; ibrutinib, n = 301) that established the non-inferiority of VenO and venetoclax/ibrutinib compared to continuous treatment with ibrutinib, PFS rates were higher for VenO for patients with IGHV-mutated CLL compared to those with IGHV-unmutated CLL (3-year PFS rates were 88% and 76%, respectively).[35]

Venetoclax/Ibrutinib

Venetoclax/ibrutinib prolonged PFS independent of IGHV mutation status in the GLOW trial. However, the estimated 60-month PFS rate was higher for IGHV mutated CLL compared to IGHV unmutated CLL (83% and 52%, respectively).[27,28]

In the FLAIR study, the PFS was significantly better for venetoclax/ibrutinib (compared to FCR) in patients with IGHV unmutated CLL; however, PFS was not significantly different between the treatment arms among patients with IGHV mutated CLL.[29] The 5-year PFS rates for venetoclax/ibrutinib were 91% and 88% for IGHV unmutated and IGHV mutated CLL, respectively compared to 57% and 76%, respectively for FCR. Venetoclax/ibrutinib was also associated with longer PFS than ibrutinib among patients with IGHV unmutated CLL, but the PFS was similar between these two treatment groups among patients with IGHV mutated CLL.

In the CAPTIVATE study, fixed-duration treatment with venetoclax/ibrutinib resulted in higher PFS rates in patients with IGHV mutated CLL than in those with IGHV unmutated CLL. The 5.5-year PFS rates were 79% and 55%, respectively.[30,31] PFS rates were also higher for patients with IGHV mutated CLL than IGHV unmutated CLL in the phase II study that evaluated venetoclax/ibrutinib as MRD-guided treatment.[32]

In CLL17 trial discussed above, venetoclax/ibrutinib resulted in similar PFS rates regardless of IGHV mutations status.[35] The 3-year PFS rate was 79% for patients with IGHV-unmutated CLL compared to 80% for those with IGHV-mutated CLL.

Venetoclax/Acalabrutinib ± Obinutuzumab
In the AMPLIFY trial, the 36-month PFS rates were 84% and 86% for venetoclax/acalabrutinib + obinutuzumab (AVO) and venetoclax/acalabrutinib, respectively in the subgroup of patients with IGHV mutated CLL and the addition of obinutuzumab to venetoclax/acalabrutinib was associated with higher survival benefit in the subgroup of patients with IGHV-unmutated CLL (36-month PFS rate was 83% for AVO compared to 69% for acalabrutinib + venetoclax).[33]

In the phase II study that evaluated AVO as MRD-guided treatment in patients with previously untreated CLL including CLL with *TP53* aberrations, AVO resulted in higher PFS rates in the subgroup of patients with IGHV-mutated CLL than in those with IGHV unmutated CLL.[34] The 4-year PFS rates were 93% and 81% in patients with IGHV mutated CLL and IGHV unmutated CLL, respectively.

Cytogenetic Abnormalities

Cytogenetic abnormalities detected by fluorescence in situ hybridization (FISH) are present in >80% of patients with CLL.[36]

Del(13q) (55%), del(11q) (18%), trisomy 12 (16%), del(17p) (7%), and del(6q) (7%) are the most common abnormalities at the time of diagnosis. Del(13q) as a sole abnormality was associated with favorable prognosis and the longest median survival (133 months) after chemoimmunotherapy. Del(11q) is often associated with extensive lymphadenopathy, disease progression, shortened TTFT, and shorter median survival (79 months) after chemoimmunotherapy. Trisomy 12 is associated with intermediate prognosis for TTFT and OS in patients with newly diagnosed CLL treated with chemoimmunotherapy.[36]

Del(17p) reflects the loss of the *TP53* gene and is frequently associated with mutation in the remaining *TP53* allele. Del(17p) is more frequently observed in patients with previously treated CLL, suggesting that acquisition and/or expansion of CLL clones with del(17p) may occur through treatment. The prognostic significance of del(17p) may be dependent on the proportion of malignant cells with this abnormality, and the prognosis is more favorable when the percentage of cells with del(17p) is low.[37,38]

TP53 Aberrations

TP53 aberrations [del(17p) or *TP53* mutation] are predictors of poor outcomes with chemoimmunotherapy. Del(17p) is associated with inferior response, shorter treatment-free interval, and inferior survival with chemoimmunotherapy.[17,36,39] *TP53* mutations (independent of 17p chromosome status) are predictors of shorter PFS and OS with fludarabine- or bendamustine-containing chemoimmunotherapy regimens.[40-43]

TP53 aberrations also remain an independent predictor of inferior PFS and OS for time-limited treatment with venetoclax-containing regimens.[44,45] Del(17p) and *TP53* mutation are independent predictors of PFS and OS, whereas del(17p) and/or *TP53* mutation with IGHV-unmutated status is associated with the shortest PFS.[44,45] Continuous treatment with a cBTKi also results in shorter PFS and OS in patients with del(17p) or *TP53*-mutated CLL.[18,19,23,46,47] However, the survival outcomes for CLL with *TP53* aberrations treated with either a BTKi-based regimen or a venetoclax-containing regimen are much better than the survival outcomes following chemoimmunotherapy.

TP53 mutations with del(17p) are associated with poor prognosis (increased resistance to chemoimmunotherapy and targeted therapies; significantly shortened TTFT and time to next treatment as well as potentially increased risk for Richter transformation). Survival outcomes (PFS and OS) for patients with *TP53*-mutated CLL in the absence of del(17p) may be more favorable than those with *TP53*-mutated CLL and del(17p).[41,48]

TP53 mutations of low variant allele frequencies (VAF; <10%) may have prognosis similar to *TP53* wild type in terms of OS.[49-52] In a retrospective analysis of 964 patients, *TP53* mutations of low VAF (<10%) and high VAF (≥10%) were found in 28% and 53% of patients, respectively.[51] *TP53* mutations of high VAF were associated with shorter TTFT (0.3 years, $P <$ .001) compared to *TP53* mutations of low VAF (1.2 years) and *TP53* wild type (4 years). The 5-year OS rates were also inferior for patients with *TP53* mutations of high VAF (80%) compared to 93% and 91% for those with*TP53* mutations of low VAF and *TP53* wild type, respectively. Other studies have shown that *TP53* mutations of low VAF (1%–10%) were associated with clonal expansion as well as short time to next treatment (TTNT) and OS following treatment with chemoimmunotherapy, but it did not have any prognostic impact on OS in patients with CLL/SLL treated with targeted therapy.[49,50,52]

Complex Karyotype

Complex karyotype (CK; ≥3 unrelated chromosomal abnormalities in more than one cell on CpG-stimulated karyotype of CLL cells) is associated with inferior clinical outcomes. A retrospective analysis of >5000 patients with available cytogenetic data indicated that CK was associated with variable clinical behavior.[53] High CK (≥5 unrelated chromosomal abnormalities) emerged as an adverse prognostic factor independent of clinical stage, IGHV mutation status, and *TP53* aberrations [del(17p) and/or *TP53* mutation], whereas low CK (three unrelated chromosomal abnormalities) and intermediate CK (four unrelated chromosomal abnormalities) were clinically relevant only if coexisting with *TP53* aberrations.

CK may be a stronger predictor of poor clinical outcomes than del(17p) or *TP53* mutation in patients with CLL treated with ibrutinib-based regimens.[54-57] It should be noted that in these studies, del(17p) often correlated with the presence of CK. Among patients with relapsed or refractory CLL treated with ibrutinib-based regimens, in a multivariable analysis, only CK was significantly associated with shorter event-free survival (EFS; $P = .006$), whereas CK ($P = .008$) and fludarabine-refractory CLL ($P = .005$) were independently associated with shorter OS.[54] In an analysis of 308 patients with CLL treated with ibrutinib on four sequential clinical trials, in a multivariable analysis, CK at baseline, presence of del(17p), and age <65 years were all independently associated with shorter time to CLL progression.[57] In patients ≥65 years without CK or del(17p), the estimated cumulative incidence of CLL progression at 4 years was 2% compared to 44% in patients <65 years with CK and del(17p). CK was not associated with worse PFS in patients with treatment-naïve CLL treated with zanubrutinib in the SEQUOIA study.[58] The result of another retrospective analysis (n = 1339) also showed that CK is a significant prognostic factor of inferior OS compared to isolated *TP53* mutation.[59] *TP53* aberration in univariate ($P < .001$) and multivariate analysis ($P = .002$) was a significant predictor of inferior OS; however, *TP53* aberration lost its significance ($P = .94$) in multivariate analysis in the presence of CK ($P < .001$).

High CK was an adverse prognostic factor in patients with CLL treated with venetoclax-containing regimens.[60] In a prospective analysis of the GAIA/CLL13 trial, CK (≥3 unrelated chromosomal abnormalities) was associated with shorter PFS (HR, 2.6; $P < .001$) and OS (HR, 3.25; $P =$ .044) after treatment with chemoimmunotherapy, whereas only high CK (≥5 unrelated chromosomal abnormalities) was an independent adverse prognosticator for PFS in the pooled venetoclax arms.[60] Chemoimmunotherapy resulted in the acquisition of additional chromosomal abnormalities whereas CK remained stable after treatment with venetoclax-containing regimens.

Beta-2 Microglobulin

Beta-2 microglobulin is readily measured by standard laboratory evaluation of blood samples, and an elevated level of serum beta-2 microglobulin was shown to be a strong independent prognostic indicator for treatment-free interval, response to treatment, and OS following treatment with first-line chemoimmunotherapy.[61,62] Beta-2 microglobulin was incorporated in prognostic models for the risk stratification of patients with CLL.[63-66] However, it is influenced in a CLL disease-independent manner by renal dysfunction.

Other Gene Mutations

In addition to *TP53* mutation, recurrent mutations with prognostic implications were identified in *ATM, NOTCH1*, *SF3B1*, and *BIRC3* genes. The incidence of these mutations is approximately 4% to 15% in patients with newly diagnosed CLL, and the incidences are much higher (15%–25%) in patients with fludarabine-refractory CLL.[67-71] *ATM*, *SF3B1*, and *NOTCH1* mutations were predictors of shorter TTFT independent of IGHV mutation status, whereas *TP53* and *NOTCH1* mutations along with IGHV-unmutated status were predictors of shorter OS.[70] In the CLL14 study, *BIRC3* and *SF3B1* mutations were independent predictors of inferior PFS after chemoimmunotherapy with chlorambucil + obinutuzumab, but these mutations had no impact on the clinical outcome after VenO; however, the follow-up was short.[44] In the MURANO study, pre-existing *TP53*, *NOTCH1*, and *BRAF* mutations were associated with lower rates of initial attainment of uMRD4 following treatment with VenR.[72] *BCOR*, *CCND2, NRAS,* and *XPO1* mutations at baseline were associated with poor MRD response as well as inferior PFS and OS following MRD-guided treatment with venetoclax/ibrutinib in relapsed/refractory CLL/SLL.[73]

An integrated prognostic model including *NOTCH1*, *SF3B1*, and *BIRC3* mutations along with the cytogenetic abnormalities detected by FISH was proposed to classify patients with newly diagnosed or previously untreated CLL who received rituximab-based chemoimmunotherapy or fludarabine or alkylating agent-based chemotherapy into four distinct prognostic subgroups: high risk (*TP53* and/or *BIRC3* abnormalities); intermediate risk (*NOTCH1* and/or *SF3B1* mutations and/or del[11q]); low risk (trisomy 12 and wild type for all genetic lesions); and very low risk (del[13q] only).[74] The 10-year survival rates for the four subgroups were 29%, 37%, 57%, and 69%, respectively. This prognostic model may have limited utility since it excludes the IGHV mutation status.

Prognostic Models

Several scoring systems and prognostic models incorporating traditional and newer prognostic markers were developed to predict the clinical course of disease and outcomes more accurately.

A prognostic nomogram and a more simplified prognostic index (based on age, beta-2 microglobulin, absolute lymphocyte count (ALC), sex, Rai stage, and number of involved lymph nodes) is useful in estimating TTFT in patients with untreated CLL, including those with early-stage disease and the utility of this prognostic index was confirmed in several studies.[64,75,76] The simplified prognostic index is also useful in estimating the survival probability and stratifies patients with untreated CLL into three different risk groups (low, intermediate, and high) with different survival outcomes.[64] The 5-year survival rates were 97% for low-risk, 80% for intermediate-risk, and 55% for high-risk groups; the 10-year survival rates were 80%, 52%, and 26%, respectively.

In another prognostic model, increased size of cervical lymph nodes, three involved nodal sites, del(17p) or del(11q), *IGHV*-unmutated status, and elevated serum lactate dehydrogenase (LDH) levels were identified as independent predictors of shorter TTFT.[77] This model may help to identify patients with newly diagnosed CLL at high risk for disease progression who may be candidates for clinical trials of interventions to delay TTFT with chemoimmunotherapy.

The Integrated CLL Scoring System (ICSS) is based on the cytogenetic abnormalities detected by FISH, IGHV mutation status, and CD38 expression.[78] ICSS stratified patients into three risk groups (low, intermediate, and high) with different TTFT and OS. ICSS is also helpful to identify patients with a high likelihood of early progression who would be candidates for clinical trials evaluating early interventions.

The International Prognostic Index for CLL (CLL-IPI) is based on *TP53* and IGHV mutation status, serum beta-2 microglobulin concentration, clinical stage, and age.[63] The CLL-IPI was validated in an independent cohort of patients with newly diagnosed CLL and is useful for predicting TTFT and risk of progression in patients receiving first-line chemoimmunotherapy.[79] CLL-IPI stratifies patients into four risk groups (low, intermediate, high, and very high) with significantly different OS. The 5-year OS rates were 93%, 79%, 63%, and 23%, for the four risk groups, respectively.

The International Prognostic Score for Early-Stage CLL (IPS-E) predicts the likelihood of disease progression to need treatment in patients with early-stage CLL and stratifies patients with early-stage CLL into three risk groups with significantly different TTFT.[80] The cumulative risk for the need of treatment after 1 and 5 years of observation was 14% and 61%, respectively, for patients with high-risk IPS-E compared to 2% and 28% for patients with intermediate-risk IPS-E and <0.1% and 8% for patients with low-risk IPS-E. These findings need to be validated in a prospective clinical trial.

Targeted therapies with small-molecule inhibitors have significantly improved survival outcomes and prognostic models were developed to predict the outcome following treatment with targeted therapies.[65,66] The first prognostic model is predictive of survival following treatment with ibrutinib and stratified patients into three risk groups (high [3–4 points]; intermediate [2 points]; and low [0 points]) based on *TP53* aberrations, prior treatment, elevated serum beta-2 microglobulin, and LDH.[65] The 3-year PFS rates were 47%, 74%, and 87% for the high-, intermediate-, and low-risk groups, respectively (P < .0001). The corresponding 3-year OS rates were 63%, 83%, and 93%, respectively (P < .0001). This model remained significant in the stratification of patients with treatment-naïve and relapsed or refractory CLL. The second prognostic model identified patients with high-risk previously treated CLL who do not achieve a good outcome with available targeted therapies (ibrutinib, idelalisib, and venetoclax).[66] This prognostic model stratified patients into three risk groups based on elevated serum beta-2 microglobulin and LDH, hemoglobin, and time from initiation of last therapy (<24 months): low (score 0–1); intermediate (score 2–3); and high risk (score 4).

Response Criteria

The response criteria developed by the International Workshop on Chronic Lymphocytic Leukemia (iwCLL) provide recommendations for the evaluations and response assessments appropriate for the general clinical practice setting versus for clinical trials.[81] These response criteria are based on the size of lymph nodes, liver and/or spleen, presence or absence of constitutional symptoms, circulating lymphocyte count and blood parameters.[81]

In clinical trials, the size of the lymph nodes is evaluated by CT scans and physical examination. There is no firmly established international consensus on the size of a normal liver. Therefore, in clinical trials, the liver size is evaluated by imaging and palpation and recorded according to the definition used in a study protocol. In the clinical practice setting, response assessment involves both physical examination and evaluation of blood parameters.

Complete Response

All of the following criteria must be met at least 2 months after treatment completion: normal circulating lymphocyte count; absence of lymphadenopathy (palpable nodes must be ≤1.5 cm in diameter), splenomegaly or hepatomegaly; absence of constitutional symptoms (weight loss, significant fatigue, fevers, night sweats); and normalization of blood counts without growth factor support.

Patients who fulfill the criteria for a complete response (CR) but present with persistent cytopenias should be considered as having achieved a CR with incomplete bone marrow recovery.

Partial Response

At least two of the following criteria must be met for at least 2 months duration: ≥50% reductions from baseline in peripheral blood lymphocyte counts, lymphadenopathy, hepatomegaly; and/or splenomegaly (if present at baseline). In addition, at least one of the blood counts should be normalized or increase by ≥50% from baseline, for at least 2 months duration.

Partial Response with Lymphocytosis

Treatment with both cBTKi (ibrutinib, acalabrutinib, zanubrutinib) and noncovalent (reversible) BTKi (ncBTKi; pirtobrutinib) and phosphatidylinositol 3-kinase inhibitors (PI3Ki; idelalisib and duvelisib) cause mobilization of lymphocytes into blood early during treatment initiation, resulting in a significant transient lymphocytosis (increase in absolute lymphocyte count [ALC]) in most patients, which does not signify disease progression. This situation may also occur with investigational kinase inhibitors that target similar pathways. Prolonged lymphocytosis following ibrutinib treatment was reported to represent the persistence of a quiescent clone, and slow or incomplete resolution of lymphocytosis does not appear to impact outcome as measured by PFS.[82]

Considering these findings, the iwCLL response criteria were revised to more precisely predict the outcome of patients with CLL treated with BTKi and PI3Ki.[83] The revised iwCLL response criteria allow for the response category, partial response (PR) with lymphocytosis (PR-L). In patients receiving BTKi (ibrutinib, acalabrutinib, zanubrutinib, or pirtobrutinib) or PI3Ki (idelalisib or duvelisib), this response category includes clinical response (reduction in lymph nodes and splenomegaly with persistent lymphocytosis, in the absence of other indicators of progressive disease). Isolated progressive lymphocytosis in the setting of reduced lymph node size or organomegaly or improvement in hemoglobin/platelets will not be considered progressive disease.

Progressive Disease

Progressive disease comprises any of the following: at least 50% increase from baseline in lymphocyte counts, lymphadenopathy, hepatomegaly, or splenomegaly; appearance of any new lesions; or occurrence of cytopenias attributable to disease (≥50% decrease from baseline in platelet count or >2 g/dL decrease from baseline in hemoglobin levels).

Stable Disease

Patients who do not meet the criteria for a CR or PR in the absence of disease progression are considered to have stable disease.

Relapse or Refractory Disease

Relapse is defined as evidence of disease progression after a period of ≥6 months following an initial CR or PR. Refractory disease is defined as lack of response to treatment or disease progression within 6 months of the last treatment.

Measurable Residual Disease

Measurable residual disease (MRD) is a highly sensitive indicator of disease burden in patients with CLL, and assessment of MRD as part of response evaluation is incorporated into some clinical trials.

MRD detection can be performed with different methods with associated sensitivities and evaluated in cells from either blood or bone marrow. A commercial next-generation sequencing (NGS)-based assay was reported to achieve a limit of CLL detection down to $\leq 10^{-6}$ (MRD6), and currently is the only assay in the United States with FDA regulatory clearance.[84-87] NGS-based assays require collection of a pretreatment sample. Multicolor (≥4) flow cytometry and allele-specific oligonucleotide polymerase chain reaction (ASO-PCR) are the two other methods used to detect MRD at the level of $\leq 10^{-4}$ (MRD4) or $\leq 10^{-5}$ (MRD5) with significantly more supporting data from clinical trials. Flow cytometry methodology was standardized by the European Research Initiative on CLL (ERIC) and is the most widely used method owing to the extensive availability and reliable detection down to MRD5 and does not require a pretreatment sample for analysis.[88] ASO-PCR is less widely used since it is expensive, requires a pretreatment sample, and is more labor intensive.[89]

Consensus recommendations for the methodology for MRD determination, assay requirements and tissue selection (blood vs. bone marrow), and the use of MRD in clinical trials were published.[90,91] Few clinical trials have evaluated MRD-guided treatment protocols to determine the duration of treatment.[29,34,92]

Continuous treatment with BTKi monotherapy does not typically result in uMRD, but the use of cBTKi in combination with anti-CD20 mAb results in higher rates of uMRD compared to monotherapy.[19,93,94] In the E1912 phase III randomized trial that compared FCR versus ibrutinib + rituximab, among patients randomized to ibrutinib + rituximab there was no significant difference in PFS rates based on uMRD status.[95] However, PFS was significantly longer in patients with MRD levels $<10^{-1}$ than those with MRD levels $>10^{-1}$ and continuous treatment with ibrutinib was necessary to maintain treatment efficacy. The prognostic value of uMRD4 has not been confirmed in the context of continuous BTKi monotherapy or in combination with anti-CD20 mAb.

Results from randomized clinical trials demonstrated that time-limited treatment (fixed-duration or MRD-guided) with BCL2i-containing regimens (venetoclax + anti-CD20 mAb or venetoclax + cBTKi with or without anti-CD20 mAb) resulted in higher rates of uMRD4 or uMRD6 (MRD levels $<10^{-6}$) in blood or bone marrow than BTKi monotherapy or chemoimmunotherapy (Table 1). uMRD4 after EOT with venetoclax + obinutuzumab containing regimens is also an independent predictor of longer PFS among patients with previously untreated as well as relapsed or refractory CLL.[24,96]

Data from clinical trials that have evaluated time-limited treatment with BCL2i-containing regimens are discussed below.

Fixed-duration Regimens

Venetoclax + anti-CD20 mAb

In the CLL14 study, deeper uMRD remissions (uMRD5 and uMRD6) were more frequent with VenO and PFS was longer in patients with uMRD6 compared to those with detectable MRD4 at EOT. The 6-year PFS rates were 62% for patients with uMRD6 and 23% for those with detectable

MRD4.[24,45] The 6-year OS rate was 87% for patients with uMRD4 and 59% for those with detectable MRD4.

In the GAIA/CLL13 phase III randomized trial, VenO (with or without ibrutinib) resulted in significantly higher uMRD4 rates (*P* < .001) compared to chemoimmunotherapy, but the uMRD4 rate was not significantly higher with venetoclax + rituximab (VenR; *P* = .32) compared to chemoimmunotherapy.[25] VenO also resulted in higher rates of uMRD (blood) at 15 months (uMRD4, 81% vs. 55%; uMRD6, 64% vs. 28%) compared to chemoimmunotherapy with FCR or bendamustine and rituximab (BR) in the phase III CRISTALLO study.[97]

Fixed-duration treatment with venetoclax + obinutuzumab also resulted in higher rates of uMRD (blood) at the EOT compared to ibrutinib in the CLL17 trial (73% compared to 0% for ibrutinib).[35] This trial established venetoclax + obinutuzumab and venetoclax/ibrutinib are non-inferior to continuous treatment with ibrutinib in terms of PFS (3-year PFS rates were 81% for both treatment arms).

In the MURANO study, uMRD as best MRD response at any time during the study was higher with VenR than BR (83% vs. 23%) and the 5-year follow-up data showed that uMRD at the EOT with VenR was associated with improved PFS and OS in patients with relapsed/refractory CLL/SLL.[96,98] The 3-year PFS rates after EOT were 61% for those with uMRD4 compared to 41% for those with low MRD-positive disease (10^{-4} to $<10^{-2}$). The 3-year OS rates after EOT were 95% for those with uMRD4 and 73% for those with low MRD-positive disease (10^{-4} to $<10^{-2}$) or high MRD-positive disease ($>10^{-2}$).[96] IGHV-unmutated, del(17p), and genomic complexity (≥3 copy number variations) were associated with higher rates of conversion to detectable MRD4 and subsequent progressive disease after attaining uMRD4 at EOT.[96]

Venetoclax/Ibrutinib

In the phase III randomized GLOW study, fixed-duration treatment with venetoclax/ibrutinib (ibrutinib x 3 cycles followed by 12 cycles of combined venetoclax/ibrutinib) resulted in higher rate of uMRD4 at 3 months after EOT (EOT+3) across all subgroups including del(11q) and IGHV-unmutated CLL.[27,28] The rate of uMRD5 was higher with venetoclax/ibrutinib (45% in blood; 40% in bone marrow) compared to chlorambucil + obinutuzumab (22% and 6%, respectively). The estimated PFS rate for patients with uMRD4 in the bone marrow at 12 months after EOT (EOT+12) was 96% for venetoclax/ibrutinib compared to 83% for chlorambucil + obinutuzumab.[27] After a median follow-up of 64 months, PFS benefit was observed, particularly in patients with IGHV-unmutated CLL, who achieved uMRD4 at EOT+3 (48-month PFS rate was 67% vs. 40% for those with detectable MRD); PFS rates at 48 months after EOT were also higher with venetoclax/ibrutinib among patients with IGHV-mutated CLL (≥84%) independent of the MRD status at EOT+3.[28]

The results of the phase II randomized CAPTIVATE study showed that fixed-duration treatment with venetoclax/ibrutinib (ibrutinib x 3 cycles followed by 12 cycles of combined venetoclax/ibrutinib) resulted in high rates of uMRD4 in all subgroups (del[17p] and/or mutated *TP53*, del[11q], and IGHV-unmutated CLL).[30,31,99,100] In the fixed-duration treatment cohort, uMRD4 rates were 83% (blood) and 45% (bone marrow) for patients with del(17p) and/or mutated *TP53*; uMRD4 rates were higher in patients with IGHV-unmutated CLL (88% in blood; 73% in bone marrow) compared to IGHV-mutated CLL (72% in blood; 60% in bone marrow).[30] The 5.5-year PFS rate was higher in patients with uMRD4 (blood) at EOT than those with detectable MRD at EOT (75% vs. 47%).[31] In the MRD cohort, patients were assigned to subsequent treatment based on the uMRD4 status at EOT.[99,100] Patients without confirmed uMRD4 were randomized to receive venetoclax/ibrutinib (n = 32) or ibrutinib (n = 31); post-randomization uMRD4 rates were higher with venetoclax/ibrutinib than with ibrutinib.[99]

The estimated 3-year PFS rates were 97% for patients in both treatment arms. Patients with confirmed uMRD4 (n = 86) were randomized to receive placebo or ibrutinib. The estimated 4-year PFS rates were 95% for those assigned to ibrutinib and 88% for those assigned to placebo. The 4-year OS rates were not significantly different for the two treatment arms (100% and 98%, respectively).[100]

Fixed-duration treatment with venetoclax/ibrutinib also resulted in higher rates of uMRD (blood) at the EOT compared to ibrutinib in the CLL17 trial (47% compared to 0% for ibrutinib).[35] This trial established venetoclax/ibrutinib and venetoclax + obinutuzumab are non-inferior to continuous treatment with ibrutinib in terms of PFS (3-year PFS rates were 79% and 81% respectively).

In the phase II single-arm CLARITY study, fixed-duration treatment with venetoclax/ibrutinib (venetoclax was added after 2 months of ibrutinib and the duration of venetoclax/ibrutinib was determined by the achievement of uMRD) resulted in high rates of uMRD4 in patients with relapsed or refractory CLL/SLL.[101,102] The duration of treatment was based on the time to achieve uMRD4 in both blood and bone marrow (14 months for patients with uMRD4 at 8 months; 26 months for those with uMRD4 at 14 months and/or at 26-month follow-up; venetoclax was discontinued and ibrutinib was given until disease progression in patients with detectable MRD at 26 months). In an exploratory analysis, the achievement of uMRD4 after 6 months or a 2-log reduction in MRD levels after 2 months of treatment with venetoclax/ibrutinib resulted in sustained uMRD4 status and ability to discontinue treatment.[102]

Venetoclax/Acalabrutinib ± Obinutuzumab

Among the patients for whom measurement of MRD were available at designated timepoint, the uMRD4 (blood) rates at the EOT and at EOT+3 was also higher after fixed-duration treatment with AVO (95% and 94%, respectively) than with chemoimmunotherapy in the AMPLIFY trial (73% and 78%, respectively).[33] The corresponding blood uMRD rates were 45% and 38%, respectively for venetoclax/acalabrutinib, suggesting that the addition of obinutuzumab increases uMRD rate. However, the lower uMRD rate did not correspond to lower PFS rate with venetoclax/acalabrutinib and longer follow-up is needed to confirm these findings. Acalabrutinib was administered for cycles 1 to 14 and venetoclax was administered from cycles 3 to 14. In the AVO arm, obinutuzumab (1000 mg IV) was administered for cycle 2 to 7 (day 1).

MRD-guided Regimens

Venetoclax/Ibrutinib

MRD-guided treatment with venetoclax/ibrutinib was evaluated in the FLAIR study (263 patients were assigned to ibrutinib monotherapy; 523 patients were randomized between venetoclax/ibrutinib, n = 260 and FCR, n = 263).[29] In the venetoclax/ibrutinib arm, ibrutinib was administered for 8 weeks before initiation of venetoclax and the duration of combined venetoclax/ibrutinib was from 2 to 6 years, determined by uMRD-guided criteria. Venetoclax/ibrutinib resulted in higher rates of uMRD compared to FCR and ibrutinib. With a median follow-up of 62 months, uMRD (bone marrow) at 60 months after randomization was observed in 72% of patients in the venetoclax/ibrutinib arm; 55% of patients in the FCR group and 0% of patients in the ibrutinib arm. The uMRD (blood) at 60 months after randomization was also higher with venetoclax/ibrutinib (93%) compared to FCR (69%) and ibrutinib (4%). At 6 years after treatment initiation with venetoclax/ibrutinib, an estimated 79% of patients had stopped treatment due to the achievement of uMRD and this percentage was higher in the IGHV unmutated group than in the IGHV mutated group (81% and 52%, respectively).The subgroup analysis suggest a benefit for MRD-guided treatment for patients with IGHV unmutated CLL.[29] FLAIR trial also identified IGHV unmutated status as a predictor of early achievement of uMRD following treatment with venetoclax/ibrutinib.[103] Large ongoing clinical trials will help to clarify the optimal duration of MRD-guided treatment with combined targeted therapy and the importance of the difference in uMRD rates between IGHV-mutated and IGHV-unmutated CLL.

The feasibility of MRD-guided treatment with venetoclax/ibrutinib given for 24 cycles was also demonstrated in a phase II study (n = 80; 18 patients had del (17p) and/or TP53 mutation).[32] Ibrutinib, was administered as monotherapy for 3 cycles and then combined with venetoclax for a total of 24 cycles of combination regimen. After a median follow-up of 39 months, uMRD (bone marrow) was 56% (n = 45) after 12 cycles and 66% (n = 53) after 24 cycles (51 out of the 53 patients who achieved uMRD after 24 cycles discontinued treatment). The estimated 3-year PFS and OS rates were 93% and 96%, respectively. The uMRD rates (bone marrow) for patients with del(17p) and/or *TP53* mutation were 69% (n = 9) and 77% (n = 10) after 12 cycles and 24 cycles, respectively.

Venetoclax/Acalabrutinib + Obinutuzumab

A phase II single center study investigated MRD-guided treatment with AVO in patients with previously untreated CLL (N = 72; 45 patients with *TP53* aberrations).[34] AVO was administered as follows: acalabrutinib followed by acalabrutinib + obinutuzumab (cycles 2 and 3) followed by AVO (cycles 4–7) and acalabrutinib + venetoclax (cycles 8–15). Patients could discontinue treatment if CR with uMRD (bone marrow) was achieved at the start of cycle 16. The CR with uMRD (bone marrow) rate was 42% (for the overall study population as well as in patients with *TP53* aberrant disease). The uMRD (bone marrow) rate was 71% for the overall study population and 78% in patients with *TP53* aberrant disease. Treatment discontinuation after achieving CR with uMRD (bone marrow) was reported in 79% of patients (58% had *TP53* aberrant disease). Forty-two percent of patients (n = 30) discontinued treatment after achieving CR with uMRD (bone marrow) before cycle 16 and an additional 38% of patients (n = 27) discontinued treatment after achieving CR with uMRD (bone marrow) after completing 24 cycles. The median duration of treatment was 25 cycles.

Venetoclax/Zanubrutinib

The SEQUOIA study evaluated MRD-guided treatment with venetoclax/zanubrutinib in the non-randomized cohort (Arm D; n = 114; zanubrutinib was administered as monotherapy for 3 cycles and then combined with venetoclax for cycles 4–28).[92] The best uMRD4 (blood) rates (percentage of patients achieving uMRD at least once) were similar in the subgroups of patients with and without *TP53* aberrations (59% and 60% respectively). The uMRD (blood) rate at the beginning of cycle 16 and 28 were 43% and 60%, respectively for patients without *TP53* aberration. The corresponding uMRD rate at cycle 16 was lower in patients with *TP53* aberrations (21%) but increased to 49% by cycle 28 suggesting that patients with *TP53* aberrations may require a longer duration of treatment to achieve uMRD4 rates like that in patients without *TP53* aberrations. Discontinuation of treatment (venetoclax after a minimum of 12 cycles or zanubrutinib after a minimum of 27 cycles) was allowed in patients meeting the uMRD-guided stopping criteria. Treatment discontinuation rate based on the achievement of uMRD criteria was 10% (most of these patients did not have *TP53* aberrations).

NCCN Recommendations

The Panel consensus supported the inclusion of the following regimens as MRD-guided treatment options for first-line therapy:

- Venetoclax/ibrutinib for CLL/SLL without del(17p) or *TP53* mutation based on the results of FLAIR study (category 1; other recommended).[29]

- Venetoclax/zanubrutinib based on the results of SEQUIOA (Arm D).[92] This combination is included with a category 2A recommendation (useful in certain circumstances for CLL/SLL without del(17p) or *TP53* mutation; preferred treatment option for CLL/SLL with del(17p) or *TP53* mutation)

- AVO (category 2A; preferred) for CLL with del(17p)/*TP53* mutation based on results of phase II single center study.[34]

NCCN Guidelines recommend serial MRD evaluations every 3 to 6 months during active treatment (using either flow cytometry or NGS assay with a sensitivity for the detection of at least 10^{-4} CLL cells in blood or bone marrow) as outlined in the published trial protocols that support MRD-guided treatment options. NGS-based clonality testing for MRD assessment should be considered if MRD-guided treatment is planned and MRD-flow is not being used for MRD assessment. It should be noted that administration of anti-CD20 mAb within the prior 3 months can lead to false blood uMRD4 status due to B-cell depletion, especially when assessing MRD in blood.[104,105]

Diagnosis

The diagnosis of CLL requires the presence of at least 5 x 10^9/L monoclonal B-lymphocytes in the peripheral blood and the clonality of B cells should be confirmed by flow cytometry.[81] Immunophenotype by flow cytometry (blood) is adequate for the diagnosis of CLL; bone marrow biopsy is generally not required.

The diagnosis of SLL requires the presence of lymphadenopathy and/or splenomegaly with <5 x 10^9/L monoclonal B lymphocytes in the peripheral blood.[81] A diagnosis of SLL should ideally be confirmed by excisional or incisional lymph node biopsy, if lymph node is accessible. Fine-needle aspiration (FNA) biopsy alone is inadequate for the diagnosis. In certain circumstances, when a lymph node is not easily accessible, a combination of core needle biopsy (multiple biopsies may be required) in conjunction with appropriate ancillary techniques may be sufficient for diagnosis. Bone marrow aspirate with biopsy can be pursued if lymph node biopsy material is nondiagnostic. B-cells with a CLL/SLL phenotype may be found in samples from patients with reactive lymph nodes; however, a diagnosis of SLL should only be made when there is effacement of the lymph node architecture by histology.

Evaluation of cyclin D1 (flow cytometry or IHC) or FISH for t(11;14), flow cytometry for CD200, and IHC for LEF1 and SOX11 may be helpful in the differential diagnosis of CLL, especially in suspected cases of mantle cell lymphoma that are cyclin D1-negative.[106-109]

FISH for the detection of del(11q), del(13q), trisomy 12, del(17p), CpG-stimulated metaphase karyotype, *TP53* sequencing, and molecular genetic analysis for IGHV mutation status can provide useful prognostic information and may guide selection of therapy.

Interphase FISH is the standard method to detect specific chromosomal abnormalities that may have prognostic significance. Conventional metaphase karyotype is difficult in CLL due to the very low in vitro proliferative activity of the leukemic cells. CpG oligonucleotide stimulation can be utilized to enhance metaphase cytogenetics.[110,111] CpG-stimulated karyotyping is useful to identify patients with high-risk CLL, particularly for treatment with targeted agents and developing a long-term treatment strategy. NGS (using an assay with a sensitivity to detect *TP53* mutations of low VAF of 5%) is preferred over Sanger sequencing for the *TP53* sequencing since the latter has a low detection limit that is variable between 10% to 20%.[112] However, a specific cutoff value for reporting VAF has not been established.

Molecular analysis for IGHV mutation testing is recommended based on reproducibility and ready availability. IGHV mutation status is helpful for the selection of initial treatment when considering time-limited treatment with BCL2i containing combination regimens. IGHV mutation status does not change over time and mutation analysis need not be repeated if previously done prior to initiation of treatment. IGHV mutation analysis is performed through amplification of IGHV transcript by polymerase chain reaction (PCR) and gene sequencing through Sanger sequencing.[113,114] Complementary DNA (cDNA) and genomic DNA (gDNA) are suitable for IGHV mutation analysis. cDNA identifies more functional rearrangements whereas gDNA can be performed on stored samples and it also avoids the need for reverse transcription.

Monoclonal B-Cell Lymphocytosis

Monoclonal B-cell lymphocytosis (MBL) is a condition in which an abnormal monoclonal B-cell population with the immunophenotype of CLL is present but does not meet the diagnostic criteria for CLL.[115,116] An ALC of <5 x 10^9/L that is stable over a 3-month period in the absence of palpable lymphadenopathy or other clinical features characteristic of a lymphoproliferative disorder (ie, anemia, thrombocytopenia, constitutional symptoms, organomegaly) is defined as MBL.[117]

MBL is further categorized into low-count MBL (<0.5 x 10^9/L) that rarely progresses to CLL and high-count MBL (0.5–4.9 x 10^9/L) that can progress to CLL requiring therapy at a rate of 1% to 2% per year.[118,119] High-count MBL is distinguished from Rai 0 CLL based on whether the monoclonal B-cell count is above or below 5 x 10^9/L.[120] A nodal variant characterized by nodal infiltration of CLL-line cells without apparent proliferation centers and absence of lymphadenopathy was also described in a subset of patients with MBL.[121]

MBL is associated with favorable molecular characteristics, including IGHV-mutated and del(13q), lower prevalence of del(11q)/del(17p) and wild-type *TP53*, slower lymphocyte doubling time, longer treatment-free survival, and very low rate of progression to CLL.[116] Observation is recommended for all individuals with MBL.

Workup

The workup of patients with CLL/SLL should include history and physical exam (measurement of size of liver and spleen and palpable lymph nodes), evaluation of performance status and B symptoms. Measurement of beta-2 microglobulin may provide useful prognostic information.[64]

CLL/SLL is diagnosed mainly in older adults, with a median age of 72 years at diagnosis. Comorbidities are frequently present in older patients, and the presence of multiple comorbidities (≥2 comorbidities) was an independent predictor of clinical outcome, independent of patients' age or disease stage. Cumulative Illness Rating Scale (CIRS), Charlson Comorbidity Index, and the NCI Comorbidity Index are some of the scoring systems that can be used to assess comorbidities in patients with CLL. CIRS in combination with creatinine clearance (CrCl) was used by the GCLLSG to assess the overall fitness of patients enrolled in clinical trials. In the CLL14 study, CIRS score >6 or an estimated CrCl <70 mL/min was used as the eligibility criteria for patients with significant comorbidities.

Quantitative immunoglobulin (Ig) levels may be informative in patients with recurrent infections. In instances when the quantitative Ig levels are abnormal, serum protein electrophoresis (SPEP) or serum immunofixation electrophoresis (SIFE) would be helpful to initiate additional evaluations (eg, *MYD88* mutation analysis or bone marrow biopsy) to rule out the diagnosis of Waldenstrom macroglobulinemia. Additionally, the presence of serum Ig monoclonal proteins or paraproteins has been associated with adverse prognostic factors in patients with CLL/SLL and detection of serum monoclonal proteins by SPEP and SIFE may also provide useful prognostic information.[122-124] Reticulocyte count, and a direct Coombs test should be performed to evaluate for the possibility of hemolysis and pure red cell aplasia (PRCA) in patients with anemia.

Bone marrow involvement (diffuse vs. nodular) is no longer a prognostic factor with the availability of more reliable prognostic markers that can be analyzed using peripheral blood (eg, IGHV mutation status and cytogenetic abnormalities detected by FISH). Thus, bone marrow biopsy ± aspirate is no longer considered essential for the diagnostic or prognostic evaluation of patients with suspected CLL, but it may be informative to confirm the presence of immune-mediated or disease-related cytopenias prior to initiation of treatment.

CT scans are not generally recommended for routine monitoring of treatment response or disease progression in asymptomatic patients. CT scans may be useful for the evaluation of symptoms of bulky disease, or for the assessment of risk for tumor lysis syndrome (TLS) prior to the initiation of venetoclax and for treatment response assessment in patients with SLL. Staging CT scan prior to initiating treatment is recommended to evaluate the extent of lymphadenopathy if using a BCL2i-based regimen or if there is concern for bulky adenopathy not accessible by physical exam.

PET scan is generally not useful in CLL but can assist in directing nodal biopsy if histologic transformation is suspected.[125,126]

First-Line Therapy

Localized SLL (Lugano stage I)

Locoregional radiation therapy (RT) is an appropriate induction therapy for patients with symptomatic localized disease. In rare patients, RT may be contraindicated or may be a suboptimal therapy due to the presence of comorbidities or the potential for long-term toxicity. Patients with localized SLL that progressed after initial RT should be treated as described below for patients with SLL (Lugano stage II–IV).

SLL (Lugano stage II–IV) or CLL (Rai stages 0–IV)

Early-stage disease in some patients may have an indolent course and in others may progress rapidly to advanced disease requiring immediate treatment. In a randomized prospective phase III study of patients with early-stage high-risk CLL, although FCR resulted in high overall response rate (ORR) (93%) and significantly prolonged EFS (median, not reached vs. 19 months; $P < .001$) compared to watch and wait, there was no significant OS benefit (5-year OS rate was 83% with FCR compared to 80% for watch and wait).[127] The results of the CLL12 trial did not demonstrate survival benefit for early treatment with ibrutinib in patients with early-stage, high-risk CLL (high risk defined according to the GCLLSG index).[128]

These results confirm that a “watch and wait” approach remains the appropriate management strategy for all patients, in the absence of disease symptoms. Treatment will be beneficial if patients become symptomatic or show evidence of progressive disease.[81] Selected patients with mild, stable cytopenia may continue to be observed and other causes of anemia or thrombocytopenia should be excluded.

Indications for initiating treatment include severe fatigue, weight loss, night sweats, and fever without infection; threatened end-organ function; progressive bulky disease (enlarged spleen or lymph nodes); progressive anemia or thrombocytopenia; or steroid-refractory autoimmune cytopenia.[81]

ALC alone is not an indication for treatment in the absence of leukostasis, which is rarely seen in patients with CLL.

In patients with indications for initiating treatment, age, functional status, comorbidities, and the presence or absence of del(17p) or *TP53* mutation should help to direct treatment options, as discussed below. Reevaluation for *TP53* mutation status and del(17p) by FISH, and IGHV mutation status (if not previously done) are recommended prior to initiating treatment. NGS-based clonality testing for MRD assessment should be considered if MRD-guided treatment is planned and MRD-flow is not being used for MRD assessment.

In addition to the aforementioned disease- and patient-specific factors, agents' toxicity profile and duration of treatment (continuous vs. fixed-duration) should also be considered for the selection of first-line therapy. cBTKis (acalabrutinib, ibrutinib, and zanubrutinib) are given continuously until disease progression, whereas venetoclax-containing regimens are time-limited with a treatment-free remission period. As discussed earlier, time-limited treatment with venetoclax-containing regimens also results in higher rates of uMRD, which is an independent predictor of improved survival.

The NCCN CLL Panel stratified all the regimens into three categories (based on the evidence, efficacy, toxicity, preexisting comorbidities, and in some cases access to certain agents): preferred, other recommended and useful in certain circumstances.

CLL/SLL without del(17p) or *TP53* Mutation

BCL2i-containing regimens (venetoclax/acalabrutinib ± obinutuzumab and VenO) and cBTKi-based regimens (acalabrutinib ± obinutuzumab, zanubrutinib) are included as preferred treatment options (with category 1 recommendation), based on the results of the phase III randomized studies (AMPLIFY, CLL14, ELEVATE-TN, and SEQUOIA).[19,21,22,33,129]

The efficacy data are discussed below and are summarized in Table 2.

Venetoclax + Obinutuzumab

The CLL14 study established VenO as an effective time-limited, chemotherapy-free, first-line treatment option with significantly improved PFS compared to chlorambucil + obinutuzumab in patients ≥65 years, or younger patients with comorbidities (CIRS score >6 or an estimated CrCl <70 mL/min).[129,130] The uMRD4 rate at the EOT was significantly higher with VenO (74% vs. 34%; $P < .0001$), and this combination was also associated with lower rate of conversion to MRD-positive status 1 year after treatment.[131]

VenO was granted broad FDA approval for the treatment of patients with CLL based on the results of the CLL14 study. The efficacy of VenO in patients <65 years of age without significant comorbidities was established in the phase III randomized GAIA–CLL13 GAIA/CLL13 trial.[25] The 4-year follow-up data confirmed that VenO with or without ibrutinib was associated with superior PFS compared to chemoimmunotherapy (FCR or BR).[26] The Panel members agreed that VenO is also an appropriate fixed-duration chemotherapy-free treatment option for younger patients without comorbidities, and the Panel consensus was to include fixed-duration VenO as a preferred treatment option (category 1) regardless of patient age or the presence of significant comorbidities.

Venetoclax/Acalabrutinib ± Obinutuzumab

In the AMPLIFY study, fixed-duration treatment with venetoclax/acalabrutinib ± obinutuzumab resulted in significantly higher ORR ($P < .0001$) and statistically significant improvement in PFS (primary endpoint) compared to investigator's choice of chemoimmunotherapy with either FCR or BR ($P = .0038$ for venetoclax/acalabrutinib vs. FCR/BR; $P < .0001$ for AVO vs. FCR/BR).[33] There was also a trend towards OS benefit with venetoclax/acalabrutinib compared to chemoimmunotherapy ($P < .0001$).

Fixed-duration treatment with venetoclax/acalabrutinib ± obinutuzumab is included as a preferred treatment option with a category 1 recommendation.

Venetoclax/Ibrutinib

The results of the CAPTIVATE study showed that fixed-duration treatment with venetoclax/ibrutinib results in improved PFS with high rates of durable response and uMRD4 across all patient subgroups.[30,31,99,100] In the fixed-duration treatment cohort (n = 159), with a median follow-up of 69 months, the 5.5 year PFS rate was 66% for the overall study population [70% for patients without del(17p)/*TP53* mutation; 55% and 79% for those with IGHV unmutated and IGHV mutated CLL respectively]. The 5.5 year OS rate was 97% for patients in the overall study population.[31]

In the GLOW trial, PFS was also significantly longer for fixed-duration treatment with venetoclax/ibrutinib compared to chlorambucil + obinutuzumab.[27,132] The 64-month follow-up data also showed OS benefit for venetoclax/ibrutinib compared to chlorambucil + obinutuzumab.[28] However, in this study, venetoclax/ibrutinib was associated with significant toxicity. Grade ≥3 treatment-emergent adverse events (TEAEs) occurred in 76% of patients and atrial fibrillation (any grade) was reported in 14% of patients) and treatment-related deaths were reported in 7% of patients.[132] Cardiac or sudden deaths during treatment occurred in patients with a CIRS score of ≥10 or an Eastern Cooperative Oncology Group performance status (ECOG PS) score of 2 and with a history of hypertension, cardiovascular disease, and/or diabetes. This increase in toxicity may be related to the advanced age of patients enrolled in the study.

The FLAIR study demonstrated that MRD-guided treatment with venetoclax/ibrutinib is superior to FCR in terms of PFS in patients without del(17p)/*TP53* mutation.[29] Long-term follow-up data (median follow-up, 62 months) showed that OS was also longer with venetoclax/ibrutinib compared to ibrutinib monotherapy in patients with IGHV unmutated CLL however the OS was similar in the two treatment groups for patients with IGHV mutated CLL.[29]

Venetoclax/ibrutinib is not approved for the treatment of CLL/SLL in the United States. Based on the safety profile reported in the GLOW trial (increased risk for cardiovascular events and infections particularly in older patients with comorbidities) and the absence of data from randomized studies comparing this regimen with approved targeted therapies for first-line therapy (other than ibrutinib), the Panel consensus was to include venetoclax/ibrutinib (fixed-duration treatment) as a category 1 recommendation under other recommended regimens. The Panel consensus also supported the inclusion of venetoclax/ibrutinib (MRD-guided treatment; other recommended regimen) as a category 1 recommendation based on the results of the FLAIR trial.

Venetoclax/Zanubrutinib

The results from the non-randomized cohort of the SEQUOIA study (Arm D) confirmed the efficacy of MRD-guided treatment with venetoclax/zanubrutinib in patients with and without *TP53* aberrations.[92] At a median follow-up of 31 months, venetoclax/zanubrutinib resulted in an ORR of 97% (96% in patients without *TP53* aberrations) and 24-month PFS rate was 92% (89% for patients without *TP53* aberrations).

Given the favorable toxicity profile of venetoclax/zanubrutinib, the Panel consensus supported its inclusion as a category 2A recommendation (useful in certain circumstances), and MRD-guided treatment with venetoclax/zanubrutinib may be an appropriate option for patients with intolerance to MRD-guided treatment with venetoclax/ibrutinib.

Acalabrutinib ± Obinutuzumab

In the phase III ELEVATE-TN trial, acalabrutinib ± obinutuzumab resulted in superior PFS compared to chlorambucil + obinutuzumab in patients with previously untreated CLL.[20] Acalabrutinib + obinutuzumab was associated with a PFS benefit in patients with IGHV-unmutated CLL as well as IGHV-mutated CLL compared to chlorambucil + obinutuzumab. At a median follow-up of 75 months, 72-month PFS rate was longer with acalabrutinib + obinutuzumab compared to acalabrutinib (78% vs. 62%). There was also a trend towards improved OS for acalabrutinib + obinutuzumab (72-month OS rate was 84% compared to 76% for acalabrutinib monotherapy), although the study was not powered to compare the PFS benefit between the two acalabrutinib arms.[133]

Acalabrutinib was granted broad FDA approval for the treatment of untreated and relapsed or refractory CLL based on the results of the ELEVATE-TN and ELEVATE-RR trials, respectively.[20,134] Acalabrutinib ± obinutuzumab is included as a preferred treatment option with a category 1 recommendation.

Zanubrutinib

Zanubrutinib is a highly selective/specific cBTKi that is FDA-approved for the treatment of CLL. In the phase III SEQUOIA study, zanubrutinib resulted in higher ORR and statistically significant improvement in PFS compared to BR in patients with untreated CLL without del(17p)/*TP53* mutation (HR, 0.29; $P = .0001$).[22] The biomarker subgroup analysis from the SEQUOIA study confirmed that PFS benefit with zanubrutinib was observed in all subgroups including patients with del(11q) ($P < .001$), IGHV-unmutated ($P < .0001$), and IGHV-mutated ($P < .01$).[58]

Based on the results of the SEQUOIA study, zanubrutinib is included as a preferred treatment option with a category 1 recommendation. The 6-year follow-up data from the SEQUOIA study also confirmed the superior efficacy and safety profile of zanubrutinib compared to BR.[135]

Ibrutinib

Ibrutinib monotherapy was approved for first-line therapy for all patients based on the results of the RESONATE-2 study that established the efficacy of ibrutinib monotherapy as first-line therapy only in patients ≥65 years without del(17p).[18,136] In the RESONATE-2 study, after a median follow-up of 5 years, ibrutinib resulted in a significantly higher ORR ($P < .0001$) and significantly longer PFS rate ($P < .0001$) compared to chlorambucil in patients ≥65 years without del(17p).[136] With 57% of patients switching to ibrutinib after disease progression on chlorambucil, the estimated 5-year OS rate was also higher with ibrutinib (without censoring for crossover from chlorambucil). Ibrutinib also improved PFS compared to chlorambucil in patients with high-risk CLL and the estimated 5-year PFS rates were 79% and 67% for patients with del(11q) and IGHV-unmutated CLL, respectively. Extended long-term data confirmed the sustained PFS benefit of ibrutinib as first-line therapy for patients with CLL, including those with high-risk genomic features of IGHV-unmutated (HR, 0.109) or del(11q) (HR, 0.033).[18]

With the inclusion of venetoclax/ibrutinib (fixed-duration or MRD-guided) as an option for first-line therapy (category 1; other recommended regimens) and randomized clinical trials demonstrating a more favorable toxicity profile for acalabrutinib and zanubrutinib (compared to ibrutinib),[134,137] the Panel acknowledged that continuous treatment with ibrutinib monotherapy would no longer be a preferred first-line therapy option for the vast majority of patients. A few Panel members also suggested ibrutinib monotherapy would be more appropriate for patients who can tolerate continuous treatment with ibrutinib and receive clinical benefit as well as for those patients with intolerance to venetoclax. The Panel consensus supported continued listing of ibrutinib monotherapy (continuous treatment) with a category 1 recommendation as an option under useful in certain circumstances.

Ibrutinib + Obinutuzumab or Rituximab

The results from randomized studies have confirmed that ibrutinib + rituximab is more effective than chemoimmunotherapy for previously untreated CLL without del(17p) or *TP53* mutation in patients ≥65 years or younger patients with comorbidities.[23,43,46,93,94,138] The Alliance North American Intergroup Study (A041202) showed primary benefit for ibrutinib and ibrutinib + rituximab in patients with IGHV unmutated CLL (61% of patients had IGHV unmutated CLL) rather than IGHV mutated CLL.[43] The presence of CK did not have an impact on PFS following treatment with ibrutinib. The estimated 2-year PFS rates were 91% and 87%, respectively, for ibrutinib and ibrutinib + rituximab among patients with CK.[93] The E1912 study and the FLAIR study showed that ibrutinib + rituximab was more effective than FCR for patients ≤70 years without del(17p)/*TP53* mutation, especially for those with IGHV unmutated CLL, indicating that ibrutinib may also be an appropriate option for younger patients with IGHV unmutated CLL.[23,46] However, the addition of rituximab to ibrutinib did not result in improved clinical outcomes compared to ibrutinib monotherapy in two randomized studies. In the Alliance North American Intergroup Study (A041202), the estimated 48-month PFS rates were 76% for both ibrutinib + rituximab and ibrutinib monotherapy.[43] In a single-center randomized study of 208 patients with high-risk CLL (27 patients with untreated CLL), at a median follow-up of 36 months, the estimated PFS rates were 86% and 87% for ibrutinib and ibrutinib + rituximab, respectively.[138]

Ibrutinib + obinutuzumab was approved by the FDA for first-line therapy based on the results of the iLLUMINATE study and there are no randomized clinical trials that compare ibrutinib versus ibrutinib + obinutuzumab.[94]

In all of the above-mentioned randomized clinical trials that evaluated ibrutinib + rituximab or obinutuzumab, ibrutinib was given continuously until disease progression or intolerance and obinutuzumab or rituximab was added to the combination arm only for the first six cycles. Therefore, the consensus was that the longer PFS was more the result of continuous treatment with ibrutinib, rather than due to the contribution of an anti-CD20 mAb (rituximab or obinutuzumab) during the first 6 months of treatment. Improved outcomes with addition of an anti-CD20 mAb may more likely be seen with time-limited treatment with this regimen.

Ibrutinib + rituximab or obinutuzumab are included as options (useful in certain circumstances) with a category 2B recommendation.

Chemoimmunotherapy

Multiple randomized trials showed the superior efficacy of cBTKi-based regimens and BCL2i-containing regimens over chemoimmunotherapy;[24,26-29,33] the Panel acknowledges that chemoimmunotherapy should no longer be recommended for first-line therapy. However, the majority of Panel members acknowledge that chemoimmunotherapy can be considered only when cBTKi and BCL2i are not available or treatment with these targeted therapies are not feasible.

FCR could be considered as an option for patients <65 years without significant comorbidities.[13,17,23,139] Bendamustine + anti-CD20 mAb (rituximab or obinutuzumab) may be a reasonable alternative for patients ≥65 years or younger patients with significant comorbidities.[139-141] High-dose methylprednisolone (HDMP) with rituximab or obinutuzumab could be considered as an option for patients with autoimmune cytopenias (category 2B recommendation for patients ≥65 years or younger patients with significant comorbidities and a category 3 recommendation for patients <65 years without significant comorbidities).[142,143] Given the favorable tolerability profile, obinutuzumab monotherapy or in combination with chlorambucil might be an acceptable treatment option for a small fraction of patients for whom more intensive chemoimmunotherapy regimens are not appropriate.[144-146] Obinutuzumab ± chlorambucil is included with a category 2A recommendation. Obinutuzumab + chlorambucil is recommended for patients ≥65 years or younger patients with significant comorbidities (CrCl <70 mL/min).

CLL/SLL with del(17p) or *TP53* Mutation

Chemoimmunotherapy is contraindicated for del(17p)/*TP53*-mutated CLL due to low response rates. There are limited data from prospective clinical studies on the efficacy of cBTKis or BCL2i as first-line therapy for patients with del(17p)/*TP53*-mutated CLL. Enrollment in an appropriate clinical trial is recommended for patients with untreated del(17p) CLL/SLL.

Patients with del(17p) CLL were not eligible for enrollment in the RESONATE-2 study, the E1912 study, the GLOW study, the GAIA/CLL13 study, the FLAIR study, or the AMPLIFY study.[18,23,26,27,29,33] Data for del(17p) or *TP53*-mutated CLL available from subgroup analyses of clinical studies that included this patient population are discussed below.

BCL2i-Containing Regimens

In the CLL14 study, the PFS benefit for VenO was seen across all patient subgroups including those with del(17p) or *TP53* mutation (del[17p] or mutated *TP53* were identified in only 8% and 12% of patients, respectively).[129]

In the CAPTIVATE study (n = 159; 27 patients had del(17p) and/or *TP53* mutation), the estimated 5.5 year PFS rate for venetoclax/ibrutinib (fixed-duration) was 36% for those with del(17p)/*TP53* mutation.[31] In the phase II study that evaluated venetoclax/ibrutinib as MRD-guided treatment (n = 80; 18 patients had del(17p) and/or *TP53* mutations), the estimated 3-year PFS rate was 86% for patients with del(17p) and/or *TP53* mutations.[32]

Patients with del(17p) or *TP53* mutation were not enrolled in the AMPLIFY study because the control arm included investigator's choice of chemoimmunotherapy.[33] In an earlier phase II study, the combination of AVO resulted in an ORR of 98% (71% CR) in patients with *TP53* aberrations (n = 45).[34] The CR rate with uMRD in the bone marrow was 58%. After a median follow-up of 55 months, the 4-year PFS and OS rates were 70% and 88%, respectively for patients with *TP53* aberrations. Therefore, the Panel agreed that AVO is an appropriate treatment option for patients with del(17p) or *TP53* mutation based on the results of this phase II study and acknowledged that there are no data to recommend venetoclax/acalabrutinib for patients with del(17p) CLL/SLL.

The results from the non-randomized cohort (Arm D) of the SEQUOIA study confirmed the efficacy of venetoclax/zanubrutinib in patients with *TP53* aberrant disease.[92] At a median follow-up of 31 months, venetoclax/zanubrutinib resulted in an ORR of 99% in patients with *TP53* aberrant disease and the 24-month PFS rate was 94% for patients with *TP53*-aberrant disease. Discontinuation of treatment was allowed in patients meeting the strict uMRD achievement criteria. Zanubrutinib was given until disease progression for patients not meeting the uMRD criteria.

The PFS benefit for VenO and venetoclax/ibrutinib for patients with del(17p)/*TP53* mutation was also established in the CLL17 trial.[35]

Given currently available data (as discussed above), VenO (fixed-duration), AVO (MRD-guided)[34] and venetoclax/zanubrutinib (MRD-guided)[92] are included as preferred treatment options with a category 2A recommendation. Venetoclax/ibrutinib (fixed-duration) is included as an option under other recommended regimens based on the results of the CAPTIVATE study.[31]

BTKi-Based Regimens

In the RESONATE-2 study, the OS benefit with ibrutinib was observed in patients with *TP53* mutation, del(11q), and/or IGHV-unmutated and the estimated 5-year PFS rate was 56% for the group of 12 patients with *TP53* mutation.[18] However, comparison between ibrutinib and chlorambucil could not be made since only three patients in the chlorambucil group had *TP53* mutation.

In a phase II trial that included 34 patients who are treatment-naïve with *TP53* aberrations (32 patients with del[17p]; 2 patients with *TP53* mutation without del[17p]; median age 62 years), ibrutinib resulted in an ORR of 97% (30% CR; 64% PR; 3% PR-L) and the estimated 6-year PFS and OS were 61% and 79%, respectively.[147]

Results from the pooled analysis of clinical trials (PCYC-1122e, E1912, RESONATE-2, and iLLUMINATE) also confirmed the long-term safety and efficacy of ibrutinib as first-line therapy in patients with *TP53* aberrations.[148] With a median follow-up of 50 months, the estimated 4-year PFS and OS rates were 79% and 88%, respectively. The ORR was 93% (CR in 39% of patients). As mentioned above, the RESONATE-2 and E1912 studies excluded patients with del(17p) CLL and *TP53* mutation was identified retrospectively. Additionally, there are also data suggesting that *TP53* mutation in the absence of del(17p) also confers increased risk. However, it may not be as notable as that associated with the concurrent presence of *TP53* mutation and del(17p).[41]

In the ELEVATE-TN study, the PFS benefit for acalabrutinib ± obinutuzumab was seen across all patient subgroups including those with del(17p) or *TP53* mutation but only 14% of patients had del(17p) CLL.[19] In patients with del(17p) and/or *TP53* mutation, the estimated 72-month PFS rate was 56% for both acalabrutinib + obinutuzumab and acalabrutinib monotherapy, indicating no benefit with the addition of obinutuzumab to acalabrutinib. The estimated 72-month OS rates were 68%, 72%, and 53% for acalabrutinib, acalabrutinib + obinutuzumab, and chemoimmunotherapy, respectively.[133]

In the phase III SEQUOIA study, patients with del(17p) were not part of the randomized cohort but were enrolled only to single-agent zanubrutinib or, subsequently, to the combination of zanubrutinib and venetoclax.[21] In the prospectively enrolled non-randomized cohort (111 patients with del[17p]/*TP53*-mutated CLL), single-agent zanubrutinib resulted in a higher ORR and statistically significant improvement in PFS.[149] The best ORR and 18-month PFS rates were 98% and 89%, respectively, for patients with high del(17p) (≥20%), and 92% and 88%, respectively, for patients with low del(17p) (>7% to <20%). The 6-year follow-up results also confirmed the efficacy of zanubrutinib in patients with del(17p)/*TP53* mutation.[135]

Acalabrutinib ± obinutuzumab and zanubrutinib are included as preferred treatment options for first-line therapy.

With the inclusion of BCL2i containing combination regimens as options for first-line therapy and randomized clinical trials demonstrating a more favorable toxicity profile for acalabrutinib and zanubrutinib (compared to ibrutinib), the Panel acknowledged that continuous treatment with ibrutinib monotherapy would no longer be a recommended first-line therapy option for the vast majority of patients. The Panel consensus supported continued listing of ibrutinib monotherapy (continuous treatment) with a category 1 recommendation as an option under useful in certain circumstances.

Other Systemic Therapy Regimens

The Panel emphasizes that the efficacy of BTKi-based regimens in del(17p) CLL exceeds that of the other regimens and BTKi-based regimens should be considered as the best choice in the absence of a contraindication to cBTKi.

HDMP + rituximab or obinutuzumab[142,143] or obinutuzumab[144] can be considered only when cBTKi and BCL2i are not available or treatment with these targeted therapies are not feasible.

Second-Line and Subsequent Therapy

The selection of second-line therapy for refractory or progressive disease should be based on the type of first-line therapy, duration of remission, and acquired resistance to treatment.

In patients with disease responding to cBTKi, treatment should be continued until progression and/or intolerance. For patients with disease responding to time-limited treatment with venetoclax-containing regimens, observation is recommended until disease relapse with indications for retreatment.

Observation until relapse with indications for treatment is also appropriate for patients with disease responding to CIT. Time-limited treatment with BCL2i-containing regimens or cBTKi-based regimens are options for relapse after a period of remission or progression on treatment.

Recommendations for the selection of second-line therapy based on outcomes after first-line therapy are outlined on CSLL-4 and CSLL-5. The efficacy data from randomized clinical trials that evaluated small-molecule inhibitors for relapsed or refractory CLL/SLL are discussed below and are summarized in Table 3.

BCL2i-Containing Regimens

VenR is approved for the treatment of relapsed or refractory CLL/SLL based on the results of the phase III randomized MURANO trial.[96,150] VenR was superior to BR with longer PFS across all subgroups of patients, including those with del(17p) or *TP53* mutation [HR, 0.21 for del(17p); HR, 0.25 for *TP53* mutation], and uMRD at the EOT was also higher for VenR (62% vs. 13% for BR).[96]

Venetoclax monotherapy is effective for relapsed or refractory CLL after prior treatment with ibrutinib or idelalisib,[151-155] although the results of a pooled analysis from four clinical trials showed that CLL refractory of BTKi or PI3Ki was significantly associated with lower CR rate and shorter duration of response (DOR).[156] Results from other retrospective analyses suggest that the use of venetoclax is associated with higher ORR and improved PFS following disease progression on ibrutinib (compared to disease progression on idelalisib) and also in patients who had received only one BTKi or PI3Ki (compared to those who had received >1 BTKi or PI3Ki).[157,158]

Venetoclax monotherapy resulted in an ORR of 77% (21% CR and 49% PR) in patients with relapsed or refractory del(17p) CLL. The ORR was 63% in patients who had received prior BTKi (ibrutinib) or PI3Ki (idelalisib).[159] The median DOR was 39 months. After a median follow-up of 70 months, the median PFS and OS for the overall study population were 28 months and 63 months, respectively. The median PFS was 15 months for patients who had received prior BTKi or PI3Ki. The 5-year PFS and OS rates were 24% and 52%, respectively, for the overall study population.

VenR is included as a treatment option for second-line and subsequent therapy with a category 1 recommendation, irrespective of the del(17p)/*TP53* mutation status. Venetoclax monotherapy is an option with a category 2A recommendation (a preferred regimen for CLL/SLL with del(17p)/TP53 mutation). The Panel consensus supported the inclusion of VenO as a preferred treatment option for second-line therapy based on the superior efficacy of obinutuzumab in first-line therapy compared to rituximab (CLL11 and GAIA/CLL13 studies) and the real world evidence demonstrating the efficacy of VenO in previously treated CLL/SLL.[25,146,160] The CLL11 and GAIA/CLL13 studies showed that obinutuzumab when used in combination with chlorambucil or venetoclax, respectively, as first-line therapy, resulted in better clinical outcomes (clinically meaningful PFS and TTNT benefit in CLL11 study; significantly higher PFS at 3 years in the GAIA/CLL13) compared to rituximab.[25,146] In a real-world study of 40 patients with relapsed or refractory CLL/SLL (including CLL with *TP53* mutation and/or del[17p] and CLL previously treated with BTKi), VenO was effective and well tolerated resulting in an ORR of 90% (63% CR) and the 2-year PFS rate was 81%.[160]

The results of the phase II CLARITY study (n = 53) showed that treatment with venetoclax/ibrutinib was effective for relapsed or refractory CLL resulting in an ORR of 89% (51% CR), and this combination also resulted in higher rates of uMRD4.[101] This study included patients with relapsed or refractory CLL/SLL after prior chemoimmunotherapy or idelalisib and patients with relapsed or refractory CLL after prior BTKi or venetoclax were excluded. The Panel consensus was to include venetoclax/ibrutinib as an option (other recommended regimens; irrespective of del(17p)/*TP53* mutation] with a category 2B recommendation based on the results of the CLARITY study.[101]

Treatment with venetoclax-containing regimens (venetoclax ± anti-CD20 mAb [VenO is preferred] or ibrutinib) is also included as an option for disease relapse after a period of remission following a time-limited treatment with venetoclax-containing regimen, irrespective of del(17p)/*TP53* mutation status.[161-166] The results of retrospective analyses suggest that treatment with the venetoclax-containing regimen (Ven2) is effective for patients with CLL/SLL previously treated with a venetoclax-containing regimen (Ven1).[161,162] An international retrospective study showed that this approach is effective in any line of therapy.[161] Among 46 patients with CLL retreated with the venetoclax-containing regimen (response data were available for 39 patients; a median of 16 months between the completion of Ven1 and initiation of Ven2), Ven2 resulted in an ORR of 80% (33% CR). At a median follow-up of 10 months, the median PFS was 25 months. In the subgroup analysis of the MURANO trial (n = 25), retreatment with VenR resulted in an ORR of 72% and the median PFS was 23 months.[162] Retreatment with VenR was also effective for disease progression while off treatment after achieving response to time-limited treatment with VenR.[163] Interim analysis of the ReVenG study demonstrated the efficacy of treatment with VenO in patients with disease relapse >2 years (n = 25) after completing first-line therapy with fixed-duration venetoclax-based regimens.[164] The feasibility of retreatment with venetoclax-based regimens in CLL without del(17p)/TP53 mutation was also confirmed in the GAIA/CLL13 study, resulting treatment-free survival rates of more than 80%.[165] Longer follow-up are needed to confirm these preliminary results.

Covalent BTK Inhibitors

Acalabrutinib, ibrutinib, and zanubrutinib are also approved for treatment of relapsed or refractory CLL/SLL based on the results of phase III randomized studies (ASCEND, ELEVATE-RR, RESONATE, and ALPINE trials).[134,167-169] The PFS benefit compared to chemoimmunotherapy was seen across all patient subgroups including those with del(17p) or *TP53* mutation.

In the ASCEND study, at a median follow-up of 47 months, the median PFS was 46 months and the 42-month PFS rate was 62% for patients with del(17p)/*TP53* mutation assigned to acalabrutinib.[167] The phase III ELEVATE-RR trial demonstrated that acalabrutinib is noninferior to ibrutinib in terms of PFS and was also associated with a more favorable safety profile in patients with relapsed or refractory del(17p) or del(11q) CLL.[134]

The final analysis of the RESONATE study showed that the presence of del(17p)/*TP53* mutation or CK was not associated with inferior PFS outcomes to ibrutinib.[168] In an exploratory analysis that combined data from patients with del(17p) and *TP53* mutation, the median PFS was 41 months for patients with del(17p) and/or *TP53* mutation versus 57 months for those without del(17p) or *TP53* mutation. Similarly, the median PFS was 41 months for patients with CK compared to 45 months for those without CK. The phase II RESONATE-17 study established the efficacy and safety of ibrutinib in patients with relapsed or refractory del(17p) CLL (n = 145), demonstrating an ORR of 83% (as assessed by the independent review committee).[170]

The randomized phase III study (ALPINE) showed that zanubrutinib resulted in a significantly higher ORR and significantly longer PFS in patients with relapsed or refractory CLL/SLL.[169] Zanubrutinib also resulted in a higher ORR and longer PFS across the major subgroups of patients, including those with a del(17p) and/or *TP53* mutation. Among patients with del(17p) and/or *TP53* mutation, the 36-month PFS rate was 59% for zanubrutinib and 39% for ibrutinib. The corresponding 36-month OS rates were 83% and 80%, respectively. The 6-year follow-up data confirmed the durable PFS benefit for zanubrutinib.[171] After a median follow-up of 74 months, the median PFS was 53 months and the 60-month PFS rate was 47% for all patients. Among patients with del17p, the median PFS and 60-month PFS rate were 50 months and 38%, respectively.

Acalabrutinib, ibrutinib, or zanubrutinib are recommended options for second-line and subsequent therapy with a category 1 recommendation, irrespective of the del(17p)/*TP53* mutation status. Acalabrutinib and zanubrutinib are listed as options for preferred regimens. Ibrutinib is included as an option under other recommended regimens based on the toxicity profile.

Non-Covalent BTK Inhibitors

Pirtobrutinib was approved for the treatment of patients with relapsed or refractory CLL/SLL who received at least two prior lines of therapy, including a BTKi and BCL2i based on the results of the BRUIN phase I–II study.[172,173]

In this study, pirtobrutinib resulted in an ORR of 73% (82% including PR-L) with a median PFS of 20 months in patients previously treated with a BTKi (n = 247). At a median follow-up of 23 months, the estimated 18-month OS rate was 81% for patients previously treated with a BTKi. In the subgroup of patients previously treated with the BTKi and venetoclax-containing regimen (n = 100), the ORR was 70% (79% including PR-L) and the median PFS was 17 months.[172] The estimated median PFS was 17 months and 19 months, respectively, for patients with del(17p) or *TP53* mutation and those with IGHV-unmutated CLL. The ORR (including PR-L) was higher irrespective of the status of prior therapy with BCL2is (83% for BCL2i-naïve and 80% for BCL2i exposed); however, PFS was longer in the BCL2i-naïve group than in the BCL2i-exposed group (23 months and 16 months, respectively).[173] The 24-month OS rates were 83% and 61%, respectively.

The phase III randomized BRUIN CLL-321 study evaluated pirtobrutinib (n = 119) and investigator's choice of treatment (BR, n = 37 or idelalisib + rituximab [IdR], n = 82) in patients with CLL/SLL previously treated with cBTKi with or without prior therapy with venetoclax (all patients had received prior cBTKi and 50% of patients had received prior therapy with venetoclax-containing regimens; cBTKi was discontinued either due to disease progression or intolerance).[174] At a median follow-up of 17 months, pirtobrutinib resulted in significant improvement in PFS compared to BR/IdR (independent review committee [IRC] assessed PFS was 14 months for pirtobrutinib vs. 9 months for BR/IdR) and the PFS benefit was observed across all subgroups of patients including those with CLL previously treated with venetoclax-containing regimens (HR, 0.54), IGHV-unmutated (HR, 0.61), *TP53* mutation and/or del(17p) (HR, 0.59), and CK (HR, 0.37). The 18-month OS rate was 73% and 71% for pirtobrutinib and BR/IdR, respectively. The OS rate, however, was confounded by a high effective crossover rate of 76%.

Based on the results of BRUIN CLL-321 study, the Panel consensus was to include pirtobrutinib as a preferred treatment option with a category 1 recommendation (irrespective of del[17p]/*TP53* mutation] for patients with intolerance to prior cBTKi or for those with disease that is resistant to cBTKi. Pirtobrutinib is also included as preferred option with a category 1 recommendation (if not previously used; irrespective of del[17p]/*TP53* mutation) for relapsed or refractory disease after prior therapy with BTKi-based regimens and venetoclax-containing regimens.[172,173]

Chimeric Antigen Receptor T-Cell Therapy

The TRANSCEND CLL 004 study evaluated the safety and efficacy of lisocabtagene maraleucel in patients with relapsed or refractory CLL/SLL after ≥2 prior lines of therapy (N = 137; 117 patients received infusion with lisocabtagene maraleucel and all had received prior BTKi).[175,176] A subset of 70 patients also received venetoclax-containing regimens after disease progression on BTKi. The primary efficacy analysis (49 patients including those who had received prior venetoclax-containing regimens after disease progression on BTKi) reported an ORR of 43% (18% CR) as assessed by an independent review committee.[175] Among the 30 patients with del(17p) and/or *TP53* mutation, the ORR was 47% (23% CR).

The DOR was longer in patients achieving a CR. At a median follow-up of 20 months, the DOR was 35 months for all patients with responding disease (not reached for patients achieving a CR and 24 months for patients achieving a PR).[175] The median PFS and OS were 12 months and 30 months, respectively. The median PFS was not reached in patients achieving a CR compared to 26 months for those achieving a PR and 4 months for those with non-responding disease. The uMRD4 rate was 63% in blood and 59% in bone marrow.[175,176] The median PFS was longer in patients who had uMRD (26 vs. 3 months for those with detectable MRD).[175] The 24-month follow-up data confirmed the high uMRD rates (64% in blood and 60% in the bone marrow) and longer DOR among patients achieving CR.[176] The median DOR was 30 months for all patients with responding disease and it was not reached for patients achieving a CR.

Results from the combination cohort of the TRANSCEND CLL004 study (n = 56) showed that ibrutinib combined with lisocabtagene maraleucel also resulted in high ORR (86%; 45% CR) and uMRD rate (86% in blood and 84% in bone marrow).[177] The median DOR for CR was not reached. The median PFS and OS were 31 months and not reached, respectively. Ibrutinib was either initiated or continued through leukapheresis and for up to 90 days after the infusion of lisocabtagene maraleucel.

Lisocabtagene maraleucel is a one-time infusion that does not require continuous treatment and is approved for relapsed or refractory CLL/SLL after at least two prior lines of therapy, including a BTKi and a BCL2i (venetoclax).

Lisocabtagene maraleucel ± ibrutinib is included as a preferred option for relapsed or refractory disease after prior therapy with BTKi-based regimens and venetoclax-containing regimens [irrespective of del(17p)/*TP53* mutation].

PI3K Inhibitor-Based Regimens

Idelalisib and duvelisib also demonstrated efficacy (in terms of median PFS) in randomized phase III studies for patients with relapsed or refractory CLL/SLL.[178-184]

In a phase III randomized trial (220 patients; CIRS >6, decreased renal function, or cumulative marrow toxicity from prior therapy; randomized to receive IdR or rituximab + placebo), IdR demonstrated efficacy in patients with relapsed or refractory CLL/SLL with and without del(17p).[179] IdR significantly prolonged survival in patients with del(17p) or *TP53* mutation compared with those treated with rituximab + placebo and there was no difference in survival benefit compared to those without del(17p).[179] The median OS was 29 months following treatment with IdR compared to 15 months following treatment with rituximab + placebo. IdR is FDA approved for relapsed or refractory CLL based on the results of this study and is available for clinical use with a black box warning regarding the risks of fatal and serious toxicities including hepatotoxicity, diarrhea, colitis, pneumonitis, and intestinal perforation. In the BRUIN CLL-321 study that evaluated pirtobrutinib in patients with previously treated CLL/SLL, IdR (n = 82) was the control arm (based on investigator's choice) along with BR (n = 37).[174] At a median follow-up of 12 months, the IRC-assessed median PFS was 9 months in the IdR/BR control arm and the estimated 18-month OS rate was 71%.

Idelalisib monotherapy also demonstrated activity in relapsed or refractory SLL.[180] The indication for idelalisib monotherapy in relapsed or refractory SLL was withdrawn by the manufacturer as they are unable to complete the required confirmatory studies following the FDA accelerated approval. While the Panel acknowledged the change in the regulatory status of idelalisib, the Panel consensus was to continue listing idelalisib monotherapy as an option for relapsed or refractory SLL, given demonstrated efficacy.[180]

In the phase DUO trial (N = 319; 160 patients were randomized to duvelisib and 159 patients were randomized to ofatumumab), duvelisib significantly extended median PFS (17 vs. 9 months) compared to ofatumumab and the PFS benefit was seen across all the subgroup of patients including those with del(17p).[181] In the final analysis, after a median follow-up of 63 months, the median OS was 52 months for patients assigned to duvelisib and 63 months for patients assigned to ofatumumab (including 90 patients who crossed over to duvelisib).[182] In the DUO crossover extension study (that evaluated the efficacy and safety of duvelisib monotherapy in patients with disease progression while receiving ofatumumab in the DUO trial), the ORR was 77% (61% PR) for the subset of 26 patients with del(17p) and/or *TP53* mutations.[183]

Duvelisib and idelalisib ± rituximab are included as options for relapsed or refractory disease after prior therapy with BTKi-based regimens and venetoclax-containing regimens, irrespective of del(17p)/*TP53* mutation status).

Other Systemic Therapy Regimens

FCR and BR have demonstrated activity in patients with relapsed or refractory disease and are included as options for relapsed or refractory CLL/SLL without del(17p) or *TP53* mutation after prior therapy with BTKi-based regimens and venetoclax-containing regimens.[185-187]

HDMP + rituximab was effective in patients with heavily pretreated CLL (including fludarabine-refractory disease), although it was associated with infectious complications (including opportunistic fungal infections) in about 30% of patients, which may necessitate adequate anti-infective prophylaxis and close monitoring for early signs of infections.[188,189] Lenalidomide ± rituximab also demonstrated activity in patients with relapsed or refractory disease.[190-192] However, the ORR was lower for lenalidomide + rituximab in the subgroup of patients with fludarabine-refractory CLL compared with those with fludarabine-sensitive CLL. Growth factors and/or dose adjustment may be needed to address cytopenias, without necessitating holding treatment. Obinutuzumab (as monotherapy) also demonstrated activity in patients with relapsed or refractory CLL/SLL.[145]

HDMP + anti-CD20 mAb (category 2B for CLL/SLL without del(17p)/*TP53* mutation), lenalidomide ± rituximab and obinutuzumab are included as options for relapsed or refractory CLL/SLL (with or without del(17p)/TP53 mutation) after prior therapy with BTKi-based regimens and venetoclax-containing regimens.

Alemtuzumab + rituximab results in a higher ORR than that observed with alemtuzumab monotherapy.[193,194] However, it should be noted that bulky lymphadenopathy does not typically respond well to alemtuzumab monotherapy in patients with refractory CLL.[195] Alemtuzumab ± rituximab is included as an option for relapsed or refractory CLL/SLL with del(17p)/*TP53* mutation after prior therapy with BTKi-based regimens and venetoclax-containing regimens.

Chemoimmunotherapy or immunotherapy are not recommended for patients who received these as first-line therapy. The recommendations for other systemic therapy regimens as listed above for CLL/SLL with del(17p)/*TP53* mutation are based on results from retrospective analyses or subgroup analyses from prospective clinical trials that had included patients with del(17p) or *TP53* mutation. However, it should be noted that these studies were not sufficiently powered to evaluate the efficacy and safety of regimens in patients with del(17p) or *TP53* mutation.

Allogeneic Hematopoietic Cell Transplant

Long-term results from several prospective studies showed that allogeneic hematopoietic cell transplant (HCT) can provide long-term disease control and also overcome the poor prognosis associated with del(17p) and *TP53* mutations.[67,196-202] Available data suggest that CK (≥5 abnormalities) is associated with inferior OS and EFS following allogeneic HCT with reduced-intensity conditioning in patients with high-risk interphase cytogenetics.[203,204]

It is understood that studies involving allogeneic HCT are subject to significant selection biases. Nonetheless, at the present time, given the favorable outcome of patients with del(17p) or *TP53* mutation treated with cBTKi as first-line therapy and the availability of venetoclax-based regimens as an effective treatment option for relapsed or refractory CLL, allogeneic HCT can be considered for relapsed or refractory disease after prior therapy with BTKi-based regimens and venetoclax-containing regimens in patients without significant comorbidities.[205] HCT-specific comorbidity index (HCT-CI) could be used for the assessment of comorbidities prior to HCT and to predict the risks of non-relapse mortality and the probabilities of survival after HCT.[206,207]

Special Considerations for Treatment

Precautions and considerations when initiating treatment or transitioning to next therapy for disease progression while on treatment are outlined in the *Supportive Care* section of the algorithm (CSLL-C 2 of 6).

Management of Resistance to Small-Molecule Inhibitors

BTK Inhibitors

Acquired resistance to BTKis is predominantly mediated by *BTK* and *PLCG2* mutations.[57,208-212] *BTK* and/or *PLCG2* mutations were detected at an estimated median of 9 months before progression in patients with CLL treated with ibrutinib, and these mutations were also detected in patients with progressive CLL during ibrutinib therapy up to 15 months before the manifestation of clinical progression.[57,208] In the randomized ELEVATE-RR trial, acquired *BTK* mutations were detected at the time of disease progression in 66% of patients with CLL treated with acalabrutinib and in 37% of patients with CLL treated with ibrutinib.[211] *BTK* C481S mutations were most common with both ibrutinib and acalabrutinib whereas the detection of other *BTK* mutations was variable (T474I and E41V mutations were detected in the acalabrutinib group; L528W and A428D were detected in the ibrutinib group). The median VAF was not significantly different (16% and 15.6% for acalabrutinib and ibrutinib, respectively). *PLCG2* mutations were detected in 6% of patients with CLL acalabrutinib and 20% of patients with CLL treated with ibrutinib, with a median VAF of 2% and 10% for acalabrutinib and ibrutinib, respectively. The reported VAF are variable, with low VAF often associated with disease progression on ibrutinib, leading to speculation that these mutations do not fully explain clinical resistance.[57,208,211] In the ALPINE trial, *BTK* C481S, L528W, A428D mutations were detected in patients with progressive disease on zanubrutinib whereas *PLCG2* mutations were not detected in patients with CLL treated with zanubrutinib.[212] *BTK* A428D, T474I, and L528W mutations also confer resistance to ncBTKi.[209]

Alternative cBTKi is not a reasonable treatment option for patients with a mutation in either *BTK* C481 or *PLCG2*. Pirtobrutinib (ncBTKi) is an effective treatment option for the management of resistance to cBTKi, including in patients with *BTK* C481 mutations.[172,173] In the BRUIN study, mutations in *BTK, TP53*, and *PLCG2* were detected at baseline in 53%, 48%, and 14% of patients, respectively. Among the patients with *BTK* C481 mutation, decrease in *BTK* C481 VAF or complete clearance of *BTK* C481 clone was observed in 86% and 55% of patients, respectively.[213] The use of pirtobrutinib for relapsed or refractory CLL following BCL2i-containing regimens without prior exposure to cBTKi has not been evaluated. Venetoclax is an also an appropriate treatment option for relapsed or refractory CLL after prior treatment with ibrutinib or idelalisib.[151-154,156-158]

Switching to alternate therapy (ncBTKi or BCL2i-containing regimen) is recommended at the time of disease progression while on treatment with a cBTKi. Treatment-free interval should be as short as possible to minimize tumor flare reactions. In patients with disease progression on cBTKi, it is recommended to continue cBTKi through venetoclax dose ramp-up to reduce the risk of tumor flare when transitioning from cBTKi to BCL2i.

Testing for *BTK* and *PLCG2* mutations may be useful to confirm resistance to BTKis in patients with disease progression or no response while on BTKi therapy, including if poor treatment adherence is considered as a possible cause. *BTK* and *PLCG2* mutation status alone is not an indication to change treatment in absence of disease progression. Testing for *BTK, PLCG2*, or *BCL2* mutations as screening for resistance to BTKi or venetoclax is not currently recommended.

BTKis are available in different formulations, dosage forms, and strengths that are subject to different administration instructions. The choice of cBTKi should be made by the treating clinician based on patient-specific factors (eg, age, functional status, comorbid conditions) and the agent's toxicity profile. These products are not interchangeable.

BCL2 Inhibitors

Acquisition of *BCL2* mutations (G101V and D103Y) was implicated in resistance to venetoclax.[214,215] *BCL2* G101V mutation (low VAF) was identified in patients with progressive CLL during venetoclax therapy up to 25 months before clinical progression.[214] *BCL2* mutations are uncommonly associated with clinical resistance to venetoclax; therefore, other resistance mechanisms must be important. Upregulation of MCL1 due to amplification of chromosome 1 as well as *NOTCH1* and *SF3B1* mutations detected during venetoclax therapy have also been associated with resistance to venetoclax.[216,217]

Mutation analysis in patients with progressive disease after fixed-duration treatment with venetoclax/ibrutinib in the CAPTIVATE study revealed the absence of *BTK, PLCG2,* and *BCL2* mutations that confer resistance to BTKi and venetoclax, suggesting that combination therapy may also mitigate the development of resistance to targeted therapies as single agents.[218]

Limited available data suggest that subsequent treatment with cBTKi or retreatment with venetoclax-containing regimens is effective in patients with relapsed CLL after a period of remission following time-limited treatment with venetoclax-containing regimen, whereas PI3Ki following time-limited treatment with venetoclax-containing regimens does not appear to result in durable remissions.[161,219-221] Pirtobrutinib also has demonstrated efficacy for the treatment of CLL/SLL previously treated with cBTKi and venetoclax.[174]

Switching to cBTKi is recommended for disease progression while on treatment with BCL2i + anti-CD20 mAb. Switching to a ncBTKi is recommended for disease progression while on treatment with BCL2i + cBTKi.

Management of Treatment-Emergent Adverse Events

BTK Inhibitors

Diarrhea, fatigue, arthralgia, infections, cytopenias, bleeding, and cardiovascular toxicities (including atrial fibrillation, ventricular arrhythmias, and hypertension) are the most common TEAEs associated with BTKis. BTKi therapy also causes mobilization of lymphocytes into blood early during treatment initiation, resulting in a significant transient lymphocytosis (increase in ALC) in most patients, which does not signify disease progression.[82,83]

TEAEs associated with BTKi are discussed below and are summarized in Table 4.

Acalabrutinib and zanubrutinib both have a more favorable toxicity profile than ibrutinib due to the more selective/specific inhibition of BTK. In the ELEVATE-RR head-to-head trial of acalabrutinib versus ibrutinib, treatment discontinuation due to TEAEs was lower with acalabrutinib (15% vs. 21% for ibrutinib).[134,222] The incidences of adverse events of special interest (AESI) were also lower with acalabrutinib compared to ibrutinib: atrial fibrillation (9% vs. 16%), hypertension (9% vs. 23%), and bleeding (38% vs. 51%).[134,222] The incidences of atrial fibrillation and hypertension were also lower for acalabrutinib when used in combination with venetoclax ± obinutuzumab in the AMPLIFY trial.[33] Atrial fibrillation (any grade) was reported in <1% of patients in the acalabrutinib + venetoclax arm and in 2% of patients in the acalabrutinib + venetoclax + obinutuzumab arm. Hypertension (grade ≥3) was reported in 3% and 2% of patients, respectively.

Acalabrutinib was associated with a higher rate of headache (35% vs. 20% for ibrutinib), with only 2% of patients experiencing grade ≥3 headache.[134] Headache is commonly observed with acalabrutinib early in the treatment course and can generally be managed with analgesics (eg, acetaminophen) and caffeine supplements and typically subsides with time on treatment.

Zanubrutinib was also associated with lower rates of cardiac events (26% vs. 36%) and lower incidence of atrial fibrillation (7% vs. 17%) compared to ibrutinib in the ALPINE trial, with no cardiac deaths reported with zanubrutinib compared to 6 cardiac deaths reported with ibrutinib.[169] In contrast, neutropenia of any grade was more frequent with zanubrutinib (32% vs. 30% for ibrutinib); however, this did not translate into a higher rate of infection (71% with zanubrutinib vs. 73% for ibrutinib). The incidences of grade ≥3 infections were 27% and 28%, respectively.

Pirtobrutinib has a favorable toxicity profile (low incidences of atrial fibrillation, major hemorrhage, and hypertension) due to more selective inhibition of BTK and the relative absence of off-target inhibition.[172-174] Longer-term follow-up data are needed to assess the incidence of these TEAEs.

The benefit and risk of BTKis should be evaluated in patients requiring anti-platelet or anticoagulant therapies. Patients requiring the use of anticoagulants including warfarin were excluded from clinical trials evaluating acalabrutinib and ibrutinib, while the use of anticoagulants including warfarin was not restricted in clinical trials evaluating zanubrutinib (except in the ALPINE trial). Zanubrutinib can be coadministered with anticoagulants including warfarin. Concomitant administration of ibrutinib or acalabrutinib with warfarin should be avoided.

A baseline assessment of cardiac function should be done prior to initiation of cBTKi.[223,224] Hypertension should be managed with antihypertensives as appropriate. Monitoring for signs of bleeding, atrial fibrillation, and hypertension along with appropriate management is recommended for patients receiving BTKis.

Acalabrutinib (tablets) and zanubrutinib can be coadministered with gastric acid-reducing agents (eg, antacids, proton pump inhibitors [PPIs], H2-receptor antagonists). Acalabrutinib tablets are the primary formulation and distribution of acalabrutinib capsules was discontinued.

Acalabrutinib and zanubrutinib were shown to be effective for the management of disease in patients with ibrutinib intolerance.[137,225,226] Pirtobrutinib is also an acceptable option for the management of intolerance to cBTKi.[172,173] Data from real-world studies suggest that dose modification of ibrutinib may resolve intolerance without compromising efficacy.[227-229] In patients with no intolerance, ibrutinib can be continued until disease progression while following recommended dose modification guidance as needed. However, the efficacy of dose modification of ibrutinib was not confirmed in prospective studies.

Switching to alternate cBTKi or ncBTKi or BCL2i-based regimen can be considered in the setting of intolerance to therapy, especially in patients with atrial fibrillation or hypertension that is not medically controllable. Alternate cBTKi should be considered for intolerance only in the absence of disease progression. Patients with CLL/SLL without del(17p)/*TP53* mutation who discontinue a cBTKi for intolerance can remain off treatment (if there is no disease progression) for an extended period of time and alternate BTKi can be reinitiated when symptomatic. When transitioning from cBTKi to BCL2i or ncBTKi in the setting of disease progression, treatment-free interval should be as short as possible to avoid a tumor flare reaction. It is recommended to continue cBTKi through venetoclax dose ramp-up to reduce the risk of tumor flare when transitioning from cBTKi to BCL2i.

BCL2 Inhibitor

TLS was an important TEAE of venetoclax in early clinical trials. Several trials including AMPLIFY suggest that debulking with a BTKi is effective in reducing the risk of TLS prior to venetoclax.[230] In the AMPLIFY trial, the incidences of grade ≥3 TLS were lower (<1%) for acalabrutinib + venetoclax and acalabrutinib + venetoclax + obinutuzumab compared to FCR/BR (3%).[33] Similarly, debulking with obinutuzumab prior to initiation of venetoclax has been suggested as a safe and effective strategy to reduce the risk of TLS in patients with previously untreated CLL, although obinutuzumab itself can also lead to TLS.[231]

Other TEAEs associated with BCL2i-containing regimens are summarized in Table 5. In the AMPLIFY trial, AVO was associated with higher rates of infections compared to other two treatment arms.[33] Growth factor support should be considered for patients with neutropenia. Dose reduction may be necessary for patients with persistent neutropenia and limited bone marrow involvement.

Initiation at lower dose (20 mg for 1 week) and gradual step-wise ramp-up over 5 weeks to target dose (400 mg daily) along with TLS prophylaxis is recommended to mitigate the risk and frequency of TLS.[232] Initiation and accelerated dose escalation (20–400 mg over 3 weeks) with close inpatient monitoring for TLS can be done in patients with high tumor burden and concern for rapid disease progression on or following BTKi therapy.[151,233,234] Administration of obinutuzumab prior to initiation of venetoclax should be considered for further cytoreduction. Switching to a cBTKi is recommended for patients with intolerance to BCL2i.

CAR T-Cell Therapy

In the primary safety analysis of the TRANSCEND CLL 004 study, cytokine release syndrome (CRS) and neurologic events were the AESI reported in 85% (grade 3, 9%) and 45% (grade 3, 18%) of patients, respectively.[175] Headache (29%), confusional state (26%), and dizziness (25%) were the most common neurologic events. Tocilizumab and/or corticosteroids were used in 67% and 33% of patients respectively, for the management of CRS and neurologic events. Neutropenia (60%), anemia (52%), thrombocytopenia (41%), and infections (17%) were the other most common grade ≥3 AEs. Second primary malignancies were reported in 9% of patients, but none were related to treatment with lisocabtagene maraleucel. The safety results after 24-month follow-up were similar to those reported in the primary safety analysis.[176]

The combination of lisocabtagene + ibrutinib was associated with lower incidences of grade ≥3 CRS (4% vs. 8%) and neurologic toxicity (11% vs.19%) compared to lisocabtagene maraleucel monotherapy.[177] Ibrutinib-related TEAEs including hypertension and atrial fibrillation were observed in 7% and 2% of patients, respectively.

CRS and neurologic toxicity should be managed based on the toxicity grade as outlined in the *Management of CAR T-Cell–Related Toxicities* section of the NCCN Guidelines for the Management of CAR T-Cell and Lymphocyte Engager-Related Toxicities (available at www.NCCN.org).

PI3K Inhibitors

Hepatotoxicity (transaminase elevations), severe diarrhea or colitis, pneumonitis, opportunistic infections, and febrile neutropenia were observed following treatment with idelalisib or duvelisib.

Hepatotoxicity is a major concern in younger patients with CLL treated with idelalisib as first-line therapy.[235] Close monitoring of transaminase levels is essential and concurrent administration of idelalisib or duvelisib with other hepatotoxic drugs should be avoided.

The addition of anti-CD20 mAb or chemoimmunotherapy to idelalisib increases the risk of febrile neutropenia.[236] Anti-infective prophylaxis for herpes simplex virus (HSV) and *Pneumocystis jirovecii* pneumonia (PJP), and monitoring for cytomegalovirus (CMV) reactivation are recommended for patients receiving idelalisib or duvelisib.

Anti-CD20 Monoclonal Antibodies

Rare complications such as mucocutaneous reactions including paraneoplastic pemphigus, Stevens-Johnson syndrome, lichenoid dermatitis, vesiculobullous dermatitis, and toxic epidermal necrolysis can occur following treatment with anti-CD20 mAb. Consultation with a dermatologist is recommended for management of these complications.

A rapid infusion over 90 minutes can be used if no severe infusion-related reactions were experienced with the prior cycle of rituximab. Obinutuzumab infusion-related reactions (initial reaction and reactions in patients with high ALC) can be severe and patients should be monitored closely. Premedication with a corticosteroid, antihistamine, and acetaminophen should be considered to mitigate infusion-related reactions. Monitoring and prophylaxis for TLS is recommended for patients with high ALC.

Re-challenge with the same anti-CD20 mAb is not recommended in patients experiencing aforementioned severe reactions to the chosen anti-CD20 mAb (rituximab or obinutuzumab). There are some data (based on clinical experience) showing that substitution with an alternative anti-CD20 mAb is tolerated in patients experiencing severe reactions to a specific anti-CD20 mAb.[237,238] However, it is unclear if such a substitution poses the same risk of recurrence.

Rituximab and hyaluronidase human injection for subcutaneous use is approved by the FDA for the treatment of patients with CLL based on the results of the SAWYER trial in which subcutaneous rituximab (rituximab with recombinant human hyaluronidase) had similar pharmacokinetic characteristics as IV rituximab when used in combination with fludarabine and cyclophosphamide.[239] Rituximab and hyaluronidase human injection for subcutaneous use may be substituted for intravenous rituximab in patients who received at least one full dose of intravenous rituximab without experiencing severe TEAEs.

Histologic Transformation

Histologic transformation of CLL/SLL to diffuse large B-cell lymphoma (DLBCL; also known as Richter transformation) or Hodgkin lymphoma (HL) occurs in about 2% to 10% of patients during the course of their disease and treatment.[240-242]

Richter transformation has a poor prognosis with moderate response rates to chemoimmunotherapy and a median survival of 5 to 12 months from diagnosis, although the median survival was significantly better for patients with previously untreated CLL.[243-247] Histologic transformation to HL is clinically less aggressive than Richter transformation but it is associated with a poorer prognosis than de novo HL.[248-251] Accelerated CLL or CLL with expanded proliferation centers may be diagnosed when proliferation centers in CLL are expanded or fused together and show a high Ki-67 proliferative rate (>40%). Progression to CLL with increased prolymphocytes may occur when there are increased prolymphocytes in the blood (>15%). Neither of these findings is considered as Richter transformation, but rather as progression of CLL, associated with a more aggressive disease course and poorer outcomes and optimal management has not been established.[241,242,252,253]

Diagnosis

The diagnosis of histologic transformation should be confirmed by excisional or incisional biopsy of the suspected lymph node/lesion with an increased standardized uptake value (SUV) on PET scan (particularly if discordant with the remaining disease). Core needle biopsy (multiple biopsies may be required) in conjunction with appropriate ancillary techniques may be sufficient for diagnosis when lymph node/lesion is not accessible for excisional or incisional lymph node biopsy. CD19, CD20, CD22, PAX5, MUM1, and LEF1 are the most commonly expressed immunohistochemical markers, whereas CD5 and CD23 are variable.[254,255]

Whole body PET/CT scan is recommended to identify the optimal site for nodal biopsy, and biopsies should be directed to lesions with highest standardized uptake value (SUV) on PET scan.[126,256-258] A maximum SUV (SUV_{max}) ≥10 on PET scan has been shown to be a valid marker to distinguish Richter transformation from CLL among patients mostly treated with chemoimmunotherapy or chemotherapy.[259,260] Among patients who developed Richter transformation after ibrutinib therapy, the median SUV_{max} was 15 for patients who developed Richter transformation compared with an SUV_{max} of 8 for those who did not develop Richter transformation.[260] However, other studies have reported that SUV_{max} ≥10 alone lacks both sensitivity and specificity to distinguish Richter transformation from CLL in patients who develop Richter transformation while on ibrutinib.[261,262] In both these studies, biopsy-proven Richter transformation was diagnosed in patients who had an SUV_{max} between 5 and 10, suggesting that PET alone is insufficient and lymph node biopsy is required for the definitive diagnosis of Richter transformation. Biopsy of lesions with SUV_{max} ≥5 is recommended if there is high clinical suspicion, or the lesion has a differentially higher uptake than the remaining disease.

Workup

The workup of patients with histologic transformation should include history and physical exam with attention to node-bearing areas, including Waldeyer's ring, and the size of liver and spleen, whole-body PET/CT scan, or chest/abdomen/pelvis CT with contrast of diagnostic quality.

Epstein-Barr virus (EBV) infection has been reported in 16% of patients with Richter transformation and is associated with a poor outcome.[263] EBV infection of CLL can produce Reed-Sternberg (RS)-like proliferations, and presence of morphologic RS cells in a CLL background should not be considered as Richter transformation. However, RS-like cells in a background of CLL may progress to classical HL in some patients.[264] Biopsy specimen should be evaluated for EBV infection using LMP1 staining or EBV-encoded RNA in situ hybridization (EBER-ISH).

Richter Transformation

Richter transformation is characterized by immunoblastic morphology and non-germinal center B-cell immunophenotype; however, cell of origin does not seem to have prognostic implications.[254] The exact mechanism of Richter transformation is not well understood; however, it has been associated with molecular characteristics of the underlying CLL/SLL. The following molecular characteristics have been associated with the risk of developing Richter transformation and may be linked to the pathogenesis of the disease[265-274]:

- IGHV-unmutated;
- Stereotyped B-cell receptor (BCR) subset 8 combined with IGHV4-39 usage;
- Cytogenetic abnormalities detected by FISH such as del(17p) and CK (≥3 clonal chromosome abnormalities); and
- Genetic abnormalities such as *TP53* and *NOTCH1* mutations, *C-MYC* activation, or inactivation of *CDKN2A/B*.

The incidence of Richter transformation increases with the number of prior chemoimmunotherapy (CIT) regimens, and the rate is higher following treatment with a combination of purine nucleoside analogues and alkylating agents.[272] Richter transformation has also been reported following treatment with ibrutinib and venetoclax.[260,275-278] Richter transformation developing after treatment with ibrutinib lacked resistance to *BTK* and *PLCG2* mutations.[277] Age <70 years, progression on treatment, *TP53* aberrations (del[17p] or *TP53* mutation), elevated LDH, and lymphadenopathy without lymphocytosis were independent prognostic variables for Richter transformation at progression in patients who received treatment with ibrutinib for CLL.[260,278] Lymph node biopsy should be considered to rule out Richter transformation in patients with disease progression on ibrutinib, an elevated LDH, or disease progression with lymphadenopathy without lymphocytosis.[260] While the rate of Richter transformation during venetoclax therapy was significantly higher among patients with heavily pretreated del(17p) CLL, it was less common among a broader group of patients with less heavily pretreated CLL/SLL.[276] Further studies are needed to determine the exact risk profile and mechanism of Richter transformation following treatment with targeted therapies.

Richter transformation can either be clonally related (78%) or clonally unrelated to underlying CLL/SLL (22%).[244,247,271,274,279-282] The majority of patients with clonally related Richter transformation carry IGHV-unmutated and have previously treated CLL/SLL.[279,282] Clonally unrelated Richter transformation is DLBCL that is more commonly seen in previously untreated CLL/SLL and at least in one study, has been found to carry mutations associated more with clonally related RT, such as *TP53, MYC, ATM,* and *NOTCH1*, in the absence of mutations frequently involved in de novo DLBCL.[281] The prevalence of *TP53* mutation, however, was significantly lower in clonally unrelated Richter transformation (23% vs. 60%).[271] Clonally unrelated Richter transformation has a significantly longer median survival than clonally related Richter transformation (62 vs.14 months).[271,281] Although clonally related Richter transformation has inferior OS compared to clonally unrelated Richter transformation,[247,271,274,280-282] clonal relatedness has not been established as an independent predictor of clinical outcomes in multivariate analysis.[283] Clonal relatedness retained its prognostic significance (along with *TP53* mutation status, international prognostic index and double-hit status of DLBCL) for OS in one series (n = 58),[282] however, in another series (n = 86), clonal relatedness lost its prognostic significance in multivariate analysis and only the lack of *TP53* aberrations was associated with a significant survival advantage.[271] Another series also reported older age, elevated LDH, Richter transformation after previously treated CLL and CK in CLL/SLL as independent predictors of shorter OS in multivariate analysis.[247]

Molecular analysis is useful to determine if there is a clonal relationship between baseline CLL/SLL tumor cells and histologically transformed tumor cells. IGHV gene rearrangement studies of CLL and histologically transformed tissue are used to establish the clonal relationship between CLL and transformed DLBCL.[271,279] Comparative clonality testing for IGHV gene rearrangement should be done on prior defined CLL sample and biopsy specimen of transformed tissue to establish clonal relatedness between CLL/SLL and DLBCL cells, using a lab-based assay for IGHV clonality. However, initiation of therapy should not be delayed for clonality testing. If the clonal relation is unknown or if clonality testing is not feasible, treatment as described for clonally unrelated Richter transformation should be considered in patients with previously untreated CLL/SLL.

Treatment Options

Enrollment in a clinical trial is recommended for all patients with Richter transformation to DLBCL. PET/CT scan is required for response assessment and should be interpreted via the 5-Point Scale (5-PS).

CIT regimens recommended for DLBCL typically result in poor outcomes.[243] In the absence of a suitable clinical trial, the following CIT regimens used at the NCCN Member Institutions are mostly based on data from single-arm phase I/II studies as summarized in Table 6.

Elevated platelet counts, higher hemoglobin levels, lower beta-2-microglobulin levels and LDH levels have been identified as independent predictors of higher response rates to CIT.[243] However, the use of these prognostic variables for the selection of optimal CIT for Richter transformation has not been established.

Although there are limited published data supporting the use of non-CIT regimens in patients with Richter transformation (and additional data will be forthcoming), given the unmet clinical need and the lack of effective treatment options, the Panel acknowledged that inclusion of non-CIT regimens as treatment options is reasonable for Richter transformation.

Immune check point inhibitor (ICI)-based regimens (nivolumab or pembrolizumab ± ibrutinib, zanubrutinib + tislelizumab, venetoclax + atezolizumab + obinutuzumab), BTKis (acalabrutinib [category 2B] and pirtobrutinib), and bispecific antibodies (epcoritamab and glofitamab) are included as options for non-CIT regimens. Chimeric antigen receptor (CAR) T-cell therapy (axicabtagene ciloleucel, tisagenlecleucel, and lisocabtagene maraleucel) is an option for eligible patients who have received at least one prior systemic therapy regimen.

ICI-based regimens nivolumab and pembrolizumab (either as monotherapy or in combination with ibrutinib) have demonstrated activity in patients with Richter transformation in phase II clinical trials and ORR are higher when used in combination with ibrutinib.[284-288]

In a phase II study of 24 patients with Richter transformation, nivolumab + ibrutinib resulted in an ORR of 42%. The median DOR and median OS were 15 months and 13 months, respectively.[287] In a phase II trial of 26 patients with Richter transformation, pembrolizumab monotherapy resulted in an ORR of 23% (8% CR and 15% PR) with a median DOR of 3 months and the median PFS was 4 months.[288] In an expanded analysis of 15 patients who received ibrutinib combination with pembrolizumab, the ORR was 63% (25% CR and 38% PR). The median DOR and PFS were 4.5 and 8 months, respectively. After a median follow-up of 50 months, the median OS and the 2-year OS rate for all 26 patients with RT was 12 months and 27%, respectively.

In a phase II trial of (N= 57; 48 patients included in the full efficacy analysis), zanubrutinib + tislelizumab was also effective resulting in an ORR of 58% (19% CR) in patients with Richter transformation.[289] After the median follow-up of 26 months, the median DOR and PFS were 17 months and 10 months, respectively. The median OS was not reached, and the estimated 24-month OS rate was 63%.

In another multicenter phase II trial of 24 patient with Richter transformation, venetoclax in combination with atezolizumab, and obinutuzumab resulted in an ORR of 68%.[290] The 12-month PFS and OS rates were 43% and 64% respectively. In the post-hoc subgroup analyses, number of prior therapies for CLL/SLL and prior treatment with BTKi were predictors of OS.

BTKi Monotherapy

In a phase I/II trial of 25 patients with Richter transformation (treatment naïve or previously treated), acalabrutinib (cBTKi) resulted in an ORR of 40% and the median PFS was 3 months.[291]

In the BRUIN phase I/II study that included 82 patients with heavily pretreated Richter transformation (including prior therapy with chemoimmunotherapy and cBTKi), pirtobrutinib (ncBTKi) resulted in an ORR of 50% (13% CR and 37% PR) and pirtobrutinib was also effective as a bridge to transplant in eight patients with ongoing response who discontinued treatment to receive HCT.[292]

Bispecific Antibody Therapy

In the EPCORE CLL-1 trial (n = 42; majority of patients had received ≥1 prior lines of therapy with BTKi and/or BCL2 inhibitor for CLL or Richter transformation), after a median follow-up of 23 months, epcoritamab (subcutaneous CD3 x CD20 bispecific antibody) resulted in an ORR of 48% (40% CR) and the median duration of CR was 10 months.[293] The median PFS and OS were 3 months and 13 months respectively for the overall study population.

In a phase I/II study of 11 patients (majority of the patients had Richter transformation that is refractory to prior therapy with anti-CD20 mAb containing regimens), fixed-duration treatment with glofitamab (CD3 x CD20 bispecific antibody) induced durable complete remissions, resulting in an ORR of 64% (46% CR).[294]

CAR T-Cell Therapy

In a real-world analysis of 30 patients with Richter transformation, at median follow-up of 12 months, lisocabtagene maraleucel resulted in an ORR of 76%; the median DOR, PFS, and OS were not reached.[295] The estimated rates of DOR, PFS, and OS were 77%, 54%, and 67%, respectively. Lisocabtagene maraleucel was also well tolerated with low incidences of grade ≥3 CRS and immune effector cell-associated neurotoxicity syndrome (ICANS).

In an international multicenter retrospective analysis (N= 69; 58 patients had received prior lines of therapy with BTKi and/or BCL2 inhibitor for CLL and/or Richter transformation), CAR T-cell therapy resulted in an ORR of 63% (46% CR) with a median duration of response of 28 months in patients achieving CR.[296] Axicabtagene ciloleucel, tisagenlecleucel, and lisocabtagene maraleucel were administered in 64% (n = 44), 25% (n = 17), and 10% (n = 7) of patients, respectively. After a median follow-up of 24 months, the median PFS and OS were 5 months and 9 months, respectively. The 2-year PFS and OS rates were 29% and 38%, respectively.

HCT

In a non-randomized comparative analysis, the estimated cumulative 3-year survival rate was significantly higher for patients who underwent allogeneic HCT after achieving a CR or PR compared with those with treatment-sensitive disease who did not undergo allogeneic HCT, or who underwent allogeneic HCT for relapsed or refractory Richter transformation (75% vs. 27% and 21%, respectively; P = .019).[243]

In a retrospective analysis that evaluated the outcome after autologous or allogeneic HCT in 59 patients with Richter transformation, the 3-year estimated OS, relapse-free survival (RFS), and cumulative incidences of relapse and non-relapse mortality rates were 36%, 27%, 47%, and 26%, respectively, for allogeneic HCT and 59%, 45%, 43%, and 12%, respectively, for autologous HCT.[297] In a multivariate analysis, treatment-sensitive disease and reduced-intensity conditioning were found to be associated with superior RFS after allogeneic HCT. In a Center for International Blood and Marrow Transplant Research registry study evaluating outcomes after HCT (autologous HCT, n = 53; allogeneic HCT, n = 118), the 3-year PFS and OS rates were 43% and 52%, respectively, for patients who underwent allogeneic HCT.[298] The corresponding 3-year PFS and OS rates were 48% and 57%, respectively, for patients who underwent autologous HCT.

Treatment-sensitive disease and ≤3 previous lines of therapy were associated with superior PFS and OS outcomes following allogeneic HCT with reduced-intensity conditioning.[299] Deeper remissions at the time of transplant was associated with better survival outcomes after allogeneic HCT.[298]

Allogeneic HCT can be considered for patients with disease responding to treatment and autologous HCT may also be appropriate for patients with who are not candidates for allogeneic HCT (due to age, comorbidities, or lack of a suitable donor).[243,297-301]

Richter Transformation from Untreated CLL or Clonally Unrelated Richter Transformation

CIT is recommended for all patients. It is reasonably safe to combine cBTKi (given continuously for disease control in patients with previously treated CLL/SLL) with CIT when given without the addition of other targeted agents.

Observation is appropriate for patients achieving CR to CIT. CAR T-cell therapy or other non-CIT regimens (as discussed above) are recommended for patients achieving a PR, refractory or progressive disease. Allogeneic HCT can be considered for patients achieving a CR to non-CIT regimens.

Richter Transformation from Previously Treated CLL (Clonally Related or Clonal Relation Unknown)

CIT or non-CIT regimens (other than CAR T-cell therapy, preferred) are included as treatment options. In patients with previously treated CLL, the selection of treatment for Richter transformation depends on treatment received for CLL/SLL prior to transformation:

- ncBTKi, ICI-based combination regimens or bispecific antibody therapy are appropriate options for transformation after prior therapy with a cBTKi and/or BCL2i-based regimens.

- cBTKi is an appropriate option for transformation after prior therapy with BCL2i-based regimens.

- ICI-based combination regimens with cBTKi used for prior therapy for CLL/SLL are not appropriate for patients who are intolerant to or have disease progression on the same cBTKi.

Allogeneic HCT can be considered for patients achieving a CR to any regimen. Continuation of treatment until progression is recommended for patients achieving CR to non-CIT regimens. Observation may also be appropriate for patients achieving CR, although allogeneic HCT is typically considered for patients who are appropriate candidates.

CAR T-cell therapy or other non-CIT regimens (not previously used) are recommended for patients achieving a PR. Continuation of non-CIT regimen until progression or switching to an alternate non-CIT regimens is recommended if they are not a candidate for HCT or CAR T-cell therapy.

Clinical trial, CAR T-cell therapy or other non-CIT regimens are recommended for patients with refractory or progressive disease.

Histologic Transformation to Hodgkin Lymphoma

ABVD (doxorubicin, bleomycin, vinblastine, and dacarbazine) was the most commonly used regimen resulting in an ORR of 68%, and achievement of CR to the ABVD regimen was the most important factor predicting survival of patients with Richter transformation to HL.[302-304]

Richter transformation to HL should be treated as outlined in the NCCN Guidelines for Hodgkin Lymphoma (available at www.NCCN.org).

Supportive Care

Infections

Infectious complications are influenced by the progressive reduction in immunoglobulin levels (hypogammaglobulinemia) and are more common in patients with previously treated CLL.[305,306] Patients with heavily pretreated fludarabine-refractory CLL have high susceptibility to developing serious infections.[307]

IVIG is associated with a significant decrease in the occurrence of infections but with no improvement in OS outcome.[308-312] Monitoring IVIG levels and monthly administration of IVIG (0.3–0.5 g/kg to maintain nadir levels of approximately 500 mg/dL) is recommended for selected patients with serum IVIG <500 mg/dL and recurrent sinopulmonary infections requiring intravenous antibiotics or hospitalization.

Anti-infective prophylaxis is also appropriate for the management of infections in patients who may be susceptible to certain infections due to a given treatment regimen. Antiinfective prophylaxis (herpes virus prophylaxis with acyclovir or equivalent), PJP prophylaxis with sulfamethoxazole trimethoprim, or equivalent is recommended for patients receiving purine-analog or bendamustine-based chemoimmunotherapy, idelalisib, corticosteroids, and/or alemtuzumab during treatment and thereafter.

Hepatitis B Virus Reactivation

Hepatitis B virus (HBV) reactivation was reported following treatment with chemotherapy ± immunotherapy agents.[313,314] HBV carriers have a high risk of HBV reactivation. Fulminant hepatitis, hepatic failure, and death associated with HBV reactivation occurred in patients receiving anti-CD20 mAb-containing regimens. Patients receiving IVIG may be HBcAb positive as a consequence of IVIG therapy.[315]

Antiviral prophylaxis and monitoring are recommended for patients receiving anti-CD20 mAb, alemtuzumab, purine analogs, and idelalisib. Prophylactic antiviral therapy with entecavir is recommended for patients who are HBsAg positive and undergoing anti-lymphoma therapy. Entecavir is more effective than lamivudine in preventing rituximab-associated HBV reactivation.[316,317] Lamivudine prophylaxis should be avoided due to the risks for the development of resistance. The appropriate duration of prophylaxis remains undefined, but the Panel recommended that surveillance and antiviral prophylaxis should be continued for up to 12 months after the completion of treatment.[318]

HBV reactivation and invasive fungal infections were rarely reported in following treatment with ibrutinib.[319,320] There currently are no sufficient data to recommend routine screening and prophylaxis.

Cytomegalovirus Reactivation

Clinicians should be aware of the high risk of CMV reactivation in patients receiving fludarabine-based chemoimmunotherapy, idelalisib, or alemtuzumab. Monitoring for the presence of CMV viremia using quantitative PCR (at least 2–3 weeks) is an effective approach to the management of CMV reactivation.[321] Current practices include the use of prophylactic ganciclovir if CMV viremia is present or the use of ganciclovir if the viral load is found to be increasing during therapy.[322,323] Consultation with an infectious disease expert may be necessary.

Autoimmune Cytopenias

Autoimmune hemolytic anemia (AIHA), immune-mediated thrombocytopenia (also known as immune thrombocytopenic purpura [ITP]), and PRCA are the most frequent autoimmune cytopenias in patients with CLL.[324,325] Bone marrow evaluation is recommended to confirm the diagnosis of autoimmune cytopenias.

Although the direct antiglobulin test (DAT) was used for the diagnosis of AIHA, some patients with AIHA have a negative DAT; additional markers such as low haptoglobin and elevated reticulocyte and LDH are required to confirm the diagnosis of AIHA.[326] Patients with advanced disease, IGHV-unmutated, increased serum beta-2 microglobulin level, and high expression of ZAP-70 are also at a higher risk of developing AIHA.[326-329] Purine analog-based therapy was associated with AIHA. Higher incidence of AIHA was reported following treatment with fludarabine or chlorambucil compared to those who received fludarabine-based combination regimens.[326,330] AIHA should not preclude the use of regimens containing fludarabine. However, patients should be observed carefully and fludarabine therapy should be avoided in those where a history of fludarabine-associated AIHA is suspected.

ITP in patients with CLL is associated with poorer survival independent of common clinical prognostic variables.[331] High white blood cell (WBC) count, *IGHV*-unmutated, positive DAT, and ZAP-70 positivity are associated with the development of ITP in patients with CLL.[331]

AIHA and ITP can be managed with corticosteroids in most cases. IVIG, cyclosporine,[332] and splenectomy should be used in steroid-refractory cases. Rituximab was also effective for the treatment of patients with autoimmune cytopenias.[333-337] Eltrombopag and romiplostim are FDA-approved for the treatment of thrombocytopenia in patients with ITP that is refractory to steroids, IVIG, and splenectomy and were also shown to be effective in the management of CLL-associated ITP that is refractory to standard therapies.[338-342]

PRCA is less common in patients with CLL. PRCA can be managed with corticosteroids, cyclophosphamide, cyclosporine, or anti-thymocyte globulin.[325] Corticosteroids tend to be less effective in PRCA than in ITP or AIHA. In very refractory cases, allogeneic HCT may be necessary. Evaluation of parvovirus B19 is also recommended for all patients with

PRCA since patients with evidence of parvovirus B19 infection usually respond well to IVIG.[325]

Tumor Flare Reactions

Tumor flare reaction associated with lenalidomide is typically observed as painful enlargement of lymph nodes, and may be accompanied by lymphocytosis, spleen enlargement, low-grade fever, rash, and/or bone pain.[343] In patients with relapsed or refractory CLL, the 25-mg initial dose of lenalidomide used in patients with multiple myeloma resulted in excessive toxicity (tumor flare, tumor lysis, and myelosuppression).[344] Initiation of lenalidomide at lower doses (5, 10, or 15 mg/day) with subsequent dose escalation by 5 mg up to a maximum of 25 mg/day is associated with an acceptable tolerability profile in patients with relapsed or refractory CLL.[345]

The Panel recommends the use of steroids to manage lymph node enlargement and inflammation, and antihistamines to manage rash/pruritus in patients who experience tumor flare reactions. Tumor flare prophylaxis with steroids may be considered for the first 10 to 14 days of therapy in patients with bulky lymph nodes (>5 cm). Severe tumor flare reaction is generally rare if an anti-CD20 mAb is initiated at least 1 week prior to the start of lenalidomide in patients with CLL treated with combination regimen.

Venous Thromboembolism

Lenalidomide may also be associated with venous thromboembolism (VTE) in patients with CLL/SLL. Prophylaxis with daily low-dose aspirin (81 mg daily) may be considered in patients with extremely high platelet counts at baseline. Patients already on anticoagulants, such as warfarin, do not need aspirin. However, it should be noted that these recommendations may differ from the NCCN Guidelines for Cancer-Associated Venous Thromboembolic Disease (available at www.NCCN.org) in which the recommendations for VTE associated with lenalidomide pertain only to patients with multiple myeloma.

Tumor Lysis Syndrome

Patients with bulky lymph nodes, progressive disease after small-molecule inhibitor therapy, and receiving chemoimmunotherapy, venetoclax, lenalidomide, and obinutuzumab are considered to be at high risk for TLS. TLS prophylaxis as noted in the *Supportive Care* section of the algorithm should be considered for these patients.

Immunizations

All live vaccines including live attenuated influenza vaccine should be avoided. Protein and conjugate vaccines were shown to induce better responses than plain polysaccharide vaccines.[346,347] Studies that evaluated the safety and efficacy of mRNA-based vaccines against the SARS-CoV-2 infection (COVID-19) in patients with hematologic malignancies reported lower seroconversion rates and decreased antibody responses in patients with CLL/SLL, regardless of their treatment status.[348-352] The correlation, if any, between antibody titers against spike protein and the protective immunity in this population was not established, and the duration of any protection is unknown. Therefore, no recommendations can be made regarding antibody testing or actions based on antibody test results. Furthermore, tests are not available to assess cellular immunity post-COVID-19 vaccination.

Annual influenza vaccine, pneumococcal vaccine, and COVID-19 vaccine are recommended for all patients as recommended by the CDC Guidelines. Zoster vaccine recombinant, adjuvanted is recommended for all patients with CLL treated with BTKi. Respiratory syncytial virus (RSV) vaccine (single dose) is recommended for all patients, including patients <60 years.

Patients with CLL tend to have poor response to influenza vaccine and should be counseled to exercise care during influenza season even with vaccination. In the absence of laboratory testing to confirm immune response to mRNA-based vaccines, patients with CLL/SLL who received COVID-19 vaccines should take precautions recommended for unvaccinated individuals, such as mask wearing, social distancing, and diligent hand hygiene, until additional data are available to further clarify their risk.

Summary

The choice of first-line treatment for CLL/SLL should be based on the disease stage, presence or absence of del(17p) or *TP53* mutation, IGHV mutation status, patient’s age, PS, comorbid conditions, duration of treatment (continuous or time-limited) and the agent’s toxicity profile. In addition to these factors, the type of prior first-line therapy, duration of remission, and acquired resistance to treatment are also important factors in the selection of treatment for relapsed or refractory CLL/SLL.

The benefit and risk of a continuous versus time-limited (fixed-duration or MRD-guided) treatment approach should also be carefully evaluated. Venetoclax-containing regimens are time-limited, whereas BTKis are given continuously until disease progression or intolerance. Chemoimmunotherapy regimens can be considered for first-line therapy only when cBTKi and BCL2i are not available or treatment with these targeted therapies are not feasible. Testing for *BTK* and *PLCG2* mutations may be useful to confirm resistance to BTKis in patients with disease progression or no response while on BTKi therapy.

Histologic transformation to DLBCL or HL is associated with a poor prognosis. Precise diagnosis and enrollment in clinical trials evaluating novel targeted agents will improve the clinical outcomes of patients with histologic transformation.

Careful monitoring of AEs after initiation of treatment and supportive care for the treatment-related complications should be an integral part in the treatment of patients with CLL/SLL and histologic transformation.

Table 1: Undetectable MRD Rates for Time-limited BCL2i-containing regimens for Treatment-Naïve CLL/SLL

Trial	Regimen	No. of Patients	Patient Characteristics	Median Follow-up	Undetectable MRD ($\leq 10^{-4}$, uMRD4)	PFS	MRD detection
GLOW (Phase III; Fixed-duration)[27]	**Ibrutinib/Venetoclax**	106	≥65 years or <65 years who also had CIRS >6 or CrCl <70 mL/min	34 months	EOT+3 (BM): 67% among patients with a CR; 64% among patients with a PR;	EOT +12: 96% (uMRD4); 93% (dMRD)	NGS
	Chlorambucil + Obinutuzumab	105			EOT+3 (BM): 31% among patients with a CR; 21% among patients with a PR;	EOT +12: 83% (uMRD4); 59% (dMRD)	
FLAIR (Phase III; MRD-guided)[29,a]	**Ibrutinib**	263	Median age 62 years (>65 years, 31%) [without del(17p)]	62 months	60 months: 4% (blood); 0% (BM)	5-year, 79%	Flow Cytometry
	Ibrutinib/Venetoclax	260			60 months: 93% (blood); 72% (BM)	5-year, 94%	
	Fludarabine, Cyclophosphamide + Rituximab (FCR)	263			60 months: 69% (blood); 55% (BM)	5-year, 58%	
CAPTIVATE (Phase II; Fixed-duration)[31]	**Ibrutinib/Venetoclax**	159	≤70 years; ECOG PS 0–1	69 months	Cycle 7: 54% (blood); EOT: 69% (blood and BM)	5.5 year, 66%; [75% (uMRD4) 47% (dMRD)]	Flow Cytometry
AMPLIFY (Phase III; Fixed-duration)[33]	**Acalabrutinib/Venetoclax**	291	Median age: 61 years [without del(17p) or *TP53* mutation])	41 months	EOT: 45% (blood); EOT+3: 38% (blood)	36-month, 77%	Flow Cytometry
	Acalabrutinib/Venetoclax + Obinutuzumab (AVO)	286			EOT: 95% (blood); EOT+3: 94% (blood)	36-month, 83%	
	FCR or Bendamustine + Rituximab (BR)	290			EOT: 73% (blood); EOT+3: 78% (blood)	36-month, 67%	
Phase II (MRD-Guided)[34,c]	**Acalabrutinib/Venetoclax + Obinutuzumab (AVO)**	72 (n = 45, *TP53* aberration)	≥18 years; ECOG PS 0–1	55 months	Cycle 16 (BM): 78% (71% for those with *TP53* aberration) Cycle 16 (blood): 83% (82% for those with *TP53* aberration)	4-year, 70% (*TP53*) 4-year, 96% (*TP53* wildtype)	NGS; Flow Cytometry
SEQUOIA (Phase III; ARM D; MRD-guided)[92,d]	**Zanubrutinib/Venetoclax** (Non-Randomized Cohort)	66 (*TP53* aberration)	≥65 years of age OR unsuitable for FCR (CIRS >6; CrCl <70 mL/min or a history of severe/multiple infections within 2 years);	39 months	59% (blood)	24-month, 94%	Flow Cytometry
		47 (without *TP53* aberration)		30 months	60% (blood)	24-month, 89%	
CLL14 (Phase III; Fixed-duration)[45]	**Venetoclax + Obinutuzumab (Veno)**	216	≥65 years (CIRS >6; CrCl <70 mL/min)	65 months	EOT+2: 75% (blood)	5-year, 63%	ASO-PCR; Flow Cytometry; NGS
	Chlorambucil + Obinutuzumab	216			EOT+2: 33% (blood)	5-year, 27%	

BM, bone marrow; dMRD, detectable MRD ($\geq 10^{-4}$); EOT, end of treatment; EOT+3, 3 months after end of treatment; EOT+12, 12 months of end of treatment; MRD, measurable residual disease; NGS, next-generation sequencing; PS, performance status;

a. The duration of combined ibrutinib/venetoclax was determined based on serial MRD monitoring and treatment discontinuation in patients meeting with the uMRD4-guided stopping criteria.
b. Discontinuation of treatment was allowed for patients achieving uMRD4 after 24 cycles.
c. Patients could discontinue treatment if CR with uMRD4 in the bone marrow was achieved at the start of cycle 16.
d. Discontinuation of treatment (venetoclax after a minimum of 12 cycles or zanubrutinib after a minimum of 27 cycles) was allowed in patients meeting with the uMRD4-guided stopping criteria

Table 2: Small-Molecule Inhibitor Therapy for Treatment-Naïve CLL/SLL

Trial	Regimen	No. of Patients	Patient Characteristics	Median Follow-up	ORR	PFS	OS
ELEVATE-TN[20]	**Acalabrutinib**	179 [del(17p) and/or mutated *TP53,* n = 23]	≥65 years or <65 years with comorbidities (CIRS >6; CrCl <70 mL/min); ECOG PS of ≤2 and adequate hematologic, hepatic, and renal function	75 months	90% (19% CR)	Median, Not reached (72-month, 62%)	Median, Not reached (72-month, 76%)
	Acalabrutinib + Obinutuzumab	179 [del(17p) and/or mutated *TP53,* n = 25]			96% (37% CR)	Median, Not reached (72-month, 78%)	Median, Not reached (72-month, 84%)
	Chlorambucil + Obinutuzumab	177 [del(17p) and/or mutated *TP53,* n = 25]			83% (14% CR)	Median, 28 months (72-month, 17%)	Median, Not reached (72-month, 75%)
RESONATE-2[18]	**Ibrutinib**	136	≥65 years [without del(17p)]	8 years	92% (34% CR)	Median, Not reached (7-year, 59%)	Median, Not reached (7-year, 78%)
	Chlorambucil	133			37%	Median, 15 months (7-year, 9%)	Not reported
Alliance North American Intergroup (A041202)[43]	**Ibrutinib**	182	≥65 years	55 months	93% (7% CR)	4-year, 76%	4-year, 85%
	Ibrutinib + Rituximab	182			94% (12% CR)	4-year, 76%	4-year, 86%
	Bendamustine + Rituximab	183			81% (26% CR)	4-year, 47%	4-year, 84%
E1912 study[23]	**Ibrutinib + Rituximab**	354	≤70 years	70 months	96% (17% CR)	5-year, 78%	5-year, 95%
	Fludarabine, Cyclophosphamide + Rituximab (FCR)	175			81% (30% CR)	5-year, 51%	5-year, 89%
SEQUOIA [without del(17p)][21,22]	**Zanubrutinib**	241 (mutated *TP53,* n = 15)	≥65 years of age OR unsuitable for treatment with FCR (CIRS >6; CrCl <70 mL/min or a history of severe/multiple infections within 2 years)	61 months	95% (7% CR)	Median, Not reached (60-month, 76%; *P* < .0001)	Median,: Not reached (60-month, 86%)
	Bendamustine + Rituximab	238 (mutated *TP53,* n = 13)			85% (15% CR)	Median, 44 months (60-month, 40%)	Median, Not reached (60-month, 85%)
SEQUOIA (ARM C) [with del(17p)][135,149]	**Zanubrutinib** (Non-Randomized Cohort)	111		77 months	97% (18% CR)*	72-month, 64%	72-month, 83%
SEQUOIA (ARM D)[92]	**Venetoclax/Zanubrutinib** (Non-Randomized Cohort)	66 (*TP53* aberration)		39 months	99% (47% CR)	24-month, 94% 36-month, 88%	Median, Not reached (24 month, 96%)
		47 (without *TP53* aberration)		30 months	96% (49% CR)	24-month, 89%	

CR, complete response; ORR, overall response rate; OS, overall survival; PS, performance status; PFS, progression-free survival
*5-year follow-up: Tam CSL et al. J Clin Oncol 2025;43:Abstract 7011.

Table 2 (continued): Small-Molecule Inhibitor Therapy for Treatment-Naïve CLL/SLL

Trial	Regimen	No. of Patients	Patient Characteristics	Median Follow-up	ORR	PFS	OS
CLL14[24,130]	**Venetoclax + Obinutuzumab**	216 [del(17p), n = 17; deleted or mutated *TP53,* n = 25]	≥65 years with comorbidities (CIRS >6; CrCl <70 mL/min)	76 months	85% (50% CR)	Median, 76 months 6-year, 53% (*P* < .0001)	6-year, 79%
	Chlorambucil + Obinutuzumab	216 [del(17p), n = 14; deleted or mutated *TP53,* n = 24]			71% (23% CR)	Median, 36 months 6-year, 22%	6-year, 69%
AMPLIFY[33]	**Venetoclax/Acalabrutinib + Obinutuzumab (AVO)**	286	Median age, 61 years [without del(17p) or *TP53* mutation])	41 months	93%	36-month, 83%	36-month, 88%
	Venetoclax/Acalabrutinib	291			93%	36-month, 77%	36-month, 94%
	Fludarabine/Cyclophosphamide + Rituximab (FCR) or Bendamustine + Rituximab (BR)	290			75%	36-month, 67%	36-month, 86%
GLOW [27,28]	**Venetoclax/ibrutinib**	106 (mutated *TP53,* n = 7)	≥65 years or <65 years who also had CIRS >6 or CrCl <70 mL/min	64 months	87% (39% CR)	60-month, 60%	60-month, 82%
	Chlorambucil + Obinutuzumab	105 (mutated *TP53,* n = 2)			36% (11% CR)	60-month, 18%	60-month, 61%
FLAIR[29]	**Ibrutinib**	263	Median age, 62 years (>65 years, 31%) [without del(17p)]	62 months	87%	5-year, 79%	5-year, 91%
	Venetoclax/ibrutinib	260			88%	5-year, 94%	5-year, 96%
	FCR	263			78%	5-year, 58%	5-year, 87%
GAIA/CLL13[25,26]	**Venetoclax/ibrutinib + Obinutuzumab**	231	≤65 years or >65 years [without del(17p) or *TP53* mutation]	51 months	94% (62% CR)	4-year, 86%	4-year, 95%
	Venetoclax + Obinutuzumab	229			96% (57% CR)	4-year, 82%	4-year, 95%
	Venetoclax + Rituximab (VenR)	237			93% (49% CR)	4-year, 70%	4-year, 96%
	Chemoimmunotherapy (FCR ≤65 Years; BR >65 Years)	229			81% (31% CR)	4-year, 62%	4-year, 94%

CR, complete response; ORR, overall response rate; OS, overall survival; PS, performance status; PFS, progression-free survival

Table 3. Phase III Randomized Studies of Small-Molecule Inhibitor Therapy for Relapsed or refractory CLL/SLL

Trial	Regimen	No. of Patients	Patient Characteristics	Median Follow-up	ORR	PFS	OS
ASCEND[167]	**Acalabrutinib**	155 [del(17p), n = 28; mutated *TP53,* n = 39]	Median age, 67–68 years with ECOG PS ≤2 and adequate hematologic, hepatic, and renal function	47 months	83%	Median, Not reached 42-month, 62% (HR, 0.28; *P* < .0001)	Median, Not reached 42-month, 78%
	Idelalisib + Rituximab (IdR) or Bendamustine + Rituximab (BR)	155 (IdR, n = 119; BR, n = 36); [del(17p), n = 21; mutated *TP53,* n = 34]			84%	Median, 17 months 42-month, 23%	Median, Not reached 42-month, 65%
ELEVATE-RR[134]	**Acalabrutinib**	268	≥18 years; ECOG PS ≤2 and the presence of del(17p) and/or del(11q)	41 months	81% (3% CR)	Median, 38 months (for both treatment arms)	Median, Not reached (in either arm)
	Ibrutinib	265			77% (4% CR)		
RESONATE[168]	**Ibrutinib**	195 [del(17p), n = 63; mutated *TP53,* n = 79]	Median age, 67 years	74 months	91% (11% CR)	Median, 44 months 60-month, 40%	Median, 68 months
	Ofatumumab	196 [del(17p), n = 64; mutated *TP53,* n = 68]			–	Median, 8 months 60-month, 3%	Median, 65 months
ALPINE[169]	**Zanubrutinib**	327 [del(17p) and/or mutated *TP53,* n = 41]	Median age, 67 years; ECOG PS ≥1; relapsed or refractory disease after ≥1 prior systemic therapy	42.5 months	86% (12% CR)	36-month, 65% (HR, 0.67; *P* = .002)	36-month, 83%
	Ibrutinib	325 [del(17p) and/or mutated *TP53,* n = 38]			75% (8% CR)	36-month, 54%	36-month, 80%
MURANO[96,98]	**Venetoclax + Rituximab**	194 [del(17p), n = 46; mutated *TP53,* n = 48]	≥18 years; ECOG PS 0–1; relapsed or refractory disease requiring therapy and adequate bone marrow, liver, and kidney function	59 months	92% (8% CR)	Median, 54 months (HR, 0.19; *P* < .0001)	5-year, 82% (HR, 0.40; *P* < .0001)
	Bendamustine + Rituximab	195 [del(17p), n = 46; mutated *TP53,* n = 51]			72% (4% CR)	Median, 17 months	5-year, 62%

CR, complete response; ORR, overall response rate; OS, overall survival; PS, performance status; PFS, progression-free survival

Table 4. Treatment-Emergent Adverse Events (TEAEs) of BTKis

TEAEs	Treatment-Naïve CLL			Relapsed or Refractory CLL				
	ELEVATE-TN[19,20]	RESONATE-2[18]	SEQUOIA[21]	ELEVATE-RR[134]		ALPINE[169]		BRUIN[172,174]
	Acalabrutinib	Ibrutinib	Zanubrutinib	Acalabrutinib	Ibrutinib	Zanubrutinib	Ibrutinib	Pirtobrutinib
Most common TEAEs (all grades)								
Diarrhea	43%	50%	14%	35%	46%	19%	26%	16%*
Headache	39%	–	11%	35%	20%	NR	NR	17%
Cough	25%	36%	11%	29%	21%	NR	NR	24%
Fatigue	24%	36%	11%	20%	17%	11%	15%	32%
Arthralgia	27%	26%	14%	16%	23%	17%	19%	–
Anemia	–	26%	4%	22%	19%	17%	19%	21%*
Neutropenia	13%	13% (Grade ≥3)	16%	21%	25%	32%	30%	16%*
Adverse events of special interest (AESI)								
Atrial fibrillation/Flutter								
Any grade	9%	16%	3%	9%	16%	7%	17%	3%*
Grade ≥3	2%	5%	<1%	5%	3%	3%	5%	<1%
Bleeding								
Any grade	45%	NR	45%	38%	51%	20%	22%	43%
Grade ≥3	5%	NR	4%	–	–	–	–	1%
Major bleeding								
Any grade	6%	11%	5%	–	–	–	–	21%
Grade ≥3	5%	7%	4%	–	–	–	–	1%
Hypertension								
Any grade	11%	23%	14%	9%	23%	27%	25%	6%*
Grade ≥3	5%	8%	6%	4%	9%	17%*	16%*	<1%
Infections								
Any grade	80%	26%	62%	–	–	29%*	20%*	71%
Grade ≥3	28%	–	16%	–	–	2%	1%	4%

*Data from BRUIN CLL-321 study

Table 5. Treatment-Emergent Adverse Events (TEAEs) of BCL2 Inhibitor-Containing Regimens

TEAEs (All Grades)	Treatment-Naïve CLL					Relapsed or Refractory CLL	
	CLL14[130]	GAIA/CLL13[25]	FLAIR[29]	AMPLIFY[33]		MURANO[150]	M13-982[159]
	VenO	VenO	Venetoclax/ Ibrutinib*	AVO	Venetoclax/ Acalabrutinib	VenR	Venetoclax
Neutropenia	58%	49%	12%	40%	31%	61%	46%
Thrombocytopenia	24%	18%	21%	5%	9%	13%	23%
Anemia	17%	8%	24%	5%	7%	16%	27%
Infusion-related reaction	45%	51%	<1%	20%	0%	8%	NR
Diarrhea	28%	33%	44%	36%	33%	40%	44%
Nausea	19%	NR	36%	22%	15%	22%	38%
Constipation	13%	NR	11%	8%	7%	14%	13%
Pyrexia	23%	24%	NR	16%	6%	15%	19%
Fatigue	15%	NR	38%	14%	15%	18%	27%
Cough	16%	NR	9%	NR	NR	18%	19%
Headache	11%	NR	15%	28%	35%	11%	18%

*Grade 1–2 TEAE; NR, Not reported

Table 6: Chemoimmunotherapy for Richter Transformation

Regimen	No. of Patients	Median Follow-up	ORR	Median PFS	OS
CHOP (Cyclophosphamide/Doxorubicin/Vincristine/Prednisone) + Rituximab[353]	15	69 months	67% (7% CR)	10 months	Median, 21 months
CHOP/Venetoclax + Rituximab[354]	42	15 months	71% (52% CR)	16 months	Median, NR; 63% at 2 years
Dose-adjusted EPOCH (Etoposide/Prednisone/Vincristine/Cyclophosphamide/Doxorubicin) + Rituximab[355]	46	39 months	39%	4 months	Median, 6 months
HyperCVAD (Cyclophosphamide/Vincristine/Liposomal Daunorubicin/Dexamethasone) + Rituximab alternating with Cytarabine + High-dose Methotrexate[356]	30	8 months	43% (27% CR)	—	1-year, 28%
OFA (Oxaliplatin/Fludarabine/Cytarabine + Rituximab)[357]	20	9 months	50%	—	6-month, 53%
Modified OFA + Rituximab[358]	35	26 months	39% (7% CR)	—	Median, 7 months 2-year: 20%

CR, complete response; ORR, overall response rate; OS, overall survival; PS, performance status; PFS, progression-free survival

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